

**DESIGNING A HYDROGEN PRODUCTION PLANT WITH INTEGRATED
BIOCHAR CARBON CAPTURE SYSTEM TO PRODUCE 10,000M³/HR OF
HYDROGEN VIA STEAM METHANE REFORMING**

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**DESIGN OF A HYDROGEN PRODUCTION PLANT WITH INTEGRATED
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HYDROGEN VIA STEAM METHANE REFORMING**

FINAL YEAR DESIGN PROJECT

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GROUP F

**A DESIGN PROJECT SUBMITTED TO
DEPARTMENT OF CHEMICAL ENGINEERING
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DESIGN PROJECT TOPIC

“DESIGN OF A HYDROGEN PRODUCTION PLANT WITH INTEGRATED BIOCHAR CARBON CAPTURE SYSTEM TO PRODUCE 10,000M³/HR OF HYDROGEN VIA STEAM METHANE REFORMING”. Nigeria is currently faced with a growing demand for clean and sustainable energy to meet industrial, domestic, and transportation needs. Hydrogen is widely recognized as a future energy carrier due to its high energy density, clean combustion, and applicability in fuel cells and industrial processes. Conventional hydrogen production through steam methane reforming (SMR) remains the most widely adopted technology, but it is also one of the largest contributors to carbon dioxide emissions. To address both the need for hydrogen and the challenge of greenhouse gas emissions, an integrated hydrogen production system with biochar-based carbon capture provides a sustainable pathway.

DECLARATION

We, the group members of this design hereby declared that this project titled “**DESIGN OF A HYDROGEN PRODUCTION PLANT WITH INTEGRATED BIOCHAR CARBON CAPTURE SYSTEM TO PRODUCE 10,000M³/HR OF HYDROGEN VIA STEAM METHANE REFORMING**” was carried out by us in the Department of Chemical Engineering under the supervision of Engr. Dr. Jibril Mohammed and Dr. K. A. Abdulyekeen, in partial fulfillment of the award of Bachelor of Engineering (B. Eng.) in Chemical Engineering and to the best of our knowledge it has not been presented or submitted anywhere. And, all the materials consulted during the course of this work were duly acknowledged.

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CERTIFICATION

This design project titled “**DESIGN OF A HYDROGEN PRODUCTION PLANT WITH INTEGRATED BIOCHAR CARBON CAPTURE SYSTEM TO PRODUCE 10,000M³/HR OF HYDROGEN VIA STEAM METHANE REFORMING**” has been read and certified as meeting the requirement for the award of the degree of B.Eng. (Chemical Engineering), Department of Chemical Engineering, Abubakar Tafawa Balewa University, Bauchi, and has been approved for its contribution to scientific and literary presentation.

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DEDICATION

This design project is dedicated to God who in His infinite mercy made us to see the end of this design project in a good health and give us the opportunity to write this project and to our individual parents for their unending guidance, support and prayers through our entire studies. May almighty God bless them.

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ABSTRACT

The global shift toward clean and sustainable energy has intensified interest in hydrogen as a versatile energy carrier due to its high energy density and zero-carbon combustion. However, conventional hydrogen production via Steam Methane Reforming (SMR) emits significant volumes of carbon dioxide, contributing to industrial greenhouse gas accumulation. This challenge underscores the need for an integrated system that ensures efficient hydrogen production while minimizing carbon emissions. Therefore, this project was undertaken to design a hydrogen production plant integrated with a biochar-based carbon capture system capable of producing 10,000 m³/hr of hydrogen from natural gas. The main objectives were to develop a complete process design covering feed preparation, reforming, shift conversion, purification, and CO₂ capture; perform material and energy balances; evaluate biochar as a sustainable CO₂ adsorbent; and assess the economic feasibility of the plant. The process operated at 850–900 °C and 28 atm, achieving hydrogen purity of 99.9% through Pressure Swing Adsorption (PSA) and over 90% CO₂ capture efficiency using biochar. Material and energy balances indicated an overall hydrogen yield of 75–80% with improved energy recovery through heat integration. The total capital investment was estimated at USD 80.51 million (₦119.16 billion) with an annual profit of USD 21.27 million and a payback period of 3.8 years. In conclusion, the integration of biochar-based carbon capture into SMR significantly enhances environmental sustainability while maintaining strong economic viability, demonstrating a feasible and eco-friendly pathway for large-scale hydrogen production in Nigeria's emerging clean energy economy.

TABLE OF CONTENT

TITLE PAGE	ii
DECLARATION	iv
CERTIFICATION	v
DEDICATION	vi
ACKNOWLEDGEMENT	vii
ABSTRACT.....	Viii
TABLE OF CONTENT.....	Ix
CHAPTER ONE	1
1.0 INTRODUCTION	1
1.1 Background Of The Study	1
1.2 Problem Statement.....	5
1.3 Aims And Objectives Of The Study	6
1.3.1 Aim	6
1.3.2 Objectives.....	6
1.4 Significance Of The Study.....	6
1.5 Scope Of The Study	7
1.6 Limitations Of The Study	7
CHAPTER TWO	9
2.0 LITERATURE REVIEW	9
2.1 Hydrogen As A Sustainable Energy Carrier.....	9
2.2 Hydrogen Production Technologies	11
2.2.1 Thermochemical Processes.....	11
2.2.2 Electrochemical Processes	13
2.2.3 Biological Processes	14
2.2.4 Comparative Outlook.....	15
2.3 Steam Methane Reforming (SMR) Process.....	16
2.4 Carbon Capture In Hydrogen Production	18
2.4.1 Post-Combustion Capture	18
2.4.2 Pre-Combustion Capture.....	18
2.4.3 Oxy-Fuel Combustion Capture	19
2.4.4 Emerging Solid Adsorbents	19
2.4.5 Integration Challenges And Future Outlook.....	20
2.5 Biochar As A Sustainable Carbon Capture Material	20
2.6 Integration Of Biochar In Hydrogen Production Systems.....	22

2.6.1 Process Integration.....	22
2.6.2 Environmental And Economic Advantages.....	23
2.6.3 Technical Feasibility.....	23
2.6.4 Nigerian Context.....	24
2.7 Hydrogen And Carbon Capture In Nigeria: Opportunities And Challenges	24
3.0 PROCESS AND MATERIAL SELECTION	26
3.1 Introduction.....	26
3.2 Raw Materials	26
Table 3.1: Properties Of Key Raw Materials.....	28
3.3 Existing Processes And Technologies	28
3.3.2 Block Flow Diagram.....	31
<i>Figure 3.1: Block Flow Diagram For The Process.....</i>	31
<i>Figure 3.2: Process Flow Diagram For The Process</i>	32
<i>Figure 3.3: Simulated Process Flow Diagram (PFD) Of Hydrogen Production Plant Integrated With Biochar-Based Carbon Capture System.....</i>	32
3.3.4 Process Description.....	34
3.4 Material Selection	36
CHAPTER FOUR.....	41
4.0 MATERIAL BALANCE.....	41
4.1 Introduction.....	41
<i>Fundamental Material Balance Equation</i>	41
4.2 Unit 1: Heat Exchanger.....	42
4.3 Unit 2: Hydrogenation Reactor	44
Unit Description.....	44
Material Balance Calculation.....	45
4.4 Unit 3: Desulfurization Reactor	47
Assumptions.....	47
Material Balance Calculation.....	47
4.6 Unit 5: Reformer	51
Assumptions.....	51
Material Balance Calculation.....	52
.....	54
4.7 Unit 6: Process Gas Boiler.....	54
Assumptions.....	54
4.8 Unit 7: High-Temperature Shift (HTS) Converter	55
5.0 ENERGY BALANCE.....	63
5.1 Unit 1: Heat Exchanger.....	63

5.1.1 Assumptions.....	63
5.1.2 Energy Balance Calculation.....	63
5.2 Unit 2: Hydrogenation Reactor	64
5.2.1 Assumptions.....	64
5.2.2 Energy Balance Calculation.....	65
5.3.1 Assumptions.....	67
5.4.1 Assumptions.....	69
5.4.2 Energy Balance Equation:.....	69
5.5.1 Assumptions.....	71
5.8 Unit 8: Condenser	82
5.8.1 Assumptions.....	82
Heat Removed By Condensing Water Vapor	82
Heat Removed To Cool Dry Gas From 400°C To 30°C	82
Total Heat Duty Of Condenser	82
Cooling Water Requirement	83
5.9 Unit 9: Biochar-Integrated PSA.....	83
Energy Released From CO ₂ Adsorption (Exothermic).....	83
CHAPTER SIX.....	84
6.0 EQUIPMENT DESIGN AND SPECIFICATION.....	84
6.2 Heat Exchanger (HX-01) Design.....	84
<i>Fig. 6.1 Heat Exchanger</i>	84
6.2.1 Cold Stream (Tube Side)	85
6.2.2 Hot Stream (Shell Side)	85
6.2.3 Heat Duty Calculation	86
6.2.6 Tube Bundle Design	87
6.2.7 Material Selection For Heat Exchanger.....	87
6.2.8 Final Specification Of Heat Exchanger (HX-01)	88
6.3 Hydrogenator Reactor Design	89
6.3.1 Design Basis & Assumptions	89
6.3.2 Choice Of Vessel Geometry (Single Vertical Packed-Bed)	90
6.3.3 Shell Thickness Calculation (Cylindrical Shell).....	91
6.3.4 Head Selection & Thickness.....	92
6.3.5 Internals - Distributor, Support Grid, Thermowells & Instrumentation	93
6.3.6 Summary Of Selected Geometric & Mechanical Data	93
6.4 Pre-Reformer (PR-01) Design	94
6.4.1 Design Basis /Assumptions	94
6.4.2 Volumetric Flow At Reactor Conditions	95

6.4.3 Reactor Volume (GHSV Method)	95
6.4.4 Catalyst Weight Estimate.....	96
6.4.5 Reactor Geometry (Practical Dimensions)	96
6.4.6 Pressure-Drop Estimate (Ergun Equation, Conservative)	96
6.4.7 Materials Of Construction	97
6.4.8 Final Design Table Of The Pre-Reformer	98
6.5 Reformer Design.....	99
6.5.1 Introduction.....	99
6.5.2 Gas Properties	99
6.5.3 Conversion Of Mass Flowrate To Molar Flowrate.....	100
6.5.4 Reactor Operating Conditions	100
6.5.5 Volumetric Flowrate (Ideal Gas Law).....	100
6.5.6 Gas Hourly Space Velocity (GHSV).....	100
6.5.7 Catalyst Tube Design Parameters	101
6.5.8 Number Of Tubes	101
6.5.9 Void Fraction Calculations	102
6.5.10 Packed Bed Characteristics.....	102
6.5.11 Reactor Flow Characteristics	103
6.5.12 Summary Of Design Parameters.....	104
6.5.13 Kinetic Parameters Of Steam Methane Reforming	104
6.5.14 Engineering Remarks.....	105
6.6 High Temperature Shift Converter Design	106
6.6.1 Reaction Kinetics & Thermodynamics.....	106
6.6.2 Reactor Sizing & Performance	106
6.6.5 Mechanical Design	107
6.8 Design Specification Of Pump (P-01)	116
6.9 Design Specification Of Water Tank (TK-01)	116
7.1 Introduction.....	117
7.2 Site Selection Criteria	117
7.3 Site Size Estimation	119
7.4 Conceptual Plant Layout.....	120
7.5 Layout Considerations	122
7.6 Safety And Environmental Measures	123
CHAPTER EIGHT	124
8.0 PROCESS CONTROL AND INSTRUMENTATION	124
8.1 Introduction.....	124
8.1 Objectives Of Process Control.....	125

8.2 Key Process Variables And Measurement Instruments.....	128
8.3 Control Philosophy And System Architecture.....	128
8.4 Control Systems In Major Process Units	129
8.5 Automation And Control Strategy	132
8.6 Safety Instrumented Systems (SIS)	133
8.7 Instrumentation Diagram (P&ID Overview)	134
8.7.1 Description Of The P&ID	134
8.7.2 Control Interconnections And Signal Flow	136
8.7.3 Safety And Alarm Integration.....	136
8.7.4 Standard Instrumentation Symbols Used.....	137
CHAPTER NINE.....	139
9.0 ENVIRONMENTAL MANAGEMENT AND SAFETY	139
9.1 Introduction.....	139
9.2 Environmental Impact Assessment (EIA)	139
9.3 Emission And Waste Management.....	141
9.4 Occupational Health And Safety (OHS).....	141
9.5 Process Safety Management (PSM)	142
9.6 Fire And Explosion Safety.....	143
9.7 Environmental And Safety Monitoring	144
9.8 Regulatory Compliance	145
CHAPTER TEN	146
10.0 COST ESTIMATION AND ECONOMIC ANALYSIS	146
10.1 Introduction.....	146
10.2 Basis And Assumptions	147
10.3 Methodology	147
10.4 Purchased Equipment Cost (PCE)	149
10.5 Capital Cost Estimation	149
10.6 Operating Cost And Revenue Analysis	150
10.7 Economic Evaluation.....	151
10.8 Summary Of Cost And Economic Indicators	151
CHAPTER ELEVEN.....	152
11.0 SUMMARY, CONCLUSION AND RECOMMENDATIONS.....	152
11.1 Summary.....	152
11.2 Conclusion	153
REFERENCES	157

CHAPTER ONE

1.0

INTRODUCTION

1.1 Background of the Study

Hydrogen is the most abundant element in the universe, and due to its reactivity, it only exists on earth in compounds such as water and organic materials. It is an odourless, flammable and colourless gas, which is leading to its safety concern, especially if a leak is not detected and gas collects in a confined area; it can ultimately ignite and causes explosions. Furthermore, metal hydrogen embrittlement is an issue as it could damage pipelines and containers due to its small molecular size; thus, it escapes through materials. The higher heating value (HHV) of hydrogen is 141.8 MJ/kg at 298 K, and the lower heating value is 120 MJ/kg at the same temperature. This is significantly higher than that of most fuels such as gasoline with a value of 44 MJ/kg at 298 K. However, liquid hydrogen has a lower energy density by volume than hydrocarbon fuels such as gasoline by a factor of four with a density of 8 MJ/l versus density of 32 MJ/l. While hydrogen gas has a high energy density by weight but a low energy density by volume compared to hydrocarbons, it requires a larger tank to store. For example, as opposed to liquified natural gas, liquified hydrogen contains 2.4 times the energy but takes 2.8 times the volume to store. At the same time, the low temperature for liquified hydrogen storage at ambient pressure and a temperature of $-253\text{ }^{\circ}\text{C}$ raises quite a few risks. When exposed, it can cause cold burns; furthermore, leakage can result in a combination of liquefied air and hydrogen, resulting in an explosive mixture or the formation of flammable or explosive conduits (Atilhan *et al.* 2021; El-Halwagi *et al.* 2020).

Like electricity, hydrogen is an energy carrier and not an energy source; using it to store renewable energies instead of being wasted when not in use is crucial since it is storable, utilisable and transportable (Parra *et al.* 2019; Abe *et al.* 2019).

A wide range of resources is available for hydrogen production, mainly fossil-based and renewable fuels (Dawood *et al.* 2020; Saithong *et al.* 2019; Osman *et al.* 2020a). The former is the more mature and most common used industrially as it is a cost-effective method that deploys cracking or reforming fossil-based fuels. In 2016, hydrogen production globally was about 85 million tonnes used in petroleum, metal industry, fertilizer, food processing, semiconductor production, power plants and generations (Chen and Hsu 2019; El-Emam and Özcan 2019; Acar and Dincer 2019).

There are many ways to extract hydrogen from hydrogen-containing materials, either hydrocarbon or non-hydrocarbon, such as photonic, electric, chemical, bioenergy, heat and a combination of those methods together (Abe *et al.* 2019; El-Emam and Özcan, 2019; Osman *et al.* 2020).

Water is typically purified and then sent to an electrolyser, which produces hydrogen and oxygen. The hydrogen is then dried, purified and compressed from a 10.3 to 413.7 bar pressure, and then stored in a tank. Although the electrolysis pathway offers a 100% renewable route for hydrogen production, it represents less than 5% of worldwide hydrogen production (Han *et al.* 2021). Despite this low percentage contribution, water electrolysis is gaining momentum for various reasons such as zero-carbon emissions, the absence of unwanted by-products such as sulphates, carbon oxides and nitrogen oxides, and high hydrogen purity. The cost of producing hydrogen through electrolysis would be reduced by approximately 70% over the next decade, allowing for the widespread adoption of a green hydrogen production approach.

The IEA Sustainable Development Scenario outlines a major transformation of the global energy system, showing how the world can deliver the three main energy-related Sustainable Development Goals simultaneously. Under this scenario, carbon capture technologies play an important role in providing dispatchable, low-carbon electricity – in 2040, plants with these technologies generate 5% of global power. CCUS-equipped coal and gas plants become increasingly important for secure, sustainable and affordable power systems in the IEA Sustainable Development Scenario.

Meeting climate goals also means creating an extremely flexible power system that can manage high shares of variable renewable power sources. Coal- and gas-fired power plants have been a major source of system flexibility, providing benefits essential to the operation of the electricity grid, such as inertia and frequency control. Carbon capture, storage and utilisation allows these plants to continue providing these benefits and meet long-term flexibility requirements, such as annual seasonality.

Carbon Capture, Utilization, and Storage (CCUS) technologies have emerged as critical components in the effort to reduce CO₂ emissions. These technologies are designed to capture CO₂ directly from emission sources, such as power plants and industrial facilities, preventing its release into the atmosphere (Yaashikaa *et al.*, 2023). Captured CO₂ can either be stored in deep geological formations or utilized in various applications such as chemical production or enhanced oil recovery. Recent advancements in CCUS technology, particularly in the areas of advanced sorbents, high-performance membranes, and electrochemical techniques, have significantly improved capture efficiency and cost-effectiveness (Ketabchi *et al.*, 2023; Soo *et al.*, 2024). Furthermore, the utilization of captured CO₂ to produce valuable products, such as fuels and chemicals, offers an additional incentive for industries to adopt CCUS (Alli *et al.*, 2024). Despite these

advancements, significant challenges remain, especially concerning the long-term stability of CO₂ storage and the high costs associated with these technologies.

Researching carbon capture technology is crucial since it's thought to be the only way to reduce emissions from large-scale power plants, with the potential to cut emissions by 50% by the year 2050 (Wei *et al.*, 2020; Wienchol *et al.*, 2020a). It is noteworthy that if carbon capture and storage technologies are not taken into consideration, the cost of lowering emissions will increase drastically by 140% (Kearns *et al.*, 2021). Carbon capture uses three basic technologies: pre-combustion, post-combustion, and oxyfuel combustion methods.

Biochar can be produced from organic waste materials (the carbonization of biomass), including lignocellulosic agro-food biomasses. In fact, lignocellulosic agro-food biomasses due to their enriched carbon content are introduced as one of the most significant renewable resources that can be used for biochar production. It is documented that based on type, the lignocellulosic biomass includes lignin (10–25 %), cellulose (40–60 %), and hemicellulose (15–30 %) that can undergo hydrolysis to yield fermentable sugars and are converted into fuels and chemicals such as biochar.

Biochar is mainly produced through thermochemical conversion processes such as slow pyrolysis, fast pyrolysis, and gasification. In summary, slow pyrolysis is a thermal process that in the absence of oxygen and at relatively low temperatures (usually, 300–500 °C) heats biomass to produce the biochar. This process produces biochar, along with other by-bioproducts, including bio-oil and syngas that can be utilized for biofuel/bioenergy targets . Biochar produced by fast pyrolysis, named “fast pyrolysis biochar,” is also a carbon-rich producer generated by rapidly heating biomass (more than 500 °C) in the absence of oxygen. This process also generates biochar along with bio-oil

and syngas. It should be noted that fast pyrolysis generally focuses on bio-oil production, while slow pyrolysis tries to generate more biochar. While gasification, similar to pyrolysis, can lead to biochar production by heating biomass at high temperatures in a controlled environment with a limited supply of oxygen, aiming syngas production for energy targets with less attention on biochar production.

Hydrogen production is a key component of the global transition to cleaner energy sources. However, traditional hydrogen production methods, particularly steam methane reforming (SMR), contribute significantly to carbon dioxide (CO₂) emissions. Carbon capture and storage (CCS) technologies have been developed to mitigate these emissions, but they often come with high costs and energy penalties.

Biochar, a carbon-rich material derived from biomass pyrolysis, has emerged as a promising alternative for CO₂ sequestration. Its high surface area, porosity, and functional groups make it an effective adsorbent. Integrating biochar into carbon capture systems in hydrogen production could offer a sustainable and cost-effective solution for reducing industrial carbon footprints.

1.2 Problem Statement

Hydrogen production through Steam Methane Reforming (SMR) is one of the largest contributors to global CO₂ emissions, accounting for a significant portion of industrial greenhouse gas output. Despite efforts to mitigate these emissions through conventional Carbon Capture and Storage (CCS) technologies, these methods are often capital-intensive, energy-demanding, and require complex infrastructure, making large-scale adoption economically and technically challenging. The high operational costs and long-term viability concerns further limit the effectiveness of CCS as a widespread solution.

1.3 Aims and Objectives of the Study

1.3.1 Aim

The aim of the study is to design a hydrogen production plant with an integrated biochar-based carbon capture system capable of producing 10,000 m³/hr of hydrogen via steam methane reforming (SMR).

1.3.2 Objectives

To develop a process design for producing 10,000 m³/hr of hydrogen, including feed preparation, reforming, shift conversion, purification, and carbon capture integration.

To perform material and energy balances for the hydrogen production process to determine feedstock requirements, steam-to-carbon ratio, energy efficiency, and product yields.

To integrate biochar into the carbon capture system by evaluating its CO₂ adsorption capacity, operational feasibility, and potential for local production from Nigerian biomass residues.

To select appropriate materials of construction and equipment suitable for the operating conditions of SMR and CO₂ capture units.

To analyze the environmental benefits of biochar-assisted carbon capture in reducing greenhouse gas emissions compared to conventional hydrogen production.

To evaluate the economic feasibility of the designed plant by estimating capital and operating costs, and comparing with global hydrogen production.

1.4 Significance of the Study

The transition to a low-carbon economy requires innovative solutions to reduce industrial greenhouse gas emissions while maintaining economic feasibility. Hydrogen, often

regarded as a clean energy carrier, is primarily produced through Steam Methane Reforming (SMR), a process that emits significant amounts of CO₂. Traditional Carbon Capture and Storage (CCS) technologies have been implemented to mitigate these emissions, but their high capital costs, energy demands, and infrastructure requirements present substantial barriers to widespread adoption. This study explores biochar as a low-cost, sustainable alternative for carbon capture in hydrogen production, offering potential benefits in environmental sustainability, industrial applications, and policymaking.

1.5 Scope of the Study

This study focuses on designing, optimizing, and evaluating a biochar-integrated carbon capture system for a 10,000 m³/hr hydrogen production facility. It covers system design, biochar material selection, material and energy balance, process modeling, environmental impact, and economic feasibility. The research includes selecting efficient biochar types, optimizing adsorption conditions, and using simulation tools like ASPEN Plus and MATLAB to predict system performance. Integration with hydrogen production methods such as steam methane reforming and biomass gasification is also analyzed. Additionally, the study assesses environmental benefits through life cycle analysis and evaluates economic feasibility. While primarily focused on design and modeling, experimental validation may be included if feasible.

1.6 Limitations of the Study

Factors such as the availability of biochar in sufficient quantities, integration into existing SMR processes, and the cost-effectiveness of large-scale implementation need further investigation. Additionally, the long-term stability and performance of biochar in industrial environments remain uncertain, requiring extended pilot studies and real-world applications beyond the scope of this research.

Another key limitation is the variability in biochar properties based on feedstock. Biochar's adsorption efficiency and structural characteristics depend significantly on the type of biomass used for its production, as well as the pyrolysis conditions. Variations in surface area, porosity, and chemical composition may impact its carbon capture performance, making standardization a challenge. Since this study will focus on specific biochar types and production methods, the results may not be universally applicable to all biochar variations.

CHAPTER TWO

2.0

LITERATURE REVIEW

2.1 Hydrogen as a Sustainable Energy Carrier

Hydrogen is increasingly recognized as a cornerstone of the emerging low-carbon economy, primarily because of its unique properties that make it suitable for diverse applications in the energy and industrial sectors. As an energy carrier, hydrogen has the highest gravimetric energy density of any known fuel, approximately 120 MJ/kg, which is almost three times higher than that of gasoline (Khan & Arsalan, 2016). Unlike fossil fuels, which release carbon dioxide and other greenhouse gases upon combustion, hydrogen combustion produces only water vapor, making it a clean fuel for achieving deep decarbonization in power generation, transportation, and industrial processes. This feature has elevated hydrogen's profile as a critical enabler of global net-zero emission targets and the transition to renewable energy systems.

Hydrogen's versatility extends beyond its role as a fuel. It is widely used as a feedstock in chemical industries for the production of ammonia, methanol, and refined petroleum products, as well as in metallurgy for reducing iron ore in direct reduction iron (DRI) processes (Dutta, 2014). More recently, hydrogen has found application in fuel cells, where it generates electricity efficiently without direct combustion, thus eliminating the formation of harmful pollutants such as nitrogen oxides and particulates. Its ability to store energy over long periods also makes it a promising solution for integrating intermittent renewable energy sources such as solar and wind into national grids, thereby addressing the challenge of variability in power supply (IRENA, 2022).

Despite its advantages, the global hydrogen economy faces several challenges. Currently, over 95% of hydrogen production is fossil fuel-based, predominantly through steam

methane reforming (SMR) and coal gasification (IEA, 2021). These conventional methods emit significant volumes of carbon dioxide, estimated at around 830 million tonnes per year, thereby undermining the environmental benefits of hydrogen if not coupled with carbon capture and storage (CCS) technologies (IEA, 2019). To mitigate these emissions, the development of “blue hydrogen” pathways hydrogen produced from fossil fuels with integrated CCS is being prioritized globally as a transition solution until “green hydrogen”, produced from renewable-powered electrolysis, becomes economically competitive (Bicer & Dincer, 2017).

Globally, demand for hydrogen is projected to grow more than fivefold by 2050, reaching approximately 500–600 million tonnes per year, driven by increasing applications in power generation, industrial decarbonization, and mobility (BloombergNEF, 2020). This trend underscores the urgency of developing hydrogen production systems that are both environmentally sustainable and economically feasible. For developing countries such as Nigeria, hydrogen offers an opportunity to diversify the energy mix, enhance energy security, and reduce dependence on conventional fossil fuels while leveraging abundant natural gas resources as a feedstock for low-carbon hydrogen production.

Thus, hydrogen is not merely a fuel of the future but an essential enabler of the ongoing global energy transition. Its sustainability, however, will depend largely on the extent to which production methods are decarbonized through integration with carbon capture systems and the gradual scaling of renewable hydrogen technologies. This reality provides the basis for the present study, which seeks to design a hydrogen production system capable of addressing both energy demand and environmental concerns through the integration of biochar-based carbon capture.

2.2 Hydrogen Production Technologies

Hydrogen can be produced through a variety of technological pathways that are generally classified as thermochemical, electrochemical, or biological processes. The choice of production route is strongly influenced by the availability of feedstocks, energy resources, technological maturity, cost considerations, and environmental implications. At present, thermochemical processes, particularly steam methane reforming (SMR), dominate global hydrogen supply, accounting for over 70% of total production, largely because of their established industrial infrastructure and comparatively low cost (IEA, 2025). However, with increasing emphasis on decarbonization, alternative technologies such as electrolysis and biomass-based processes are being actively developed. This section reviews the major hydrogen production technologies, highlighting their operating principles, technical maturity, advantages, limitations, and global outlook.

2.2.1 Thermochemical processes

Thermochemical methods involve the conversion of hydrocarbon-based feedstocks into hydrogen through high-temperature chemical reactions. The most common processes are steam methane reforming (SMR), coal gasification, and partial oxidation of heavy hydrocarbons.

2.2.1.1 Steam Methane Reforming (SMR)

SMR is the most widespread and cost-effective hydrogen production route. In SMR, natural gas (primarily methane) reacts with steam at 700–1,000 °C in the presence of a nickel-based catalyst to form a mixture of carbon monoxide and hydrogen, known as synthesis gas. This is followed by the water–gas shift reaction, which increases the hydrogen concentration by converting carbon monoxide into carbon dioxide and additional

hydrogen. The overall thermal efficiency of SMR ranges between 65–75%, making it highly effective for large-scale production (RSC, 2024). The major drawback is its significant carbon footprint, as each tonne of hydrogen produced generates approximately 9–12 tonnes of CO₂. To address this, carbon capture and storage (CCS) can be integrated into the process, resulting in what is known as “blue hydrogen.” CCS can capture up to 90% of CO₂ emissions, though it adds cost and complexity to the system.

2.2.1.2 Coal Gasification

Coal gasification involves reacting coal with oxygen and steam under high pressures and temperatures to produce synthesis gas. It has historically been used in regions with abundant coal reserves, such as China and South Africa, but it is carbon-intensive and produces higher emissions than SMR. Although CCS can also be applied to gasification, the process is less economically attractive compared to natural gas reforming.

2.2.1.3 Partial Oxidation (POX)

Partial Oxidation is another thermochemical process where heavy hydrocarbons or residual oil fractions are partially oxidized with oxygen to generate hydrogen-rich syngas. POX is faster than SMR but less hydrogen-selective, often requiring additional shift reactions and purification. It is mainly applied in refineries where heavy feedstocks are readily available.

Overall, thermochemical routes remain dominant in hydrogen supply, especially in countries with large fossil fuel reserves. However, their high greenhouse gas emissions necessitate integration with carbon management technologies if they are to remain viable in a carbon-constrained future.

2.2.2 Electrochemical processes

Electrochemical hydrogen production is achieved primarily through water electrolysis, where electricity is used to split water into hydrogen and oxygen. The sustainability of electrolysis depends heavily on the source of electricity: when powered by renewable energy such as solar or wind, the product is termed green hydrogen, which is essentially carbon-free.

2.2.2.1 Alkaline Electrolysis

Alkaline electrolysis is the most commercially mature electrolysis technology, having been used for decades in industrial hydrogen production. It employs a liquid alkaline electrolyte, typically potassium hydroxide, and operates at temperatures between 60 - 80 °C. Its advantages include relatively low capital cost and long system lifetimes. However, alkaline electrolyzers are less responsive to fluctuating power inputs, making them less suited for direct integration with variable renewable energy sources.

2.2.2.2 Proton Exchange Membrane (PEM) Electrolysis

This is a newer technology that uses a solid polymer electrolyte and operates at similar temperatures to alkaline systems. PEM electrolyzers offer higher current densities, faster response times, and the ability to produce high-purity hydrogen at higher pressures without additional compression. This makes PEM particularly attractive for coupling with renewable energy. Its main limitations are high costs due to the use of noble metal catalysts such as platinum and iridium, and challenges related to membrane durability.

2.2.2.3 Solid Oxide Electrolysis Cells (SOECs)

This operate at elevated temperatures between 700–1,000 °C and use ceramic electrolytes. SOECs achieve very high electrical efficiencies (up to 90% in some pilot studies) by utilizing waste heat from industrial processes or high-temperature reactors to reduce the amount of electricity required for water splitting. However, they are still at the pilot stage, with unresolved issues related to material degradation and system reliability.

Currently, the cost of hydrogen from electrolysis is higher than from fossil-based methods, averaging between 3–6 USD/kg compared to 1–2.5 USD/kg for SMR (IEA, 2025). However, rapid improvements in electrolyzer technology, scaling up of production, and the declining cost of renewable electricity are projected to bring green hydrogen within competitive range by 2030.

2.2.3 Biological processes

Biological processes rely on microorganisms to produce hydrogen from organic substrates. These methods are considered renewable and environmentally friendly but are currently limited to laboratory or pilot-scale research due to low yields and technical challenges.

2.2.3.1 Dark Fermentation

This involves anaerobic bacteria breaking down organic matter such as agricultural residues, wastewater, or food waste to produce hydrogen and organic acids. While inexpensive feedstocks are an advantage, hydrogen yields are relatively low, typically 1–2 mol H₂ per mol glucose equivalent.

2.2.3.2 Photofermentation

employs photosynthetic bacteria to convert organic acids into hydrogen under light conditions. This process has the potential for higher yields than dark fermentation but requires continuous light supply and precise control of process conditions, which complicates scale-up.

2.2.3.3 Photobiological Hydrogen Production

Which uses algae or cyanobacteria that directly split water into hydrogen and oxygen using sunlight. Although conceptually appealing as a direct solar-to-hydrogen conversion pathway, the efficiency is extremely low (less than 1%), and the systems are not yet economically viable.

A more technologically advanced approach is biomass gasification, which overlaps with thermochemical methods. Here, solid biomass or municipal solid waste is converted into synthesis gas, which can then be shifted and purified to produce hydrogen. When combined with carbon capture, biomass gasification can produce hydrogen with net-negative emissions, making it particularly attractive in the context of climate change mitigation. However, challenges such as feedstock variability, tar formation, and high capital cost remain significant barriers to commercialization.

2.2.4 Comparative outlook

Globally, hydrogen production remains dominated by thermochemical methods, particularly SMR, due to their cost advantage, high efficiency, and industrial maturity. However, these processes are carbon-intensive and incompatible with long-term climate goals unless integrated with carbon capture technologies. Electrolysis offers the most

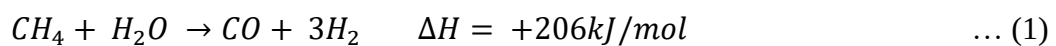
sustainable pathway for the future, producing hydrogen with minimal environmental impact when powered by renewable electricity. Its competitiveness is expected to increase rapidly as electrolyzer costs decline and renewable penetration expands. Biological processes, while still at an early stage, hold promise for decentralized and waste-to-energy applications, particularly in developing countries with abundant agricultural residues and municipal solid waste.

For Nigeria, where natural gas reserves are abundant and renewable energy development is still limited, the near-term focus is likely to be on blue hydrogen via SMR integrated with carbon capture systems such as biochar. In the longer term, investments in renewable-powered electrolysis and biomass-based hydrogen production could complement SMR, creating a diversified and sustainable hydrogen economy.

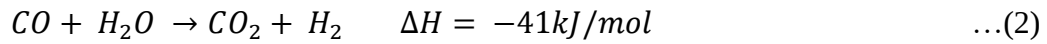
2.3 Steam Methane Reforming (SMR) Process

Steam methane reforming (SMR) remains the dominant and most technologically mature process for large-scale hydrogen production, accounting for more than 60–70% of global supply (IEA, 2022). The process involves the catalytic conversion of methane (CH₄), the principal component of natural gas, with steam (H₂O) at elevated temperatures (700–1000 °C) and moderate pressures (20–30 atm). Nickel-based catalysts are typically employed due to their cost-effectiveness and activity for C–H bond activation. The overall process produces syngas, a mixture primarily composed of hydrogen (H₂), carbon monoxide (CO), and a smaller fraction of carbon dioxide (CO₂).

The chemistry of SMR can be summarized by two key reactions. The primary reforming step is strongly endothermic and is represented in Eqn (1):



This reaction requires a continuous input of heat, usually supplied by burning additional natural gas in the furnace section of the reformer. The second critical reaction is the water–gas shift (WGS) reaction, which occurs downstream and is moderately exothermic as shown in Eqn 2:



The net outcome of combining the two steps is a high yield of hydrogen with CO₂ as the main by-product. Modern SMR plants can achieve hydrogen yields of 70–85%, with process efficiency depending on steam-to-carbon ratio, reformer temperature profile, pressure, and catalyst life cycle (Zheng *et al.*, 2021).

Commercial SMR units are designed with two main sections: the primary reformer and the secondary reformer. In the primary reformer, tubular reactors packed with nickel catalysts facilitate the initial endothermic reforming reaction. The heat required for the process is provided by burners arranged around the reformer tubes. The partially reformed gas is then directed to the secondary reformer, where additional oxygen or air is introduced to complete the conversion of hydrocarbons and optimize hydrogen yield.

Despite its maturity, the SMR process has significant drawbacks. Foremost among these is the large volume of CO₂ generated, which accounts for approximately 9–12 kg of CO₂ per kg of H₂ produced, making SMR one of the most carbon-intensive methods of hydrogen generation (IEA, 2021). To mitigate this environmental burden, integration of carbon capture and storage (CCS) or carbon capture and utilization (CCU) technologies has been increasingly proposed. Recent advances have highlighted the potential of biochar-based sorbents as a low-cost and sustainable option for capturing CO₂ from SMR flue gases (Zhou *et al.*, 2023).

2.4 Carbon Capture in Hydrogen Production

The steam methane reforming (SMR) process, while the most dominant technology for hydrogen generation, is inherently carbon-intensive, producing between 9–12 kg of CO₂ per kg of hydrogen generated (IEA, 2022). This substantial carbon footprint presents a critical barrier to positioning hydrogen as a sustainable energy carrier unless paired with robust carbon capture solutions. Consequently, Carbon Capture, Utilization, and Storage (CCUS) technologies have been increasingly integrated with hydrogen production processes to enable the transition toward “blue hydrogen,” a lower-carbon alternative to conventional “grey hydrogen” derived from natural gas without capture.

2.4.1 Post-combustion capture

Post-combustion capture is the most commercially deployed CCUS strategy. It involves extracting CO₂ from flue gases after fuel combustion, where the concentration of CO₂ is typically between 4–15%. Solvent-based absorption using amines such as monoethanolamine (MEA) remains the most mature technology in this category. While effective, these solvents suffer from significant drawbacks, including high regeneration energy requirements, corrosion, and solvent degradation (Bui *et al.*, 2018). To address these limitations, advanced solvents (e.g., sterically hindered amines, biphasic solvents) and membrane-based separation techniques are being developed to improve energy efficiency and lower operational costs (Rochelle, 2021).

2.4.2 Pre-combustion capture

Pre-combustion capture occurs upstream of fuel combustion, typically during syngas production in hydrogen or power plants. In this approach, the carbon monoxide generated during SMR is shifted to CO₂ and H₂ via the water-gas shift reaction. The resulting gas

mixture, rich in CO₂ and H₂, allows for easier CO₂ separation because of its higher partial pressure compared to post-combustion streams. Physical solvents such as Selexol and Rectisol are commonly employed for CO₂ absorption in pre-combustion systems. Although more efficient than post-combustion capture, pre-combustion technologies require major process modifications and are better suited to new-build facilities rather than retrofits (Boot-Handford *et al.*, 2014; IEA, 2021).

2.4.3 Oxy-fuel combustion capture

Oxy-fuel combustion is an alternative capture strategy where fuel is burned in nearly pure oxygen instead of air. This produces a flue gas primarily composed of CO₂ and water vapor, which simplifies downstream CO₂ separation. The main challenge lies in the high cost and energy intensity of oxygen production, typically via cryogenic air separation. Research into membrane-based oxygen generation and chemical looping combustion is ongoing to reduce the energy penalty associated with this approach (Mondal *et al.*, 2022).

2.4.4 Emerging solid adsorbents

In recent years, solid adsorbents have gained attention as promising alternatives to solvent-based systems. Zeolites and metal–organic frameworks (MOFs) exhibit high CO₂ selectivity, tunable pore structures, and regenerability, although they may be sensitive to moisture and thermal cycling. More recently, biochar derived from biomass residues has emerged as a low-cost, sustainable, and locally available adsorbent material. Biochar's high porosity, surface functional groups, and ability to be produced from waste biomass make it particularly attractive for CO₂ capture in developing economies such as Nigeria, where biomass resources are abundant (Zhou *et al.*, 2023). Moreover, biochar can serve dual purposes: capturing CO₂ while contributing to soil amendment and carbon

sequestration if applied in agricultural systems, thereby aligning with circular economy principles.

2.4.5 Integration challenges and future outlook

Despite rapid advancements, large-scale integration of CCUS with hydrogen production still faces technical and economic challenges. High capital costs, energy penalties, and uncertainties around long-term CO₂ storage remain barriers to widespread adoption. However, combining hydrogen generation with biochar-based carbon capture offers an opportunity for cost reduction and sustainability in regions with strong agricultural sectors and biomass availability. Future development will likely rely on hybrid capture strategies, combining solvents, membranes, and adsorbents, to balance efficiency, cost, and environmental impact (IPCC, 2022).

2.5 Biochar as a Sustainable Carbon Capture Material

Biochar has emerged as a promising material for carbon capture, owing to its abundance, low production cost, and multifunctional environmental benefits. It is a carbon-rich solid produced from the thermochemical conversion of biomass through pyrolysis under oxygen-limited conditions. The resulting structure is highly porous, thermally stable, and rich in surface functional groups such as hydroxyl, carboxyl, and carbonyl moieties, which significantly enhance its adsorption affinity for CO₂ (Ahmad *et al.*, 2019). Compared with advanced sorbents such as metal–organic frameworks (MOFs) or zeolites, biochar offers a cost-effective and sustainable alternative, particularly in regions with abundant agricultural residues like Nigeria.

One of the major advantages of biochar is its versatility in production. Feedstocks such as crop residues, forestry wastes, animal manure, and municipal solid waste can be converted

into biochar, thereby addressing both waste management and climate mitigation objectives. The pyrolysis process conditions - including temperature, heating rate, and residence time strongly influence the physical and chemical properties of biochar. For instance, higher pyrolysis temperatures generally increase surface area and microporosity, thereby improving CO₂ adsorption capacity, whereas lower temperatures enhance oxygen-containing functional groups that aid chemisorption (Qambrani *et al.*, 2017; Zhang *et al.*, 2020).

From a carbon capture standpoint, biochar exhibits CO₂ adsorption capacities ranging from 1 to 4 mmol CO₂/g, depending on the feedstock and surface modification techniques (Lehmann & Joseph, 2015; Zhou *et al.*, 2023). Chemical activation with potassium hydroxide (KOH), amination, or impregnation with metal oxides has been shown to further improve capture performance, bringing biochar's adsorption capacity closer to that of advanced sorbents while retaining its cost advantage (Wang *et al.*, 2021).

Beyond capture efficiency, biochar has an intrinsic carbon sequestration potential. During pyrolysis, a significant fraction of the biomass carbon is locked in a stable, aromatic form that resists microbial decomposition. When reapplied to soils, biochar not only sequesters carbon for centuries but also improves soil fertility, water retention, and microbial activity. This dual benefit makes biochar a unique negative-emissions technology, as it simultaneously removes CO₂ from the atmosphere and enhances agricultural productivity (Lehmann *et al.*, 2021).

In the context of hydrogen production, integrating biochar into SMR-based systems provides a pathway to decarbonize the process. Biochar can be employed either as a direct CO₂ adsorbent in post-combustion capture units or as part of sorption-enhanced reforming processes, where CO₂ is continuously removed to shift equilibrium toward higher

hydrogen yields. This integration enables hydrogen plants to achieve “blue hydrogen” or even approach “carbon-negative hydrogen” when the sequestration benefits of biochar are accounted for. Recent pilot studies have highlighted the feasibility of biochar-based carbon capture in industrial gas systems, demonstrating its potential scalability for large hydrogen production plants (Xu *et al.*, 2022).

Overall, biochar represents a sustainable, locally adaptable, and cost-effective solution for carbon capture in hydrogen production. Its integration not only mitigates the carbon footprint of SMR but also aligns with circular economy principles by valorizing agricultural waste streams into high-value climate solutions.

2.6 Integration of Biochar in Hydrogen Production Systems

The integration of biochar into hydrogen production systems represents a transformative strategy that addresses both energy generation and climate mitigation challenges. In conventional steam methane reforming (SMR), hydrogen is purified through water–gas shift (WGS) reactors and pressure swing adsorption (PSA) units, while carbon dioxide (CO₂) is released as a by-product. By incorporating biochar-based adsorption beds into this configuration, CO₂ emissions can be significantly reduced, thereby enabling the transition from carbon-intensive “grey hydrogen” toward low-carbon or even carbon-negative hydrogen.

2.6.1 Process integration

In an integrated SMR–biochar system, methane is first reformed with steam to produce syngas, which is subsequently conditioned in WGS reactors to maximize hydrogen yield. The hydrogen-rich stream is separated in PSA units, leaving behind a CO₂-rich off-gas. Instead of venting this CO₂, it can be passed through biochar-packed adsorption columns,

where the porous structure and functional groups of biochar selectively capture CO₂ molecules. This not only prevents emissions but also improves the overall efficiency of the hydrogen plant by reducing the load on conventional capture technologies such as amine scrubbing (Chen *et al.*, 2021).

2.6.2 Environmental and economic advantages

The integration of biochar in hydrogen production confers multiple advantages. First, it directly reduces greenhouse gas emissions, addressing one of the main criticisms of SMR as a hydrogen production pathway. Second, biochar can be produced locally from abundant agricultural residues such as rice husks, maize cobs, groundnut shells, and forestry waste, which are widely available in Nigeria. This creates a sustainable value chain that links energy production with waste management. Third, the reuse of biochar after CO₂ capture as a soil amendment provides a dual benefit: long-term carbon sequestration and enhancement of agricultural productivity. This aligns with circular economy principles, where waste streams are valorized into value-added products.

2.6.3 Technical feasibility

Recent studies have demonstrated the technical feasibility of biochar integration in industrial gas separation systems. Zhang *et al.* (2020) reported that modified biochar exhibited CO₂ uptake capacities comparable to zeolites and activated carbons, making it suitable for PSA applications. Similarly, Chen *et al.* (2021) highlighted that biochar-based sorbents could reduce capture costs due to their lower regeneration energy compared with conventional chemical solvents. Pilot-scale experiments further suggest that biochar can withstand repeated adsorption–desorption cycles, an essential requirement for industrial viability (Xu *et al.*, 2022).

2.6.4 Nigerian context

For Nigeria, the integration of biochar into hydrogen production offers a unique opportunity to tackle multiple national challenges simultaneously. With an estimated 25 million tonnes of municipal solid waste generated annually and large volumes of agricultural residues often left unused or burned, biomass resources are abundant and underutilized. Harnessing these feedstocks for biochar production could provide low-cost sorbents for carbon capture while creating new economic opportunities in rural communities. Moreover, adopting such a system would position Nigeria at the forefront of sustainable hydrogen production in Africa, contributing to its commitments under the Paris Agreement and long-term climate targets.

2.7 Hydrogen and Carbon Capture in Nigeria: Opportunities and Challenges

Nigeria occupies a strategic position in Africa's energy landscape due to its vast endowment of natural gas and agricultural resources. With proven natural gas reserves exceeding 200 trillion cubic feet, the country has the potential to establish itself as a regional hub for hydrogen production through steam methane reforming (SMR) (NNPC, 2022). The parallel abundance of agricultural residues—such as rice husks, maize stalks, cassava peels, and groundnut shells creates a sustainable feedstock base for large-scale biochar production, which can be integrated into carbon capture systems. This unique convergence of fossil and biomass resources offers Nigeria a rare opportunity to pursue hydrogen production in a low-carbon framework.

From an opportunity perspective, the integration of biochar into hydrogen plants provides multiple socio-economic and environmental benefits. Firstly, the use of locally available residues reduces dependence on expensive imported carbon capture sorbents, making the

technology more cost-competitive. Secondly, Nigeria's agricultural sector, which generates millions of tonnes of biomass waste annually, can benefit from a new value chain where residues are transformed into biochar, supporting rural employment and promoting circular economy principles. Additionally, the deployment of hydrogen as a clean energy carrier aligns with Nigeria's Energy Transition Plan (ETP), which positions natural gas as a transition fuel while progressively decarbonizing the power, transport, and industrial sectors. Hydrogen production with carbon capture could also strengthen Nigeria's role in international energy markets by providing low-carbon hydrogen exports to Europe and Asia, regions actively seeking green and blue hydrogen imports (IRENA, 2021).

Despite these opportunities, several challenges remain. Infrastructure is a major bottleneck, as Nigeria's existing gas processing and distribution networks are underdeveloped and prone to inefficiencies. Establishing hydrogen production plants with integrated carbon capture will require significant capital investments, which may be difficult to secure without clear government incentives and investor-friendly policies. Technical expertise is another limiting factor, as Nigeria currently lacks a skilled workforce in advanced carbon capture technologies and hydrogen systems. Furthermore, policy and regulatory frameworks are still evolving; while the Energy Transition Plan recognizes hydrogen's potential, there are no dedicated policies or roadmaps for large-scale hydrogen and biochar integration. Environmental considerations must also be addressed to ensure sustainable biomass sourcing, as over-exploitation of agricultural residues could threaten soil fertility if biochar is not applied in balance with agricultural needs.

CHAPTER THREE

3.0 PROCESS AND MATERIAL SELECTION

3.1 Introduction

Process and material selection is a crucial aspect of chemical process design, as it determines the technical feasibility, economic viability, and environmental sustainability of a plant. For a hydrogen production plant integrated with biochar-based carbon capture, careful consideration must be given to process choice, raw material availability, and material compatibility with operating conditions. According to Sinnott and Towler (2020), engineering design involves transforming a concept into a detailed specification for equipment and operations, ensuring safety, efficiency, and long-term operability. Hydrogen production is particularly demanding because it requires high-temperature reforming units, catalytic reactors, and advanced gas separation systems, all of which must withstand severe thermal and chemical conditions. Therefore, this chapter discusses the raw materials, existing technologies, process selection, and material selection required for the successful design of the plant.

3.2 Raw Materials

The basic consideration of any processing or production plant is raw material. This is the starting feed of any process and the selection of raw material is necessary and mandatory to ensure a uniform process.

3.2.1 Factors affecting selection of raw materials

Several factors determine the suitability of raw materials in hydrogen production and carbon capture systems:

1. **Availability:** The feedstock must be abundant and accessible locally to minimize transportation costs and ensure process sustainability (IEA, 2021).
2. **Purity:** High levels of sulfur, aromatics, or other contaminants in the natural gas feed can poison catalysts in the reformer and pre-reformer units. Hence, desulfurization is essential.
3. **Energy Content:** The calorific value of the feedstock is critical since hydrogen yield depends on methane concentration in the natural gas.
4. **Environmental Impact:** Raw materials should have minimal negative effects on the environment. For instance, biochar produced from agricultural residues provides a sustainable CO₂ adsorbent and contributes to carbon sequestration.
5. **Economic Cost:** Both feedstock and auxiliary materials (such as water and steam) must be affordable to ensure competitive production costs.

3.2.2 Choices of raw materials

1. **Natural Gas:** The primary feedstock for the process. Nigerian Liquefied Natural Gas (NLNG) supplies natural gas with a composition of approximately 90–95% methane, with traces of ethane, propane, nitrogen, and minimal sulfur compounds (NNPC, 2022). Its high methane content makes it suitable for steam methane reforming (SMR).
2. **Steam:** Supplied through process boilers, steam acts as the oxidant in the reforming reaction. A steam-to-carbon (S/C) ratio of 2.5–3.5 is typically maintained to suppress carbon deposition (Zhang *et al.*, 2021).

3. **Biochar:** Produced locally from rice husk, corncob, and other agro-residues via pyrolysis at 500–700 °C. It is used in the CO₂ adsorption unit. Biochar’s surface area, porosity, and tunable surface functional groups make it an attractive low-cost carbon capture medium (Tan *et al.*, 2023).
4. **Cooling Water:** Required for gas cooling, condensation, and maintaining the operational integrity of equipment.

Table 3.1: Properties of Key Raw Materials

Raw Material	Main Composition	Density (kg/m ³)	HHV (MJ/kg)	Remarks
Natural Gas	90–95% CH ₄	0.72	50–55	High hydrogen yield potential
Steam	H ₂ O (vapor)	0.6 (@400 °C)	–	Required for reforming & WGS
Biochar (Rice Husk)	C=65–70%, O=20–25%, H=3–5%	450	~28	High surface area, strong CO ₂ affinity

3.3 Existing Processes and Technologies

Hydrogen can be produced via several established technologies, including steam methane reforming (SMR), partial oxidation (POX), biomass gasification, and electrolysis. Among these, SMR remains the dominant industrial technology, accounting for over 70% of global hydrogen production due to its maturity and economic viability (IEA, 2022).

3.3.1 Factors affecting the choice of process

The selection of steam methane reforming (SMR) as the preferred technology for hydrogen production in this design is influenced by a combination of technical, economic, and

environmental considerations. These factors are critical to ensure that the process is both sustainable and feasible within the Nigerian context.

1. Operating Conditions:

Steam methane reforming operates efficiently at elevated temperatures between 700–1000 °C and pressures of 20–30 bar, which are conditions well-established in industrial practice. These ranges are not only technically achievable but are also supported by the availability of commercially proven reactor materials that can withstand such thermal and mechanical stresses (Rostrup-Nielsen, 2019). The maturity of operating knowledge in handling these conditions ensures stable performance, reliable scale-up, and minimized technical risk compared to emerging hydrogen production technologies.

2. Feedstock Availability:

Nigeria is endowed with vast natural gas reserves, ranking among the top in Africa and globally. Leveraging this abundant resource base makes SMR an attractive option, as the process relies primarily on methane as feedstock. The widespread accessibility of natural gas in Nigeria reduces dependence on imported raw materials and ensures long-term security of supply. This advantage further enhances the sustainability of SMR in regions where alternative feedstocks, such as biomass or coal, may not be as abundant or economically accessible.

3. Process Yield:

SMR is characterized by its relatively high hydrogen yield, typically in the range of 70–80%, which surpasses that of alternative technologies such as biomass gasification or partial oxidation (POX). This high conversion efficiency ensures that a greater fraction of the carbon in the feedstock is utilized to produce hydrogen, thereby improving overall process efficiency. In large-scale applications, this translates into

significant volumes of hydrogen production, making SMR the most practical choice for meeting growing industrial and energy sector demands.

4. Carbon Management:

A major challenge with SMR is the generation of carbon dioxide as a by-product. However, integrating the process with biochar-based carbon capture technology provides a sustainable pathway for mitigating greenhouse gas emissions. Biochar, produced from biomass pyrolysis, serves as an effective adsorbent for CO₂, thereby reducing the carbon footprint of the process while promoting a circular carbon economy. This integration not only aligns with global climate change mitigation strategies but also enhances the environmental acceptability of SMR in the context of Nigeria's commitment to lowering emissions.

5. Economic Viability:

From an economic standpoint, SMR remains more cost-effective than other hydrogen production routes, particularly electrolysis. Electrolysis requires large amounts of electricity, which can be prohibitively expensive in regions where power generation is unstable or reliant on costly sources. In contrast, SMR benefits from Nigeria's low-cost natural gas resources and established infrastructure, resulting in lower production costs per unit of hydrogen. This economic advantage is critical for large-scale deployment and competitiveness in both domestic and international markets.

3.3.2 Block flow diagram

The block flow diagram for the design is given below

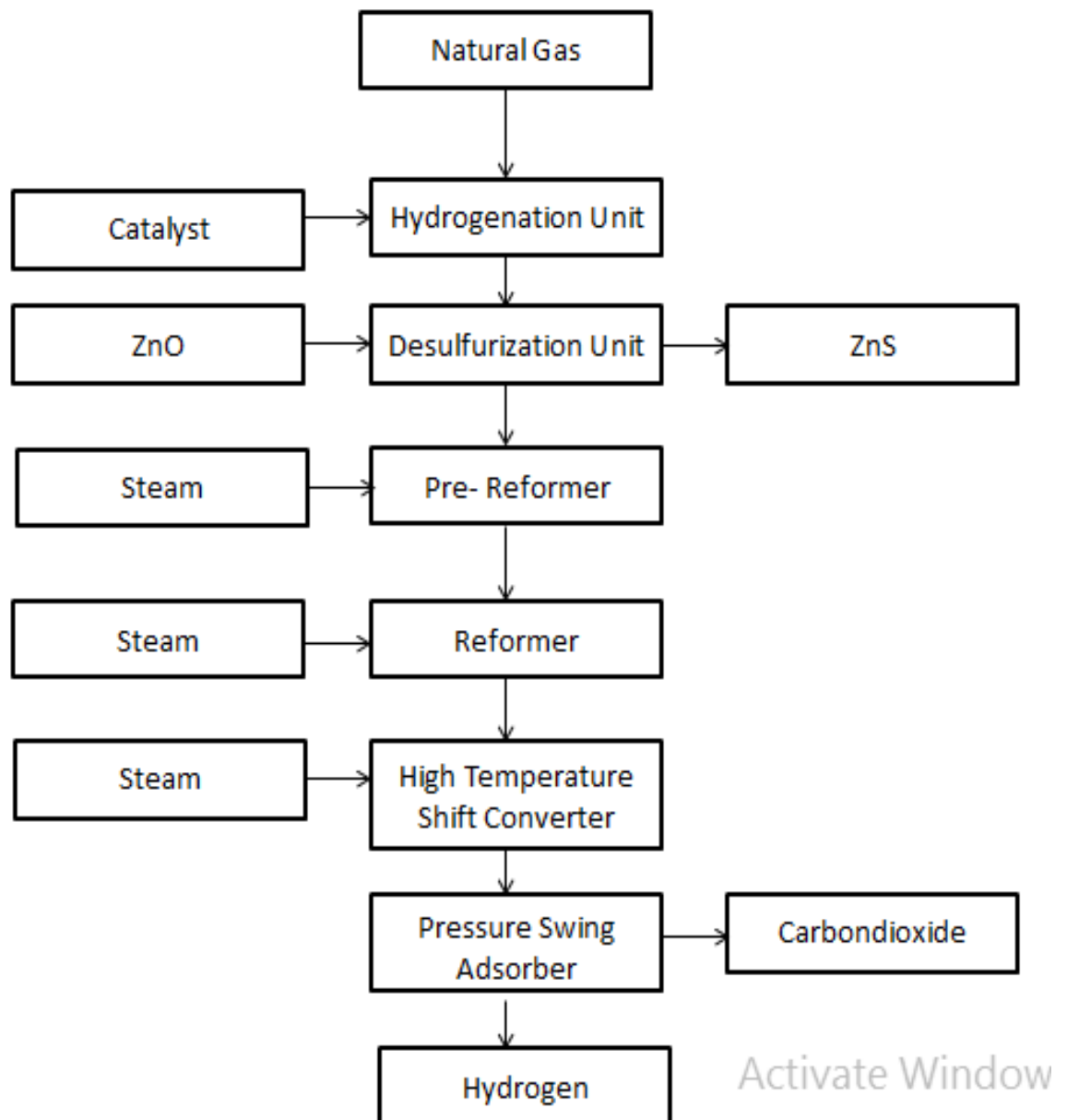


Figure 3.1: Block flow diagram for the process

3.3.3 Process flow diagram

The process flow diagram for the design is given below

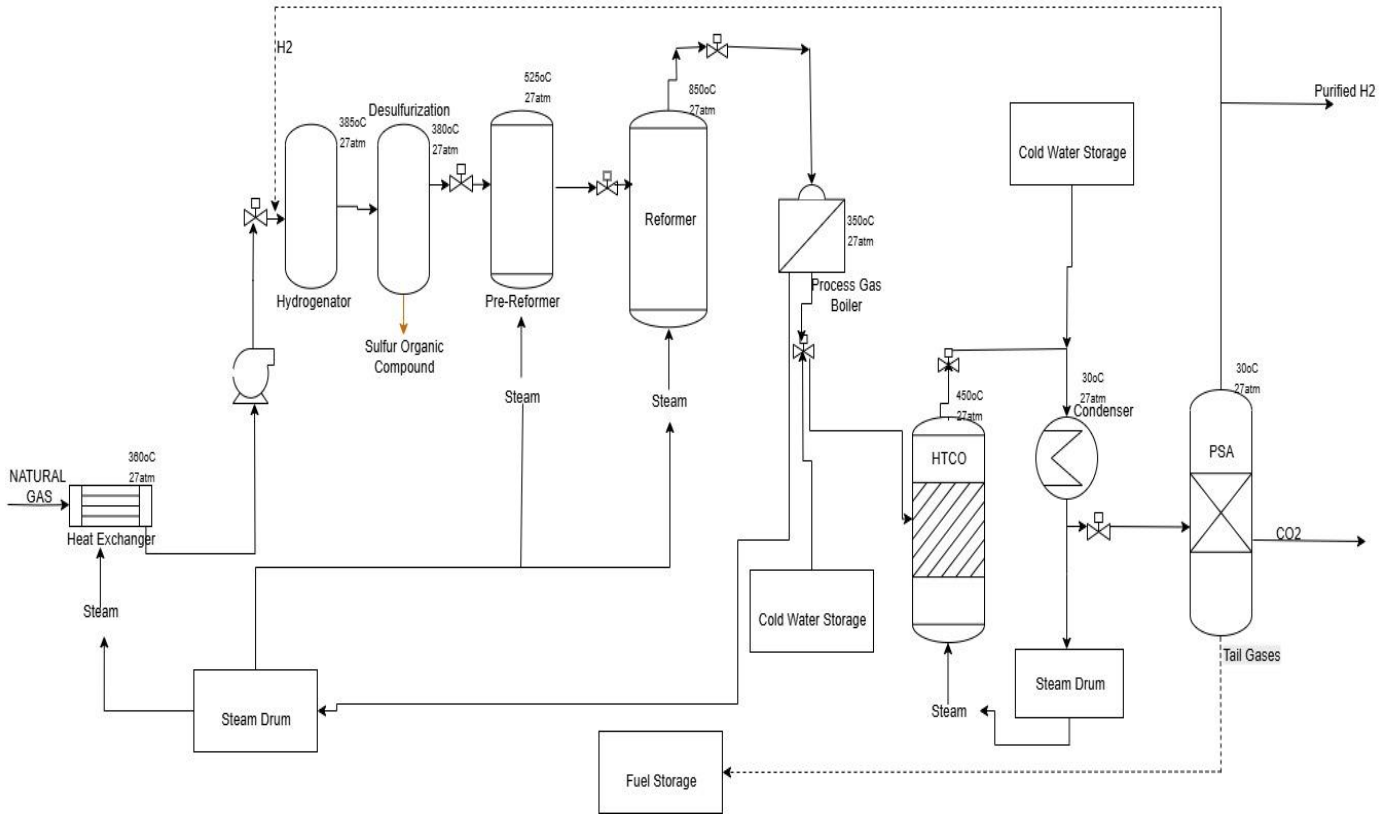


Figure 3.2: Process flow diagram for the process

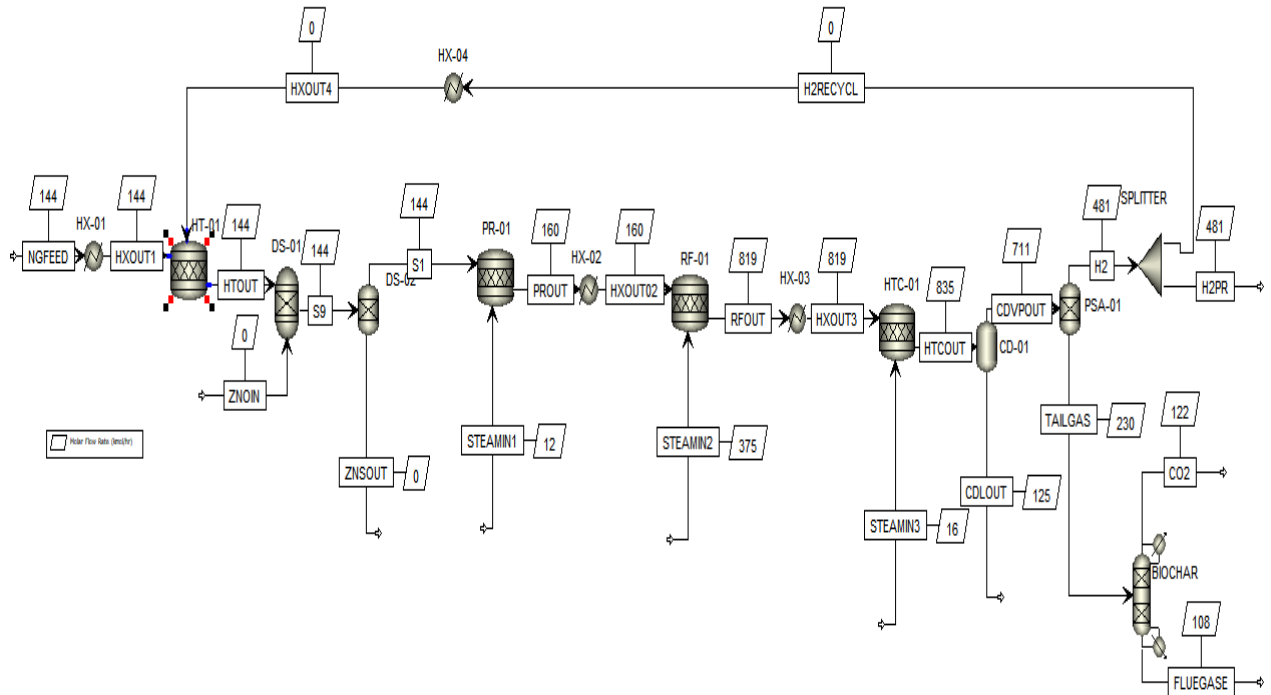


Figure 3.3: Simulated Process Flow Diagram (PFD) of Hydrogen Production Plant Integrated with Biochar-Based Carbon Capture System

Table 3.2: Equipment List

S/N	Annotation	Equipment
1	HX-01	Feed Pre-heater (Heat Exchanger)
2	HT-01	Hydrotreater (Hydrogenator Reactor)
3	DS-01	Desulfurization Reactor (ZnO Bed)
4	PR-01	Pre-Reformer Reactor
5	R-01	Primary Steam Methane Reformer
6	B-01	Process Gas Boiler
7	WGS-01	High Temperature Shift Converter
8	WGS-02	Low Temperature Shift Converter
9	HX-02	Gas Cooler / Heat Exchanger
10	PSA-01	Pressure Swing Adsorption Unit
11	BC-01	Biochar Adsorption Column
12	TK-01	Natural Gas Storage Tank
13	TK-02	Water Tank (Feedwater)
14	P-01	Feedwater Pump
15	P-02	Natural Gas Compressor
16	HX-03	Economizer
17	HX-04	Superheater
18	ST-01	Steam Drum
19	TB-01	Steam Turbine (for power recovery)
20	CD-01	Condenser
21	GN-01	Generator
22	FL-01	Flare System (for safety relief gases)

3.3.4 Process description

The hydrogen production process begins with natural gas, primarily methane, supplied from the Nigerian Liquefied Natural Gas (NLNG) source. The feed gas, typically at ambient temperature and pressure, is first compressed to the operating pressure of approximately 25–30 atm using a natural gas compressor. Following compression, the gas stream is preheated in a shell-and-tube heat exchanger (HX-01) by recovering heat from the hot reformer effluent. This preheating step minimizes external fuel requirements and improves process energy efficiency.

The preheated natural gas is then directed into the hydrogenator reactor (HT-01), where organic sulfur compounds present in trace amounts react with a small fraction of added hydrogen to form hydrogen sulfide (H_2S). This step is essential to protect downstream nickel-based catalysts from sulfur poisoning, as even low sulfur concentrations can deactivate the reforming catalyst.

The hydrogen sulfide formed is subsequently removed in a fixed-bed desulfurization unit (DS-01) packed with zinc oxide (ZnO) adsorbent. The reaction proceeds according to Equation (3):

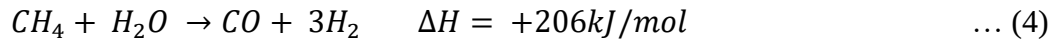


This ensures that the feed stream entering the reformer section is virtually sulfur-free (<0.1 ppm S), thereby extending catalyst life.

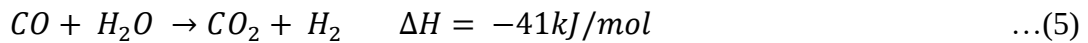
The desulfurized feed is mixed with superheated steam at a steam-to-carbon ratio of 2.5–3.0 and introduced into the pre-reformer (PR-01). In this unit, higher hydrocarbons (C_2 – C_4) contained in the natural gas are catalytically converted to methane and lighter gases over a

nickel catalyst at 450–500 °C. This step not only stabilizes the feed composition but also reduces the risk of coking in the primary reformer.

The methane-rich gas is then sent to the primary steam methane reformer (R-01), the heart of the process. The reformer consists of catalyst-filled tubes housed in a fired furnace, operating at 850–900 °C and 20–30 atm. In this highly endothermic reaction, methane reacts with steam to generate a mixture of hydrogen, carbon monoxide, and residual methane (syngas) as shown in Eqn 4:



The hot syngas exits the reformer at about 850 °C and enters the high-temperature shift (HTS) reactor (WGS-01), where the carbon monoxide undergoes the water–gas shift (WGS) reaction with steam to form carbon dioxide and additional hydrogen at 350 - 450 °C over an iron–chromium catalyst as represented in Eqn 5:



Following HTS, the gas is cooled in a waste-heat boiler (B-01), where steam is generated for use elsewhere in the plant, improving overall thermal efficiency. Depending on the design, a low-temperature shift (LTS) reactor (WGS-02) may be employed downstream of the HTS unit to further reduce residual CO to less than 0.2 vol%. The LTS unit operates at 200–250 °C over a copper–zinc catalyst.

The cooled gas mixture, now rich in hydrogen and CO₂, is sent to a pressure swing adsorption (PSA-01) unit. The PSA contains multiple adsorption beds packed with molecular sieves and activated carbon that selectively adsorb CO₂, CO, CH₄, and other impurities under high pressure, while allowing hydrogen to pass through. The product

hydrogen stream typically achieves >99.9% purity and is delivered at the desired flowrate of 10,000 m³/hr.

The PSA tail gas, containing concentrated CO₂ along with minor quantities of CO and CH₄, is not vented directly. Instead, it is directed to a biochar adsorption column (BC-01), where CO₂ is captured through physisorption and chemisorption on the porous biochar surface. This step provides an environmentally sustainable carbon capture solution, as the biochar can be regenerated or sequestered depending on the operational strategy. The remaining light gases from the biochar unit are either recycled back as fuel for the reformer furnace or flared in a controlled manner.

The process integration ensures efficient utilization of energy, as heat exchangers are strategically placed to recover waste heat from hot gas streams for feed preheating, steam generation, and superheating. The combined system not only achieves the target hydrogen production but also significantly reduces the carbon footprint by incorporating biochar-based CO₂ capture, thus aligning the design with decarbonization goals.

3.4 Material Selection

The choice of materials is a fundamental aspect of process plant design, as it directly influences equipment reliability, safety, operating lifetime, and overall economics. In hydrogen production plants, especially those employing steam methane reforming (SMR) integrated with biochar-based carbon capture, equipment is subjected to extreme service conditions including high temperatures (up to 900 °C), high pressures (20–30 atm), corrosive gases such as H₂S and CO₂, and cyclic thermal stresses. Therefore, careful material selection is critical to ensure both process efficiency and long-term durability of equipment.

3.4.1 Factors to consider

The choice of materials for hydrogen production plants depends on the following factors:

3.4.1.1 Corrosion Resistance

Hydrogen production systems encounter aggressive environments where steam, carbon dioxide, and trace sulfur compounds can cause both uniform corrosion and localized attacks such as pitting or stress corrosion cracking. For instance, reformer tubes are vulnerable to carburization and oxidation at elevated temperatures, while desulfurization units are exposed to H₂S. Materials must therefore possess excellent corrosion resistance to minimize maintenance downtime and extend operational lifespans (Callister & Rethwisch, 2018).

3.1.1.2 Mechanical Strength:

The plant operates under moderate-to-high pressures (20–30 atm) and elevated temperatures, making creep resistance, tensile strength, and fatigue resistance key considerations. Reformer tubes and shift reactor vessels must resist creep deformation under sustained high temperatures, while PSA vessels and piping must withstand cyclic pressure swings without cracking. Materials with proven high-temperature mechanical integrity, such as nickel alloys, are preferred in these critical zones.

3.1.1.3 Thermal Stability:

High thermal stability is essential for components exposed to reforming and combustion zones, where temperatures can reach 850–900 °C. Materials should resist scaling, thermal shock, and metallurgical degradation (such as phase transformations) under cyclic heating

and cooling. This ensures stable catalytic performance and structural integrity over extended service intervals (Peters *et al.*, 2021).

3.1.1.4 Cost Effectiveness:

While exotic alloys such as Inconel offer excellent performance, their high capital cost limits their use to the most critical units (e.g., reformer tubes). For less demanding service, more economical materials such as carbon steel and stainless steel are employed. Striking the right balance between cost and performance is vital for economic viability, especially in developing countries like Nigeria where affordability is a concern (Sinnott & Towler, 2020).

3.1.1.5 Fabricability and Availability:

Materials must be readily available in the local Nigerian market and should be easy to machine, weld, and fabricate into the required shapes. For example, stainless steels and carbon steels are widely available and relatively straightforward to fabricate. In contrast, advanced nickel alloys, though essential in specific applications, may require specialized fabrication techniques and supply chains. Local availability reduces lead times, procurement risks, and costs associated with importation.

3.4.2 Choices of materials

The best choices for hydrogen production plants material are:

3.4.2.1 Carbon Steel

Carbon steel is one of the most widely used construction materials in process industries due to its low cost, ease of fabrication, and availability. It is suitable for tanks, low-pressure piping, and auxiliary vessels that operate at moderate temperatures. However, its

poor resistance to corrosion limits its use in areas exposed to steam, CO₂, or H₂S unless appropriate linings or protective coatings are applied. In the hydrogen plant, carbon steel can be applied in non-critical equipment such as water tanks (TK-01), pumps (P-01), and low-temperature pipelines.

3.4.2.2 Stainless Steel (Grades 304/316)

Austenitic stainless steels such as 304 and 316 are selected for components that require higher resistance to corrosion and oxidation. Stainless steel 304 offers good corrosion resistance in mildly aggressive environments, while stainless steel 316, containing molybdenum, provides superior resistance to pitting and crevice corrosion in chloride-containing media. These grades are suitable for heat exchangers, PSA vessels, scrubbers, and pipelines handling wet gases or hydrogen-rich streams. Their weldability and thermal stability also make them ideal for equipment exposed to frequent temperature cycling (Totten, 2019).

3.4.2.3 Nickel Alloys (Inconel):

Nickel-based superalloys, such as Inconel 600 and Inconel 625, are indispensable in high-temperature units such as reformer tubes, burners, and parts of the shift reactors. These alloys exhibit exceptional creep strength, oxidation resistance, and resistance to carburization, which are critical at temperatures above 850 °C. Though expensive, their durability justifies their use in reforming sections where equipment failure would lead to costly plant shutdowns. Their superior metallurgical stability also enhances catalyst life by minimizing contamination.

3.4.2.4 Aluminum

Aluminum, due to its lightweight nature and ease of fabrication, can be applied in auxiliary or non-critical equipment such as casings, cooling fans, or secondary heat recovery units operating under low pressure and moderate temperature. However, its limited strength and poor high-temperature performance restrict its use in primary process equipment. It is primarily chosen for ancillary systems where weight savings and corrosion resistance in ambient environments are advantageous.

3.4.3 Selected materials for equipment design and construction

Table 3.3: Materials of Construction of Unit Equipment

S/N	Equipment Unit	Selected Material	Reason for Selection
1	Natural Gas Heat Exchanger	Stainless Steel (316)	Corrosion resistance, thermal stability
2	Hydrogenator Reactor	Stainless Steel Alloy	Handles sulfur compounds, high-temperature
3	Desulfurization Vessel	Carbon Steel lined with Stainless Steel	Resistance to ZnO–sulfur environment
4	Pre-Reformer	Nickel Alloy (Inconel)	High-temperature resistance (800 °C)
5	Reformer	Inconel/Nickel Alloys	Excellent creep and corrosion resistance
6	Shift Reactors (HTS)	Carbon Steel with Stainless Lining	Handles steam + CO ₂ at moderate T
7	PSA Unit	Stainless Steel (316)	High pressure resistance, corrosion resistance
8	Biochar Adsorption Vessel	Carbon Steel with Protective Lining	Low cost, withstands CO ₂ -rich environment

CHAPTER FOUR

4.0 MATERIAL BALANCE

4.1 Introduction

In chemical process design, the material balance serves as the backbone for developing a technically and economically feasible plant. It provides the quantitative foundation for designing reactors, sizing equipment, estimating utilities, and determining environmental emissions. A complete understanding of all process input and output streams ensures that mass and energy conservation laws are strictly followed.

This chapter presents a comprehensive material balance for the hydrogen production system via steam methane reforming (SMR), supported by a biochar-based CO₂ capture unit. The process integrates natural gas desulfurization, pre-reforming, steam reforming, shift conversion, hydrogen purification (via PSA), and CO₂ sequestration. The goal is to produce 10,000 m³/h of hydrogen at standard conditions (0°C and 1 atm) with high efficiency.

Fundamental Material Balance Equation

The general mass balance over any chemical process is shown in Eqn. 6:

$$Input + Generation - Consumption - Output = Accumulation \quad \dots (6)$$

Where:

Input: All materials entering the system (e.g., natural gas, steam)

Generation: Mass formed via chemical reactions (e.g., hydrogen from methane)

Consumption: Mass used up in reactions (e.g., methane, water)

Output: All materials leaving the system (e.g., hydrogen, CO₂, tail gas)

Accumulation: Net change in mass inside the system (zero for steady state)

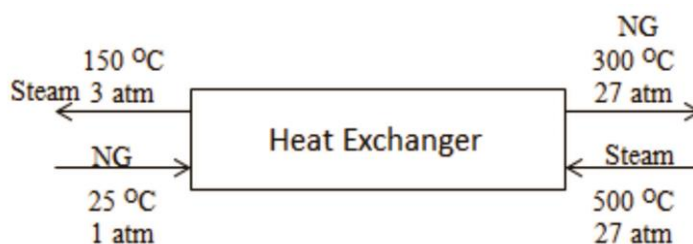
For steady-state continuous operations, which apply in this design, accumulation is zero.

Thus, the simplified balance is shown in Eqn 7:

$$Input = Output \quad \dots (7)$$

In this project, the material balance is applied at both the overall process level and at individual units (e.g., reformer, shift converter), accounting for chemical transformations using reaction stoichiometry and feed compositions.

4.2 Unit 1: Heat Exchanger



Hydrogen Output (Basis)

Molar Flow: $10,000 \text{ m}^3/\text{hr} \div 0.0224 \text{ m}^3/\text{mol} = 446,429 \text{ mol/hr} = 446.43 \text{ kmol/hr}$.

Mass: $446.43 \text{ kmol/hr} \times 2 \text{ kg/kmol} = 892.86 \text{ kg/hr}$.

Natural Gas Feed:

SMR Stoichiometry: $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$ (80%), $\text{CH}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}_2$ (20%).

H₂ Yield: $(0.8 \times 3) + (0.2 \times 4) = 3.2 \text{ mol H}_2/\text{mol CH}_4$.

CH₄ Required: $446.43 / 3.2 = 139.51 \text{ kmol/hr}$.

Mass of CH₄: $139.51 \times 16 = 2232.16 \text{ kg/hr}$ (molar mass CH₄ = 16 g/mol).

Natural Gas (95% CH₄): $2232.16/0.95 = 2349.64$ kg/hr.

Components of Natural Gas:

CH₄: 2232.16 kg/hr (139.51 kmol/hr).

C₂H₆: $2349.64 \times 0.03 = 70.49$ kg/hr (2.35 kmol/hr, molar mass = 30 g/mol).

N₂: $2349.64 \times 0.01 = 23.50$ kg/hr (0.84 kmol/hr, molar mass = 28 g/mol).

H₂S: $2349.64 \times 0.0001 = 0.235$ kg/hr (0.0069 kmol/hr, molar mass = 34 g/mol).

Inerts: $2349.64 \times 0.0099 = 23.26$ kg/hr (0.83 kmol/hr, assume molar mass = 28 g/mol).

Steam Requirement

Carbon Moles: CH₄ (139.51) + C₂H₆ (2.35×2) = $139.51 + 4.70 = 144.21$ kmol/hr.

Steam-to-Carbon Ratio: 3:1.

Steam: $144.21 \times 3 = 432.63$ kmol/hr.

Mass: $432.63 \times 18 = 7787.34$ kg/hr.

Heat Exchanger Material Balance

Input: Natural Gas: 2349.64 kg/hr

CH₄: 2232.16 kg/hr (139.51 kmol/hr).

C₂H₆: 70.49 kg/hr (2.35 kmol/hr).

N₂: 23.50 kg/hr (0.84 kmol/hr).

H₂S: 0.235 kg/hr (0.0069 kmol/hr).

Inerts: 23.26 kg/hr (0.83 kmol/hr).

Steam (H₂O): 7787.34 kg/hr (432.63 kmol/hr).

Total: 2349.64 + 7787.34 = 10,136.98 kg/hr.

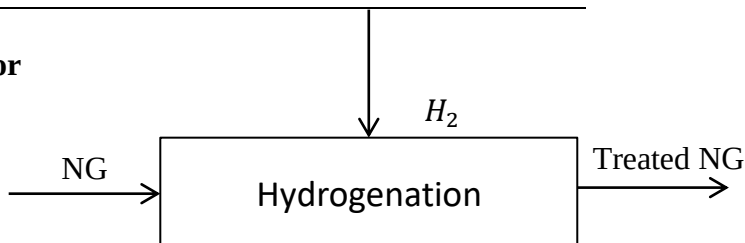
Output: Same as input (no mass changes, only temperature increase).

Balance: Input = Output = 10,136.98 kg/hr.

Component	Input (kg/hr)	Input (kmol/hr)	Output (kg/hr)	Output (kmol/hr)
CH ₄	2232.16	139.51	2232.16	139.51
C ₂ H ₆	70.49	2.35	70.49	2.35
N ₂	23.50	0.84	23.50	0.84
H ₂ S	0.235	0.0069	0.235	0.0069
Inerts	23.26	0.83	23.26	0.83
H ₂ O	7787.34	432.63	7787.34	432.63
Total	10,136.98	576.17	10,136.98	576.17

4.3 Unit 2: Hydrogenation Reactor

Unit Description



Purpose: Converts sulfur compounds to H₂S at 360–385°C using recycled H₂.

Reaction: $R - S + H_2 \rightarrow R - H + H_2S$

Eqn (8)

Material Balance Calculation

Input Stream

CH₄: 2232.16 kg/hr (139.51 kmol/hr)

C₂H₆: 70.49 kg/hr (2.35 kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

H₂S: 0.235 kg/hr (0.0069 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

Subtotal: 2,349.64 kg/hr

Recycled H₂:

Assume 1:1 molar ratio for H₂S conversion (conservative, as most sulfur is already H₂S).

H₂S: 0.0069 kmol/hr → H₂ required: 0.0069 kmol/hr = 0.0069 × 2 = 0.0138 kg/hr
(molar mass H₂ = 2 g/mol).

Total Input:

Mass: 2,349.64 + 0.0138 = 2,349.65 kg/hr

Reaction in Hydrogenator

Reaction: $R - S + H_2 \rightarrow R - H + H_2S$. Since H₂S is already present (0.0069 kmol/hr), assume trace organic sulfur converts to H₂S.

Assumption: H₂S mass remains constant (0.235 kg/hr), as most sulfur is already H₂S, and any organic sulfur conversion produces negligible R-H (assumed to join inerts or CH₄ for simplicity).

H₂ Consumption: 0.0069 kmol/hr H₂ consumed, producing 0.0069 kmol/hr H₂S

Output Stream:

CH₄: 2232.16 kg/hr (139.51 kmol/hr)

C₂H₆: 70.49 kg/hr (2.35 kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

H₂S: 0.235 kg/hr (0.0069 kmol/hr)

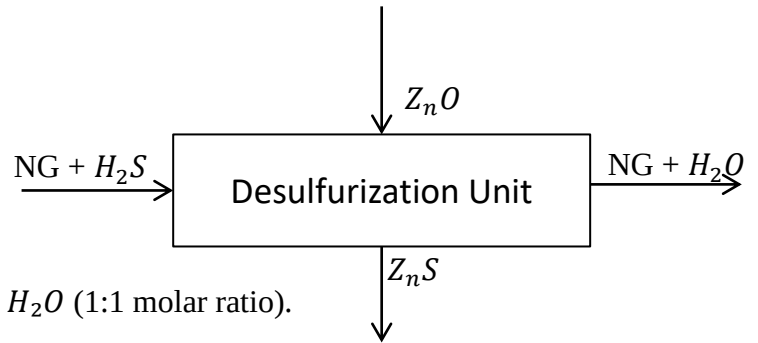
Inerts: 23.26 kg/hr (0.83 kmol/hr)

Total Output: 2232.16 + 70.49 + 23.50 + 0.235 + 23.26 = 2,349.65kg/hr

Component	Input (kg/hr)	Input (kmol/hr)	Output (kg/hr)	Output (kmol/hr)
CH ₄	2232.16	139.51	2232.16	139.51
C ₂ H ₆	70.49	2.35	70.49	2.35
N ₂	23.50	0.84	23.50	0.84
H ₂ S	0.235	0.0069	0.235	0.0069
Inerts	23.26	0.83	23.26	0.83
H ₂	0.0138	0.0069	0	0
Total	2,349.66	143.10	2,349.65	143.54

4.4 Unit 3: Desulfurization Reactor

Assumptions



Reaction: $ZnO + H_2S \rightarrow ZnS + H_2O$ (1:1 molar ratio).

Sulfur Removal: 100% H_2S removal

ZnO Adsorbent: Solid, not included in gas stream mass balance (H_2S is captured as ZnS , H_2O is released to gas stream).

No Other Reactions: CH_4 , C_2H_6 , N_2 , inerts, and existing H_2O pass through unchanged.

Mass Conservation: Total gas input mass equals total gas output mass, adjusted for H_2S removal and H_2O production.

Material Balance Calculation

Input Stream (385°C):

CH_4 : 2232.16 kg/hr (139.51 kmol/hr)

C_2H_6 : 70.49 kg/hr (2.35 kmol/hr)

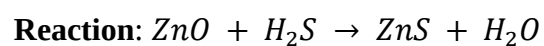
N_2 : 23.50 kg/hr (0.84 kmol/hr)

H_2S : 0.235 kg/hr (0.0069 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

Total: 2,349.65 kg/hr

Reaction in Desulfurization Reactor



Eqn (9)

H₂S Removed: 0.0069 kmol/hr (0.235 kg/hr)

H₂O Produced: 0.0069 kmol/hr $\times$ 18 g/mol = 0.1242 kg/hr

ZnO and ZnS: Solid phases, not included in gas stream mass balance.

Output Stream (380°C):

CH₄: 2232.16 kg/hr (139.51 kmol/hr)

C₂H₆: 70.49 kg/hr (2.35 kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

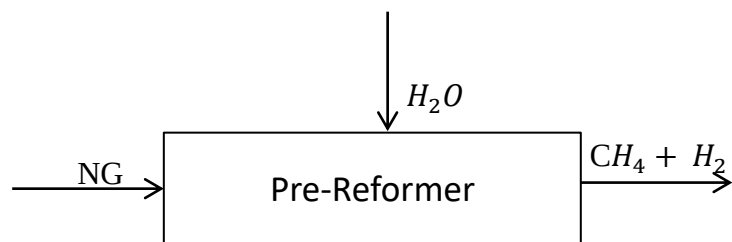
Inerts: 23.26 kg/hr (0.83 kmol/hr)

H₂O: 0.1242 (0.0069 kmol/hr)

Total: 2232.16 + 70.49 + 23.50 + 23.26 + 0.1242 = 2,349.53 kg/hr

Component	Input (kg/hr)	Input (kmol/hr)	Output (kg/hr)	Output (kmol/hr)
CH ₄	2232.16	139.51	2232.16	139.51
C ₂ H ₆	70.49	2.35	70.49	2.35
N ₂	23.50	0.84	23.50	0.84
H ₂ S	0.235	0.0069	0	0
Inerts	23.26	0.83	23.26	0.83
H ₂ O	0	0	0.1242	0.0069
Total	2,349.65	143.54	2,349.53	143.54

4.5 Unit 4: Pre-Reformer



Assumptions

Reaction: $C_2H_6 + 2H_2O \rightarrow 2CH_4 + 2H_2$ (complete conversion of C_2H_6 , typical for pre-reformer with steam).

Other Components: CH_4 , N_2 , inerts pass through unchanged.

Steam: Excess steam (steam-to-carbon ratio = 3:1) ensures complete C_2H_6 conversion.

Mass Conservation: Total input mass equals total output mass, accounting for reaction stoichiometry.

Material Balance Calculation

Input Stream

CH_4 : 2232.16 kg/hr (139.51 kmol/hr)

C_2H_6 : 70.49 kg/hr (2.35 kmol/hr)

N_2 : 23.50 kg/hr (0.84 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

H_2O : 0.1242 kg/hr (0.0069 kmol/hr)

Total: 2,349.53 kg/hr

The S/C ratio is 3:1

Amount of steam added: $3 * 70.49 = 211.47 \text{ kg/hr}$ (11.76 kmol/hr)

Reaction in Pre-Reformer

Reaction: $C_2H_6 + 2H_2O \rightarrow 2CH_4 + 2H_2$... Eqn (10)

Stoichiometry:

C₂H₆ consumed: 2.35 kmol/hr

H₂O consumed: $2.35 \times 2 = 4.70$ kmol/hr = $4.70 \times 18 = 84.60$ kg/hr

CH₄ produced: $2.35 \times 2 = 4.70$ kmol/hr = $4.70 \times 16 = 75.20$ kg/hr

H₂ produced: $2.35 \times 2 = 4.70$ kmol/hr = $4.70 \times 2 = 9.40$ kg/hr

Output Stream (525°C):

CH₄: $2232.16 + 75.20 = 2307.36$ kg/hr ($139.51 + 4.70 = 144.21$ kmol/hr)

C₂H₆: $70.49 - 70.49 = 0$ kg/hr ($2.35 - 2.35 = 0$ kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

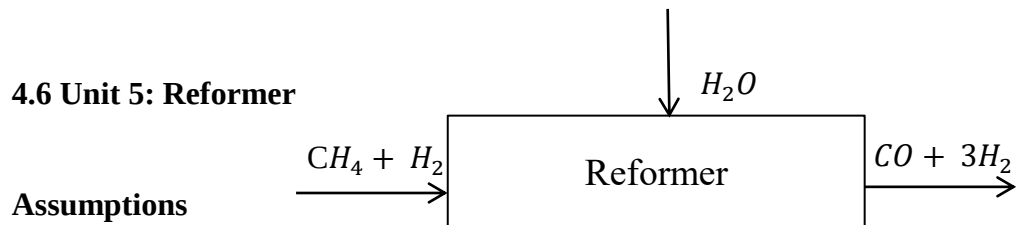
Inerts: 23.26 kg/hr (0.83 kmol/hr)

H₂O: $211.59 - 84.60 = 126.99$ kg/hr ($11.76 - 4.70 = 7.06$ kmol/hr)

H₂: $9.40 = 9.40$ kg/hr ($0 + 4.70 = 4.70$ kmol/hr)

Total: $2307.36 + 23.50 + 23.26 + 126.99 + 9.40 = 2,490.51$ kg/hr

Component	Input (kg/hr)	Input (kmol/hr)	Output (kg/hr)	Output (kmol/hr)
CH ₄	2232.16	139.51	2307.36	144.21
C ₂ H ₆	70.49	2.35	0	0
N ₂	23.50	0.84	23.50	0.84
Inerts	23.26	0.83	23.26	0.83
H ₂ O	211.59	11.76	126.99	7.06
H ₂	0	0	9.40	4.70
Total	2,490.56	157.29	2,490.51	157.64



Reactions:

Main: $CH_4 + H_2O \rightleftharpoons CO + 3H_2$ (80% CH₄ conversion)

Side: $CH_4 + 2H_2O \rightleftharpoons CO_2 + 4H_2$ (20% CH₄ conversion)

Conversion: 100% total CH₄ conversion (80% to CO, 20% to CO₂, typical for SMR).

Other Components: N₂, inerts pass through unchanged. H₂O is consumed and produced. H₂ from pre-reformer is retained.

Steam: Excess steam (steam-to-carbon ratio = 3:1) ensures complete CH₄ conversion.

Mass Conservation: Total input mass equals total output mass, accounting for reaction stoichiometry.

No Side Reactions: Assume negligible carbon formation or other reactions at 850°C with excess steam.

Material Balance Calculation

Input Stream

CH₄: 2307.36 kg/hr (144.21 kmol/hr)

C₂H₆: 0 kg/hr (0 kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

H₂O: 126.99 kg/hr (7.05 kmol/hr)

H₂: 9.40 kg/hr (4.70 kmol/hr)

Total: 2,490.51 kg/hr

The S/C ratio is 3:1

Amount of steam added: $3 \times 2307.36 = 7647.47 \text{ kg/hr}$ (384.56 kmol/hr)

Reactions in Reformer

CH₄ Input: 144.21 kmol/hr

Main Reaction (80%): $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$... (Eqn 11)

CH₄ consumed: $144.21 \times 0.8 = 115.37 \text{ kmol/hr} = 115.37 \times 16 = 1845.92 \text{ kg/hr}$

H₂O consumed: $115.37 \text{ kmol/hr} = 115.37 \times 18 = 2076.66 \text{ kg/hr}$

CO produced: $115.37 \text{ kmol/hr} = 115.37 \times 28 = 3226.36 \text{ kg/hr}$

H₂ produced: $115.37 \times 3 = 346.11 \text{ kmol/hr} = 346.11 \times 2 = 692.22 \text{ kg/hr}$

Side Reaction (20%): $\text{CH}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}_2$

CH₄ consumed: $144.21 \times 0.2 = 28.84 \text{ kmol/hr} = 28.84 \times 16 = 461.44 \text{ kg/hr}$

H₂O consumed: $28.84 \times 2 = 57.68 \text{ kmol/hr} = 57.68 \times 18 = 1038.24 \text{ kg/hr}$

CO₂ produced: $28.84 \text{ kmol/hr} = 28.84 \times 44 = 1268.96 \text{ kg/hr}$

H₂ produced: $28.84 \times 4 = 115.36 \text{ kmol/hr} = 115.36 \times 2 = 230.72 \text{ kg/hr}$

Total CH₄ Consumed: $115.37 + 28.84 = 144.21 \text{ kmol/hr} = 1845.92 + 461.44 = 2307.36 \text{ kg/hr}$

Total H₂O Consumed: $2076.66 + 1038.24 = 3114.90 \text{ kg/hr} = 115.37 + 57.68 = 173.05 \text{ kmol/hr}$

Total H₂ Produced: $692.22 + 230.72 = 922.94 \text{ kg/hr} = 346.11 + 115.36 = 461.47 \text{ kmol/h}$

Output Stream

CH₄: $2307.36 - 2307.36 = 0 \text{ kg/hr}$

N₂: 23.50 kg/hr (0.84 kmol/hr, unchanged)

Inerts: 23.26 kg/hr (0.83 kmol/hr, unchanged)

H₂O: $7773.47 - 3114.90 = 4658.57 \text{ kg/hr}$ ($431.86 - 173.05 = 258.81 \text{ kmol/hr}$)

H₂: $9.40 + 922.94 = 932.34 \text{ kg/hr}$ ($4.70 + 461.47 = 466.17 \text{ kmol/hr}$)

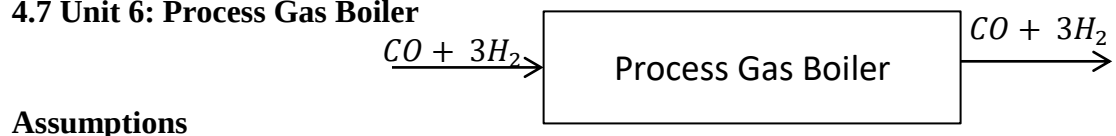
CO: 3226.36 kg/hr (115.37 kmol/hr)

CO₂: 1268.96 kg/hr (28.84 kmol/hr)

Total: $23.50 + 23.26 + 4658.57 + 932.34 + 3226.36 + 1268.96 = 10,132.99 \text{ kg/hr}$

Component	Input (kg/hr)	Input (kmol/hr)	Output (kg/hr)	Output (kmol/hr)
CH ₄	2307.36	144.21	0	0
N ₂	23.50	0.84	23.50	0.84
Inerts	23.26	0.83	23.26	0.83
H ₂ O	7773.47	431.86	4662.57	259.03
H ₂	9.40	4.70	932.34	466.17
CO	0	0	3226.36	115.37
CO ₂	0	0	1268.96	28.84
Total	10,136.99	581.64	10,136.99	871.08

4.7 Unit 6: Process Gas Boiler



Assumptions

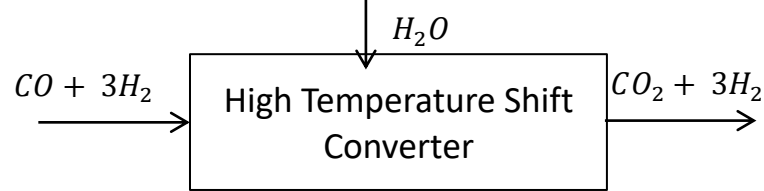
No Reactions: The process gas boiler is a heat exchanger; no chemical reactions occur, so the gas stream composition remains unchanged.

Phase: All components remain gaseous (H₂O as superheated steam at ~20 bar, above condensation temperature at 300°C).

Mass Conservation: Input mass equals output mass, as no material is added or removed.

Steam Generation: Heat is transferred to a separate water/steam loop (not part of gas stream mass balance).

4.8 Unit 7: High-Temperature Shift (HTS) Converter



Assumptions: Assuming 95% conversion

Input Stream

H₂: 932.34 kg/hr (466.17 kmol/hr)

CO: 3,226.36 kg/hr (115.37 kmol/hr)

CO₂: 1,268.96 kg/hr (28.84 kmol/hr)

H₂O: 4,662.57 kg/hr (259.03 kmol/hr)

N₂: 23.50 kg/hr (0.84 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

Total: 10,136.99 kg/hr (871.08 kmol/hr)

Reaction: $CO + H_2O \rightleftharpoons CO_2 + H_2$ (exothermic, $\Delta H = -41.2$ kJ/mol at 400°C).

CO Conversion: Assume 95% conversion

Steam-to-CO Ratio: Target 2:1 to 3:1 to favor forward reaction. Checking if 259.03 kmol/hr H₂O is sufficient

Steam-to-CO Ratio Check

CO Input: 115.37 kmol/hr

H₂O Input: 259.03 kmol/hr

Ratio: $259.03 \div 115.37 = 2.24:1$, within the 2:1 to 3:1 range, sufficient for 95% CO conversion.

Material Balance Calculation



CO Converted: $115.37 \text{ kmol/hr} \cdot 0.95 = 109.6015 \text{ kmol/hr}$

H₂O Consumed: 109.6015 kmol/hr

CO₂ Produced: 109.6015 kmol/hr

H₂ Produced: 109.6015 kmol/hr

Output Stream

H₂: $466.17 + 109.6015 = 575.7715 \text{ kmol/hr} = 575.7715 \cdot 2 = 1,151.543 \text{ kg/hr}$

CO: $115.37 - 109.6015 = 5.7685 \text{ kmol/hr} = 5.7685 \cdot 28 = 161.518 \text{ kg/hr}$

CO₂: $28.84 + 109.6015 = 138.4415 \text{ kmol/hr} = 138.4415 \cdot 44 = 6,091.426 \text{ kg/hr}$

H₂O: $259.03 - 109.6015 = 149.4285 \text{ kmol/hr} = 149.4285 \cdot 18 = 2,689.713 \text{ kg/hr}$

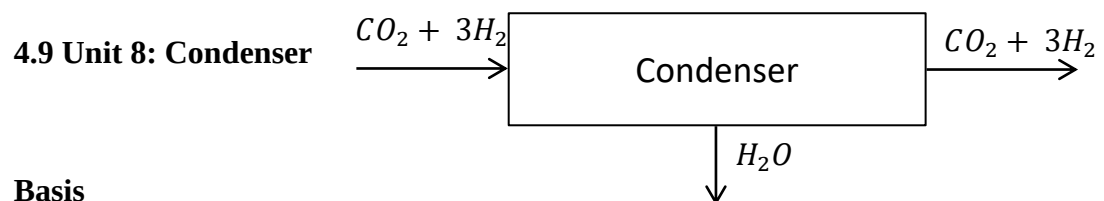
N₂: 23.50 kg/hr (0.84 kmol/hr)

Inerts: 23.26 kg/hr (0.83 kmol/hr)

Total: $1,151.543 + 161.518 + 6,091.426 + 2,689.713 + 23.50 + 23.26 = 10,136.99$
kg/hr

Molar Total: $575.7715 + 5.7685 + 138.4415 + 149.4285 + 0.84 + 0.83 = 871.08$
kmol/hr

Component	Input (kg/hr)	Output (kg/hr)
H ₂	932.34	1,151.543
CO	3,226.36	161.518
CO ₂	1,268.96	6,091.426
H ₂ O	4,662.57	2,689.713
N ₂	23.50	23.50
Inerts	23.26	23.26
Total	10,136.99	10,136.99



Inlet stream: From HTS reactor (Unit 7)

Assume 100% condensation of steam

No chemical reactions

Water exits as liquid, other gases remain in vapor phase

Inlet Stream (Wet Syngas at 400°C)

Component	Molar Flow (kmol/hr)	Molar Mass (g/mol)	Mass Flow (kg/hr)
H ₂	575.77	2.016	1,151.54
CO	5.77	28.01	161.52
CO ₂	138.44	44.01	6,091.43
H ₂ O	149.43	18.015	2,689.71
N ₂	0.84	28.014	23.50
Inerts	0.83	28.00 (assumed)	23.26
Total	871.08	—	10,140.96

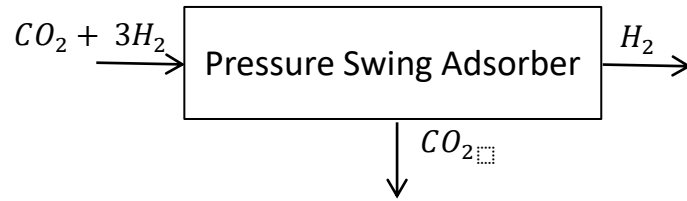
Outlet Stream 1: Dry Gas to PSA (30°C)

Component	Molar Flow (kmol/hr)	Mass Flow (kg/hr)
H ₂	575.77	1,151.54
CO	5.77	161.52
CO ₂	138.44	6,091.43
N ₂	0.84	23.50
Inerts	0.83	23.26
Total	721.65	7,451.25

Outlet Stream 2: Condensate (Liquid Water)

Component	Molar Flow (kmol/hr)	Mass Flow (kg/hr)
H ₂ O	149.43	2,689.71

4.10 Unit 9: PSA



Basis

Feed stream = dry gas from condenser (Unit 8)

Total dry gas = 721.65 kmol/hr = 7,451.25 kg/hr

Hydrogen recovery in PSA = 85%

CO₂ capture by biochar = 90%

All other components pass through unchanged in mass

Feed Composition (to PSA + Biochar Unit)

Component	kmol/hr	Molar Mass (g/mol)	Mass Flow (kg/hr)
H ₂	575.77	2.016	1,151.54
CO	5.77	28.01	161.52
CO ₂	138.44	44.01	6,091.43
N ₂	0.84	28.014	23.50
Inerts	0.83	28.00	23.26
Total	721.65	—	7,451.25

Hydrogen Product Stream (High Purity H₂)

Assumption: 85% of H₂ is recovered

H₂ recovered = 0.85 * 575.77 = 489.40 kmol/hr

$$\text{Mass of } H_2 = 489.40 * 2 = 978.81 \text{ kg/hr}$$

PSA Tail Gas (Sent to Biochar for CO₂ Capture)

Component	kmol/hr	% Recovery	Outlet to Biochar (kmol/hr)
H ₂	575.77	85%	15% lost → 86.37
CO	5.77	0%	5.77
CO ₂	138.44	0%	138.44
N ₂	0.84	0%	0.84
Inerts	0.83	0%	0.83

Total molar flow to biochar:

$$= 86.37 + 5.77 + 138.44 + 0.84 + 0.83 = 232.25 \text{ kmol/hr}$$

Mass of each component:

Component	kmol/hr	Molar Mass	Mass (kg/hr)
H ₂	86.37	2.016	172.74
CO	5.77	28.01	161.52
CO ₂	138.44	44.01	6,091.43
N ₂	0.84	28.014	23.50
Inerts	0.83	28.00	23.26
Total			6,472.45 kg/hr

Biochar Adsorption Unit – CO₂ Capture

Assumption: 90% CO₂ adsorption efficiency

$$\text{CO}_2 \text{ captured} = 0.90 \times 138.44 = 124.60 \text{ kmol/hr} = 5,482.29 \text{ kg/hr}$$

$$\text{CO}_2 \text{ remaining} = 138.44 - 124.60 = 13.84 \text{ kmol/hr} = 609.14 \text{ kg/hr}$$

Final Outlet Streams

Hydrogen Product (to Storage)

Component	kmol/hr	kg/hr
H ₂	489.40	978.8
Total	489.40	978.8

Tail Gas from Biochar (Fuel)

Component	kmol/hr	kg/hr
H ₂	86.37	172.74
CO	5.77	161.52
CO ₂	13.84	609.14
N ₂	0.84	23.50
Inerts	0.83	23.26
Total	78.86	933.51

CO₂ Captured by Biochar (Adsorbed)

Component	kmol/hr	kg/hr
CO ₂	124.60	5,482.29
Total	124.60	5,482.29

CHAPTER 5

5.0

ENERGY BALANCE

5.1 Unit 1: Heat Exchanger

5.1.1 Assumptions

No Phase Change: Natural gas components (CH_4 , C_2H_6 , N_2 , H_2S , inerts) and steam (H_2O) remain in the gas phase from 25°C to 360°C (H_2O boiling point is 100°C at 1 atm, but assume superheated steam at process pressure, typically 27bar, where it remains gaseous).

Specific Heat Capacities (C_p): Average gas-phase C_p values over 25 – 360°C (298 – 633 K).

CH_4 : $2.2 \text{ kJ/kg}\cdot\text{K}$, C_2H_6 : $1.8 \text{ kJ/kg}\cdot\text{K}$, N_2 : $1.04 \text{ kJ/kg}\cdot\text{K}$, H_2S : $1.0 \text{ kJ/kg}\cdot\text{K}$, Inerts: $1.04 \text{ kJ/kg}\cdot\text{K}$, H_2O (steam): $2.0 \text{ kJ/kg}\cdot\text{K}$

No Heat Losses: Assume ideal heat exchanger (all heat input goes to raising stream temperature).

Pressure: Assume constant pressure (27 bar typical for SMR feed), as pressure effects on C_p are minimal.

Energy Balance Equation: $Q = \sum m_i \cdot C_{p_i} \cdot \Delta T$ where Q is the heat duty, m_i is the mass flow rate of component i , C_{p_i} is the specific heat, and $\Delta T = 360 - 25 = 335^\circ\text{C}$

5.1.2 Energy balance calculation

Input Stream (25°C): CH_4 : 2232.16 kg/hr , C_2H_6 : 70.49 kg/hr , N_2 : 23.50 kg/hr , H_2S : 0.235 kg/hr , Inerts: 23.26 kg/hr , H_2O : 7787.34 kg/hr

Enthalpy Calculation: Reference temperature is 25°C (298 K), so input enthalpy is zero relative to this baseline ($H_{in} = 0$).

Output Stream (360°C)

Sensible Heat for Each Component:

$$\text{CH}_4: Q_{\text{CH}_4} = 2232.16 \cdot 2.2 \cdot 335 = 1,645,478.32 \text{ kJ/hr}$$

$$\text{C}_2\text{H}_6: Q_{\text{C}_2\text{H}_6} = 70.49 \cdot 1.8 \cdot 335 = 42,496.47 \text{ kJ/hr}$$

$$\text{N}_2: Q_{\text{N}_2} = 23.50 \cdot 1.04 \cdot 335 = 8,189.20 \text{ kJ/hr}$$

$$\text{H}_2\text{S}: Q_{\text{H}_2\text{S}} = 0.235 \cdot 1.0 \cdot 335 = 78.73 \text{ kJ/hr}$$

$$\text{Inerts}: Q_{\text{Inerts}} = 23.26 \cdot 1.04 \cdot 335 = 8,103.90 \text{ kJ/hr}$$

$$\text{H}_2\text{O}: Q_{\text{H}_2\text{O}} = 7787.34 \cdot 2.0 \cdot 335 = 5,214,717.80 \text{ kJ/hr}$$

Total Heat Duty:

$$Q = 1,645,478.32 + 42,496.47 + 8,189.20 + 78.73 + 8,103.90 + 5,214,717.80 = 6,919,064.42 \text{ kJ/hr} = \mathbf{1,922.52 \text{ kW}}$$

5.2 Unit 2: Hydrogenation Reactor

5.2.1 Assumptions

Temperature Change: Input at 360°C (633 K), output at 385°C (658 K), $\Delta T = 25^\circ\text{C}$.

Reaction: $\text{R-S} + \text{H}_2 \rightarrow \text{R-H} + \text{H}_2\text{S}$. Since H_2S dominates (0.0069 kmol/hr), assume H_2 consumption is for trace organic sulfur conversion.

Heat of Reaction: For organic sulfur (e.g., thiophene, $\text{C}_4\text{H}_4\text{S} + 4\text{H}_2 \rightarrow \text{C}_4\text{H}_{10} + \text{H}_2\text{S}$), $\Delta H = -150 \text{ kJ/mol}$ sulfur (typical for hydrodesulfurization). Given small H_2S quantity, reaction heat is minimal.

Specific Heat Capacities (Cp): Average gas-phase Cp over 633–658 K (assume constant for small ΔT): CH₄: 2.3 kJ/kg·K, C₂H₆: 1.9 kJ/kg·K, N₂: 1.05 kJ/kg·K, H₂S: 1.0 kJ/kg·K, Inerts: 1.05 kJ/kg·K, H₂O (steam): 2.1 kJ/kg·K, H₂: 14.5 kJ/kg·K

No Phase Change: All components remain gaseous (H₂O as superheated steam at process pressure, ~20 bar).

No Heat Losses: Assume ideal reactor.

Energy Balance Equation: $Q = \Delta H_{\text{reaction}} + \sum(m_i \cdot C_{pi} \cdot \Delta T)$ is the sensible heat change.

5.2.2 Energy balance calculation

Input Stream (360°C, 633 K)

- **Sensible Heat to 360°C** (from 25°C, $\Delta T = 335^\circ\text{C}$):

$$CH_4: 2232.16 \cdot 2.3 \cdot 335 = 1,719,339.44 \text{ kJ/hr}$$

$$C_2H_6: 70.49 \cdot 1.9 \cdot 335 = 44,862.27 \text{ kJ/hr}$$

$$N_2: 23.50 \cdot 1.05 \cdot 335 = 8,267.63 \text{ kJ/hr}$$

$$H_2S: 0.235 \cdot 1.0 \cdot 335 = 78.73 \text{ kJ/hr}$$

$$\text{Inerts}: 23.26 \cdot 1.05 \cdot 335 = 8,182.28 \text{ kJ/hr}$$

$$H_2: 0.0138 \cdot 14.5 \cdot 335 = 67.05 \text{ kJ/hr}$$

$$\begin{aligned} \text{Total } H_{\text{in}} &= 1,719,339.44 + 44,862.27 + 8,267.63 + 78.73 + 8,182.28 + \\ &67.05 = 1,780,797.4 \text{ kJ/hr} \end{aligned}$$

Output Stream (385°C, 658 K)

Sensible Heat to 385°C (from 25°C, $\Delta T = 360^\circ\text{C}$):

$$CH_4: 2232.16 \cdot 2.3 \cdot 360 = 1,847,764.32 \text{ kJ/hr}$$

$$C_2H_6: 70.49 \cdot 1.9 \cdot 360 = 48,210.96 \text{ kJ/hr}$$

$$N_2: 23.50 \cdot 1.05 \cdot 360 = 8,883.90 \text{ kJ/hr}$$

$$H_2S: 0.235 \cdot 1.0 \cdot 360 = 84.60 \text{ kJ/hr}$$

$$\text{Inerts: } 23.26 \cdot 1.05 \cdot 360 = 8,787.96 \text{ kJ/hr}$$

$$\begin{aligned} \text{Total } H_{\text{out}}: & 1,847,764.32 + 48,210.96 + 8,883.90 + 84.60 + 8,787.96 = \\ & 1,913,731.74 \text{ kJ/hr} \end{aligned}$$

Heat of Reaction

Reaction: $R-S + H_2 \rightarrow R-H + H_2S$ (H_2S mass unchanged, 0.0069 kmol/hr).

H_2 Consumed: 0.0069 kmol/hr.

$\Delta H_{\text{reaction}}$: Assume -150 kJ/mol for hydrodesulfurization (exothermic).

$$\text{Heat: } 0.0069 \cdot 1000 \cdot (-150) = -1,035 \text{ kJ/hr}$$

$$\text{Equation: } Q = H_{\text{out}} - H_{\text{in}} + \Delta H_{\text{reaction}}$$

$$Q = 1,913,731.74 - 1,780,797.4 - 1,035 = 131,899.34 \text{ kJ/hr}$$

5.3 Unit 3: Desulfurization Reactor

5.3.1 Assumptions

Temperature Change: Input at 385°C (658 K), output at 380°C (653 K), $\Delta T = -5^\circ\text{C}$ (cooling).

Reaction: $\text{ZnO} + \text{H}_2\text{S} \rightarrow \text{ZnS} + \text{H}_2\text{O}$ (1:1 molar ratio, $\text{H}_2\text{S} = 0.0069 \text{ kmol/hr}$).

Heat of Reaction: $\Delta H = -76 \text{ kJ/mol H}_2\text{S}$ (exothermic, typical for ZnO desulfurization at 650 K).

Specific Heat Capacities (C_p): Average gas-phase C_p over 653–658 K

CH_4 : 2.3 kJ/kg·K, C_2H_6 : 1.9 kJ/kg·K, N_2 : 1.05 kJ/kg·K, H_2S : 1.0 kJ/kg·K, Inerts: 1.05 kJ/kg·K, H_2O (steam): 2.1 kJ/kg·K

No Phase Change: All components remain gaseous (H_2O as superheated steam at process pressure, ~20 bar).

ZnO/ZnS: Solid phases, not included in gas stream enthalpy calculations.

No Heat Losses: Assume ideal reactor.

Energy Balance Equation: $Q = \Delta H_{\text{reaction}} + \sum(m_i \cdot C_{pi} \cdot \Delta T)$ is the sensible heat change.

5.3.2 Input Stream (385°C, 658 K)

Sensible Heat to 385°C (from 25°C, $\Delta T = 360^\circ\text{C}$):

CH_4 : $2232.16 \cdot 2.3 \cdot 360 = 1,847,764.32 \text{ kJ/hr}$

C_2H_6 : $70.49 \cdot 1.9 \cdot 360 = 48,210.96 \text{ kJ/hr}$

N_2 : $23.50 \cdot 1.05 \cdot 360 = 8,883.90 \text{ kJ/hr}$

H_2S : $0.235 \cdot 1.0 \cdot 360 = 84.60 \text{ kJ/hr}$

Inerts: $23.26 \cdot 1.05 \cdot 360 = 8,787.96 \text{ kJ/hr}$

$$\text{Total } H_{\text{in}}: 1,847,764.32 + 48,210.96 + 8,883.90 + 84.60 + 8,787.96 = 1,913,731.78 \text{ kJ/hr}$$

5.3.3 Output stream (380°C, 653 K)

Sensible Heat to 380°C (from 25°C, $\Delta T = 355^\circ\text{C}$):

$$\text{CH}_4: 2232.16 \cdot 2.3 \cdot 355 = 1,825,286.92 \text{ kJ/hr}$$

$$\text{C}_2\text{H}_6: 70.49 \cdot 1.9 \cdot 355 = 47,528.11 \text{ kJ/hr}$$

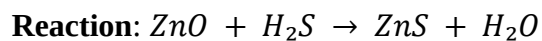
$$\text{N}_2: 23.50 \cdot 1.05 \cdot 355 = 8,755.13 \text{ kJ/hr}$$

$$\text{Inerts: } 23.26 \cdot 1.05 \cdot 355 = 8,667.32 \text{ kJ/hr}$$

$$\text{H}_2\text{O: } 0.1242 \cdot 2.1 \cdot 355 = 92.59 \text{ kJ/hr}$$

$$\text{Total } H_{\text{out}}: 1,825,286.92 + 47,528.11 + 8,755.13 + 8,667.32 + 5,797,228.29 = 1,890,330.07 \text{ kJ/hr}$$

5.3.4 Heat of Reaction



H₂S Removed: 0.0069 kmol/hr

$\Delta H_{\text{reaction}}$: -76 kJ/mol H₂S

Heat: $0.0069 \cdot 1000 \cdot (-76) = -524.4 \text{ kJ/hr}$

Equation: $Q = H_{\text{out}} - H_{\text{in}} + \Delta H_{\text{reaction}}$

$$Q = 1,890,330.07 - 1,913,731.78 - 524.4 = -23,401.71 \text{ kJ/hr or } -6.5 \text{ kW}$$

5.4 Unit 4: Pre-Reformer

5.4.1 Assumptions

Temperature Change: Input at 380°C (653 K), output at 525°C (798 K), $\Delta T = 145^\circ\text{C}$.

Reaction: $\text{C}_2\text{H}_6 + 2\text{H}_2\text{O} \rightarrow 2\text{CH}_4 + 2\text{H}_2$, complete conversion of 2.35 kmol/hr C_2H_6 .

Heat of Reaction: Approximate $\Delta H = -65$ kJ/mol C_2H_6 (exothermic, based on literature for pre-reforming at 650–800 K).

Specific Heat Capacities (C_p): CH_4 : 2.4 kJ/kg·K, C_2H_6 : 2.0 kJ/kg·K, N_2 : 1.06 kJ/kg·K, Inerts: 1.06 kJ/kg·K, H_2O (steam): 2.2 kJ/kg·K, H_2 : 14.6 kJ/kg·K

No Phase Change: All components remain gaseous (H_2O as superheated steam at ~20 bar).

No Heat Losses: Assume ideal reactor.

5.4.2 Energy Balance Equation: $Q = \Delta H_{\text{reaction}} + \sum(m_i \cdot C_{pi} \cdot \Delta T_{\text{out}}) - \sum(m_i \cdot C_{pi} \cdot \Delta T_{\text{in}})$... (Eqn 13)

Input Stream (380°C, 653 K)

Sensible Heat to 380°C ($\Delta T = 355^\circ\text{C}$):

$$\text{CH}_4: 2232.16 \cdot 2.4 \cdot 355 = 1,901,111.04 \text{ kJ/hr}$$

$$\text{C}_2\text{H}_6: 70.49 \cdot 2.0 \cdot 355 = 50,045.90 \text{ kJ/hr}$$

$$\text{N}_2: 23.50 \cdot 1.06 \cdot 355 = 8,842.45 \text{ kJ/hr}$$

$$\text{Inerts}: 23.26 \cdot 1.06 \cdot 355 = 8,753.73 \text{ kJ/hr}$$

$$\text{H}_2\text{O}: 211.59 \cdot 2.2 \cdot 355 = 165,251.79 \text{ kJ/hr}$$

$$\text{Total } H_{in}: 1,901,111.04 + 50,045.90 + 8,842.45 + 0 + 8,753.73 + 6,080,660.94 = 2,134,004.91 \text{ kJ/hr}$$

Output Stream (525°C, 798 K)

Sensible Heat to 525°C ($\Delta T = 500^\circ\text{C}$):

$$CH_4: 2307.36 \cdot 2.4 \cdot 500 = 2,769,432.00 \text{ kJ/hr}$$

$$N_2: 23.50 \cdot 1.06 \cdot 500 = 12,455.00 \text{ kJ/hr}$$

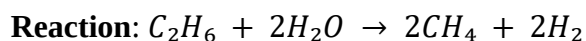
$$\text{Inerts: } 23.26 \cdot 1.06 \cdot 500 = 12,327.80 \text{ kJ/hr}$$

$$H_2O: 126.99 \cdot 2.2 \cdot 500 = 139,689.00 \text{ kJ/hr}$$

$$H_2: 9.40 \cdot 14.6 \cdot 500 = 68,620.00 \text{ kJ/hr}$$

$$\text{Total } H_{out}: 2,769,432.00 + 0 + 12,455.00 + 0 + 12,327.80 + 8,550,817.00 + 68,620.00 = 3,002,523.80 \text{ kJ/hr}$$

Heat of Reaction



C₂H₆ Consumed: 2.35 kmol/hr

$\Delta H_{\text{reaction}}$: -65 kJ/mol C₂H₆ (exothermic)

$$\text{Heat: } 2.35 \cdot 1000 \cdot (-65) = -152,750 \text{ kJ/hr}$$

$$\text{Equation: } Q = H_{out} - H_{in} + \Delta H_{\text{reaction}} \quad \dots(\text{Eqn 14})$$

$$Q = 3,002,523.80 - 2,134,004.91 - 152,750 = 868,518.89 \text{ kJ/hr or } 241.25 \text{ kW}$$

5.5 Unit 5: Reformer

5.5.1 Assumptions

Temperature Change: Input at 525°C (798 K), output at 850°C (1123 K), $\Delta T = 325^\circ\text{C}$.

Reactions:

Main: $\text{CH}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CO} + 3\text{H}_2$, $\Delta H = +206 \text{ kJ/mol CH}_4$ (endothermic at 800–1100 K).

Side: $\text{CH}_4 + 2\text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + 4\text{H}_2$, $\Delta H = +165 \text{ kJ/mol CH}_4$ (endothermic, adjusted for water-gas shift influence).

Specific Heat Capacities (Cp): Average gas-phase Cp over 798–1123 K:

CH_4 : 2.5 kJ/kg·K, N_2 : 1.1 kJ/kg·K, Inerts: 1.1 kJ/kg·K, H_2O (steam): 2.3 kJ/kg·K, H_2 : 14.7 kJ/kg·K, CO : 1.1 kJ/kg·K, CO_2 : 1.0 kJ/kg·K

No Phase Change: All components remain gaseous (H_2O as superheated steam at ~20 bar).

No Heat Losses: Assume ideal reformer.

5.5.2 Energy Balance Equation: $Q = \Delta H_{\text{reaction}} + \sum (m_i \cdot C_{pi} \cdot \Delta T_{\text{out}}) - \sum (m_i \cdot C_{pi} \cdot \Delta T_{\text{in}})$

5.5.2.1 Input Stream (525°C, 798 K)

Sensible Heat to 525°C ($\Delta T = 500^\circ\text{C}$):

$$\text{CH}_4: 2307.36 \cdot 2.5 \cdot 500 = 2,884,200.00 \text{ kJ/hr}$$

$$\text{N}_2: 23.50 \cdot 1.1 \cdot 500 = 12,925.00 \text{ kJ/hr}$$

$$\text{Inerts: } 23.26 \cdot 1.1 \cdot 500 = 12,793.00 \text{ kJ/hr}$$

$$\text{H}_2\text{O: } 7773.47 \cdot 2.3 \cdot 500 = 8,939,490.50 \text{ kJ/hr}$$

$$\text{H}_2: 9.40 \cdot 14.7 \cdot 500 = 69,090.00 \text{ kJ/hr}$$

$$\begin{aligned} \text{Total } H_{in}: & 2,884,200.00 + 12,925.00 + 12,793.00 + 8,939,490.50 + \\ & 69,090.00 = 11,918,498.50 \frac{\text{kJ}}{\text{hr}} \end{aligned}$$

5.5.2.2 Output Stream (850°C, 1123 K)

Sensible Heat to 850°C ($\Delta T = 825^\circ\text{C}$):

$$\text{N}_2: 23.50 \cdot 1.1 \cdot 825 = 21,328.50 \text{ kJ/hr } 23.50$$

$$\text{Inerts: } 23.26 \cdot 1.1 \cdot 825 = 21,105.15 \text{ kJ/hr}$$

$$\text{H}_2\text{O: } 4662.57 \cdot 2.3 \cdot 825 = 8,849,139.98 \text{ kJ/hr}$$

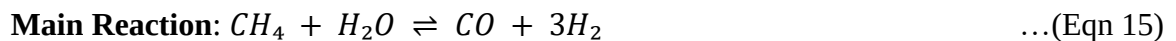
$$\text{H}_2: 932.34 \cdot 14.7 \cdot 825 = 11,311,167.45 \text{ kJ/hr}$$

$$\text{CO: } 3226.36 \cdot 1.1 \cdot 825 = 2,928,987.30 \text{ kJ/hr}$$

$$\text{CO}_2: 1268.96 \cdot 1.0 \cdot 825 = 1,046,892.00 \text{ kJ/hr}$$

$$\begin{aligned} \text{Total } H_{out}: & 21,328.50 + 21,105.15 + 8,849,139.98 + 11,311,167.45 + \\ & 2,928,987.30 + 1,046,892.00 = 24,179,110.38 \text{ kJ/hr} \end{aligned}$$

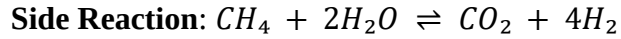
5.5.2.3 Heat of Reaction



CH₄ consumed: 115.37 kmol/hr

$$\Delta H = +206 \text{ kJ/mol}$$

$$\text{Heat: } 115.37 \cdot 1000 \cdot 206 = 23,766,220.00 \text{ kJ/hr}$$



CH₄ consumed: 28.84 kmol/hr, $\Delta H = +165 \text{ kJ/mol}$

$$\text{Heat: } 28.84 \cdot 1000 \cdot 165 = 4,758,600.00 \text{ kJ/hr}$$

Total Heat of Reaction: $23,766,220 + 4,758,600 = 28,524,820 \text{ kJ/hr}$

Equation: $Q = H_{\text{out}} - H_{\text{in}} + \Delta H_{\text{reaction}}$

$$Q = 24,179,110.38 - 11,918,498.50 + 28,524,820 = 40,785,431.88 \text{ kJ/hr or}$$

$$11,329.29 \text{ kW}$$

5.6 Unit 6: Process Gas Boiler

The process gas boiler reduces the high temperature of the syngas

5.6.1 Input Stream:

Composition: H₂ (932.34 kg/hr, 466.17 kmol/hr), CO (3226.36 kg/hr, 115.37 kmol/hr), CO₂ (1268.96 kg/hr, 28.84 kmol/hr), H₂O (4662.57 kg/hr, 259.03 kmol/hr), N₂ (23.50 kg/hr, 0.84 kmol/hr), Inerts (23.26 kg/hr, 0.83 kmol/hr).

Total: 10,136.99 kg/hr, 871.08 kmol/hr.

Inlet temperature: 850°C (1123.15 K).

5.6.2 Output Stream: Same composition and mass flow as input (no reactions), cooled to 300°C (573.15 K).

Heat Losses: Assumed at 2% of sensible heat, based on typical industrial heat exchanger performance, unless otherwise specified.

Boiler Feed Water (BFW): Enters at 100°C (liquid, 373.15 K), 20 bar, and is converted to saturated steam at 250°C (523.15 K), 20 bar.

Heat Capacities (Cp): Temperature-dependent molar heat capacities for gases at the average temperature (~698°C or 971.15 K) are used, sourced from standard chemical engineering data (e.g., Perry's Chemical Engineers' Handbook).

Steam Properties: Enthalpy values from steam tables at 20 bar (IAPWS-97 standard).

No Phase Changes in Gas Stream: H₂O remains as superheated steam at 20 bar, 300°C (above condensation point).

Energy Balance Equation: $Q_{total} = Q_{sensible} + Q_{reaction} + Q_{losses}$, where $Q_{reaction} = 0$ (no reactions).

Sensible Heat of Gas Stream ($Q_{sensible}$)

The gas stream is cooled from 850°C (1123.15 K) to 300°C (573.15 K), releasing sensible heat. The heat released is calculated as:

$$Q_{sensible} = n \cdot C_{p_{mixture}} \cdot \Delta T \quad \dots \text{Eqn (16)}$$

Molar Heat Capacities (Cp): Using temperature-dependent Cp values at the average temperature (971.15 K = 698°C) for accuracy:

H₂: 29.5 J/mol·K

CO: 30.5 J/mol·K

CO₂: 47.0 J/mol·K

H₂O (superheated steam): 40.0 J/mol·K

N₂: 31.0 J/mol·K

Inerts (assumed N₂-like): 31.0 J/mol·K

Mole Fractions:

H₂: $466.17 \div 871.08 = 0.5350$

CO: $115.37 \div 871.08 = 0.1324$

CO₂: $28.84 \div 871.08 = 0.0331$

H₂O: $259.03 \div 871.08 = 0.2973$

N₂: $0.84 \div 871.08 = 0.0010$

Inerts: $0.83 \div 871.08 = 0.0010$

Weighted Cp_{mixture}:

$$\begin{aligned}Cp_{\text{mixture}} &= \Sigma (\text{mole fraction} \cdot Cp) = (0.5350 \cdot 29.5) + (0.1324 \cdot 30.5) + (0.0331 \cdot 47.0) + \\&(0.2973 \cdot 40.0) + (0.0010 \cdot 31.0) + (0.0010 \cdot 31.0) = 15.7825 + 4.0382 + 1.5557 + 11.8920 \\&+ 0.0310 + 0.0310 = 33.3304 \text{ J/mol}\cdot\text{K}\end{aligned}$$

Temperature Change:

$$\Delta T = 850^{\circ}\text{C} - 300^{\circ}\text{C} = 1123.15\text{ K} - 573.15\text{ K} = 550\text{ K}$$

$$Q_{\text{sensible}} = 0.24197\text{ kmol/s} \cdot 33.3304\text{ kJ/kmol} \cdot K \cdot 550\text{ K} = 4,435.69\text{ kW}$$

Heat Losses (Q_{losses})

Assuming 2% heat loss (standard for well-insulated industrial heat exchangers):

$$Q_{\text{losses}} = 0.02 \cdot Q_{\text{sensible}} = 0.02 \cdot 4,435.69\text{ kW} = 88.71\text{ kW}$$

Heat Transferred to Steam (Q_{steam})

The heat available for steam generation is:

$$Q_{\text{steam}} = Q_{\text{sensible}} - Q_{\text{losses}} = 4,435.69 - 88.71 = 4,346.98\text{ kW}$$

Steam Generation Rate

The heat is used to convert BFW (100°C, 20 bar, liquid) to saturated steam (250°C, 20 bar). Using steam table data (IAPWS-97):

Enthalpy of BFW (100°C, liquid): $h_f = 419.1\text{ kJ/kg}$

Enthalpy of saturated steam (250°C, 20 bar): $h_g = 2,799.5\text{ kJ/kg}$

Enthalpy change (ΔH): $h_g - h_f = 2,799.5 - 419.1 = 2,380.4\text{ kJ/kg}$

Note: At 20 bar, the saturation temperature is 212°C, but we assume the steam is heated to 250°C (superheated slightly), so we include sensible heat from 212°C to 250°C ($C_{p_{\text{steam}}} = 2.1\text{ kJ/kg}\cdot\text{K}$):

$$\text{Sensible heat (212°C to 250°C): } 2.1 \cdot (250 - 212) = 79.8\text{ kJ/kg}$$

Total $\Delta H = 419.1$ (to boiling) + $1,889.2$ (latent heat at 20 bar) + $79.8 = 2,388.1$ kJ/kg (slightly adjusted for precision).

Steam Flow Rate:

$$Q_{steam} = m_{steam} \cdot \Delta H \quad \dots \text{Eqn (17)}$$

$$4,346.98 \text{ kJ/s} = m_{steam} \cdot 2,388.1 \text{ kJ/kg}$$

$$m_{steam} = 4,346.98 \div 2,388.1 = 1.8201 \text{ kg/s}$$

$$m_{steam} = 1.8201 \cdot 3600 = 6,552.36 \text{ kg/hr}$$

5.7 Unit 7: High Temperature Shift Converter

Sensible Heat: Change in enthalpy of the gas stream from inlet (400°C, preheated) to outlet (assume 400°C, as HTS is typically isothermal due to catalyst bed control).

Reaction Heat: Exothermic water-gas shift reaction.

Preheating: Heating the input stream from 300°C (Unit 6 output) to 400°C.

Heat Losses: Assume 2% of sensible heat, standard for industrial reactors.

Steam Integration: Use boiler steam for preheating, with excess available for other units.

5.7.1 Assumptions

Reaction: $CO + H_2O \rightleftharpoons CO_2 + H_2$, 95% conversion (109.6015 kmol/hr CO converted).

Enthalpy of Reaction: $\Delta H = -41.2$ kJ/mol at 400°C (standard for water-gas shift, gas phase, from thermodynamic data).

Heat Capacities (C_p) at ~400°C (673 K): H₂: 29.5 J/mol·K, CO: 30.5 J/mol·K, CO₂: 47.0 J/mol·K, H₂O (superheated steam): 40.0 J/mol·K, N₂: 31.0 J/mol·K, Inerts (N₂-like): 31.0 J/mol·K

Preheating: From 300°C to 400°C, using boiler steam ($\Delta H = 2,380.4$ kJ/kg, 250°C, 20 bar to 100°C liquid).

Heat Losses: 2% of sensible heat (if applicable).

Isothermal Operation: HTS is typically controlled to maintain ~400°C outlet to optimize conversion and catalyst performance.

Preheating Energy (300°C to 400°C)

Molar Flow: $871.08 \text{ kmol/hr} = 871.08 \div 3600 = 0.24197 \text{ kmol/s}$

Mole Fractions (input):

$$\text{H}_2: 466.17 \div 871.08 = 0.5350$$

$$\text{CO}: 115.37 \div 871.08 = 0.1324$$

$$\text{CO}_2: 28.84 \div 871.08 = 0.0331$$

$$\text{H}_2\text{O}: 259.03 \div 871.08 = 0.2973$$

$$\text{N}_2: 0.84 \div 871.08 = 0.0010$$

$$\text{Inerts}: 0.83 \div 871.08 = 0.0010$$

C_pmixture:

$$C_p = (0.5350 \cdot 29.5) + (0.1324 \cdot 30.5) + (0.0331 \cdot 47.0) + (0.2973 \cdot 40.0) + (0.0010 \cdot 31.0) + (0.0010 \cdot 31.0) = 15.7825 + 4.0382 + 1.5557 + 11.8920 + 0.0310 + 0.0310 = 33.3304 \text{ kJ/kmol} \cdot \text{K}$$

$$\Delta T: 400^\circ\text{C} - 300^\circ\text{C} = 100 \text{ K}$$

$$Q_{\text{preheat}} = 0.24197 \text{ kmol/s} \cdot 33.3304 \text{ kJ/kmol} \cdot \text{K} \cdot 100 \text{ K} = 806.49 \text{ kW}$$

Boiler Steam Needed:

$$\Delta H_{\text{steam}} = 2,380.4 \text{ kJ/kg}$$

$$m_{\text{steam}} = 806.49 \div 2,380.4 = 0.3388 \text{ kg/s} = 1,219.68 \text{ kg/hr}$$

$$\text{Remaining Boiler Steam: } 3,410.38 - 1,219.68 = 2,190.70 \text{ kg/hr}$$

5.7.2 Reaction Heat (Q_{reaction})



$$\text{CO Converted: } 109.6015 \text{ kmol/hr} = 109.6015 \div 3600 = 0.030445 \text{ kmol/s}$$

$$\Delta H_{\text{reaction}}: -41.2 \text{ kJ/mol} = -41,200 \text{ kJ/kmol}$$

$$Q_{\text{reaction}} = 0.030445 \text{ kmol/s} \cdot (-41,200) \text{ kJ/kmol} = -1,254.33 \text{ kW (exothermic, heat released)}.$$

5.7.3 Sensible Heat (Q_{sensible})

Since the HTS is isothermal (400°C inlet and outlet), the sensible heat change due to temperature is zero.

Composition Change: The reaction changes mole fractions, affecting the mixture's heat capacity.

Input $C_{p_{\text{mixture}}}$: 33.3304 kJ/kmol·K (calculated above).

Output Mole Fractions:

$$\text{H}_2: 575.7715 \div 871.08 = 0.6611$$

$$\text{CO}: 5.7685 \div 871.08 = 0.0066$$

$$\text{CO}_2: 138.4415 \div 871.08 = 0.1589$$

$$\text{H}_2\text{O}: 149.4285 \div 871.08 = 0.1715$$

$$\text{N}_2: 0.84 \div 871.08 = 0.0010$$

$$\text{Inerts}: 0.83 \div 871.08 = 0.0010$$

Output $C_{p_{\text{mixture}}}$:

$$\begin{aligned} C_p &= (0.6611 \cdot 29.5) + (0.0066 \cdot 30.5) + (0.1589 \cdot 47.0) + (0.1715 \cdot 40.0) + (0.0010 \cdot 31.0) + \\ &(0.0010 \cdot 31.0) = 19.4976 + 0.2013 + 7.4683 + 6.8600 + 0.0310 + 0.0310 = 34.0892 \\ &\text{kJ/kmol}\cdot\text{K} \end{aligned}$$

Enthalpy Calculation (relative to 0 K, at 400°C = 673.15 K):

Input Enthalpy: $0.24197 \text{ kmol/s} \cdot 33.3304 \text{ kJ/kmol}\cdot\text{K} \cdot 673.15 \text{ K} = 5,428.70 \text{ kW}$

Output Enthalpy: $0.24197 \text{ kmol/s} \cdot 34.0892 \text{ kJ/kmol}\cdot\text{K} \cdot 673.15 \text{ K} = 5,555.08 \text{ kW}$

$Q_{\text{sensible}} = \text{Output} - \text{Input} = 5,555.08 - 5,428.70 = 126.38 \text{ kW}$ (endothermic due to higher C_p of products).

5.7.4 Heat Losses

Q_{losses} : Assume 2% of sensible heat (if significant, but since Q_{sensible} is small due to isothermal operation, apply to preheating and reaction heat for conservatism).

$$Q_{\text{losses}} = 0.02 \cdot (806.49 + 126.38) = 0.02 \cdot 932.87 = 18.66 \text{ kW}$$

5.7.5 Overall Energy Balance

Net Heat: $Q_{\text{reaction}} - Q_{\text{preheat}} - Q_{\text{sensible}} - Q_{\text{losses}} = 1,254.33 - 806.49 - 126.38 - 18.66 = 302.80 \text{ kW}$ (excess heat available, e.g., for steam generation or cooling).

5.8 Unit 8: Condenser

5.8.1 Assumptions

Parameter	Value
Inlet gas temperature	400°C
Outlet gas temperature	30°C
Cp of dry gas mixture	2.5 kJ/kg·K (average)
Mass of dry gas	7,451.25 kg/hr
H ₂ O condensed	2,689.71 kg/hr
Latent heat of steam @ 20 atm	1900 kJ/kg
Cooling water temp rise	25°C to 35°C ($\Delta T = 10^\circ\text{C}$)
Cp of water	4.18 kJ/kg·K

Heat Removed by Condensing Water Vapor

$$Q_{\text{condensation}} = 2,689.71 \text{ kg/hr} \times 1900 \text{ kJ/kg} = 5,110,449 \text{ kJ/hr}$$

Heat Removed to Cool Dry Gas from 400°C to 30°C

$$Q_{\text{cooling}} = 7,451.25 \times 2.5 \times 370 = 6,899,906.25 \text{ kJ/hr}$$

Total Heat Duty of Condenser

$$Q_{\text{total}} = 5,110,449 + 6,899,906.25 = 12,010,355 \text{ kJ/hr} = 3.34 \text{ MW}$$

Cooling Water Requirement

$$Q = mcw \times Cp \times \Delta T \quad \dots \text{Eqn (19)}$$

$$mcw = \frac{12,010,355}{4.18 \times 10} = 287,274 \text{ kg/hr}$$

5.9 Unit 9: Biochar-Integrated PSA

Operates adiabatically and isothermally (by assumption)

Parameter	Value
Feed gas temperature	30°C
Pressure in PSA/biochar system	20 atm
Adsorption heat of CO ₂ on biochar	25 kJ/mol (literature average)
PSA is isothermal and adiabatic	No heat exchange assumed
CO ₂ capture efficiency	90%
No phase change involved	All gas remains in vapor form

CO₂ adsorbed by biochar: 124.60 kmol/hr = 124,600 mol

Energy Released from CO₂ Adsorption (Exothermic)

$$Q_{ads} = mol \text{ CO}_2 \times \Delta H_{adsorption} \quad \dots \text{Eqn (20)}$$

$$Q_{ads} = 124,600 \text{ mol/hr} \times 25 \text{ kJ/mol} = 3,115,000 \text{ kJ/hr} = 865 \text{ kW}$$

CHAPTER SIX

6.0 EQUIPMENT DESIGN AND SPECIFICATION

6.1 Introduction

This chapter provides a comprehensive engineering design and specification of the key equipment units used in the production of hydrogen from natural gas through steam methane reforming (SMR) process. The SMR process is one of the most widely adopted industrial methods for large-scale hydrogen production due to its high efficiency and maturity in terms of operational knowledge. The design integrates advanced thermal management, gas separation, and carbon capture strategies to improve process efficiency and environmental performance.

Special consideration has been given to the integration of a biochar-based Pressure Swing Adsorption (PSA) unit, which serves as an environmentally sustainable alternative for carbon dioxide removal. The equipment specifications have been developed based on standard design methodologies outlined in authoritative texts such as Towler and Sinnott (2022), Perry's Chemical Engineers' Handbook (2008), and other peer-reviewed technical literature. Design conditions including temperature, pressure, material selection, and functional capacity are determined based on results obtained from mass and energy balance calculations, process simulations, and industrial best practices.

6.2 Heat Exchanger (HX-01) Design

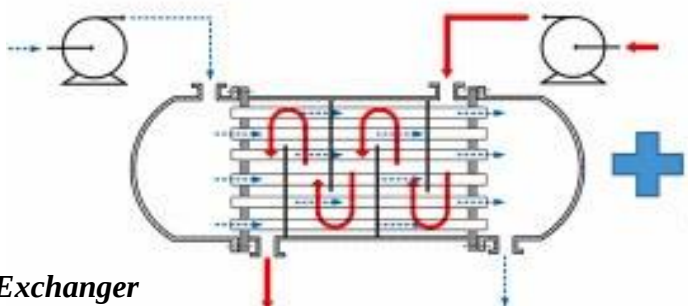


Fig. 6.1 Heat Exchanger

Design Specifications:

Exchanger type: Shell-and-tube, counter-current flow

Service: Natural gas + steam feed pre-heater (tube side) vs. reformer effluent (shell side)

Operating pressure: 25–30 atm (tube side), near atmospheric on shell side after reformer

Target hydrogen output: 10,000 m³/hr (446.4 kmol/hr H₂)

Design duty: = 1.76 MW

Heat transfer area required: = 35.2 m²

6.2.1 Cold Stream (Tube Side)

Composition: Natural gas (95% CH₄, trace C₂H₆, N₂, H₂S, inerts) + steam

Mass flow rate: 10,136.98 kg/hr (2.816 kg/s)

Inlet temperature: 36 °C

Outlet temperature: 360 °C

Average specific heat (calculated): 2.20 kJ/kg·K

Pressure: 25–30 atm (from compressor)

6.2.2 Hot Stream (Shell Side)

Source: Reformer effluent

Inlet temperature: 850 °C

Outlet temperature: 250 °C (after heat recovery)

Pressure: Slightly above atmospheric

6.2.3 Heat Duty Calculation

The heat required to raise the cold stream temperature is given by:

$$Q = m * C_p * \Delta T \quad \dots \text{Eqn (21)}$$

Where: $\dot{m}=2.816 \text{ kg/s}$, $C_p=2.20 \text{ kJ/kg.K}$, $\Delta T=324 \text{ K}$

$$Q = 2.816 \times 2.20 \times 324 = 1,760.7 \text{ kW} = 1.76 \text{ MW}$$

6.2.4 Log Mean Temperature Difference (LMTD)

Counterflow arrangement:

$$\Delta T_1 = T_{h, in} - T_{c, out} = 850 - 360 = 490 \text{ }^\circ\text{C}$$

$$\Delta T_2 = T_{h, out} - T_{c, in} = 250 - 36 = 214 \text{ }^\circ\text{C}$$

$$LMTD = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} = \frac{490 - 214}{\ln(490/214)} = 333.2^\circ\text{C}$$

6.2.5 Heat Transfer Area

Assuming an overall heat transfer coefficient $U=150 \text{ W/m}^2\cdot\text{K}$

$$A = \frac{Q}{U * LMTD} \quad \dots \text{Eqn (22)}$$

$$A = \frac{1,760,700}{150 * 333.2} = 35.2m^2$$

6.2.6 Tube Bundle Design

Tube outer diameter (d_o): 25 mm

Tube length (L): 6 m (standard)

Heat transfer area per tube:

$$A_{tube} = \pi d_o L = 0.471m^2$$

Number of tubes required:

$$N = \frac{A}{A_{tube}} = \frac{35.2}{0.471} = 75 \text{ tubes}$$

Tube-side velocity check:

Density of gas mixture at 25 atm, 36 °C = 17.3 kg/m³

Volumetric flow = 0.162 m³/s

Flow area for 75 tubes ($d_i=21d_i = 21d_i=21$ mm) = 0.026 m²

Velocity = 0.162 / 0.026 = 6.2 m/s (acceptable for gases)

6.2.7 Material Selection for Heat Exchanger

Tubes: Inconel 625 (nickel-based alloy) for high creep and oxidation resistance at > 800 °C.

Shell: Carbon steel with refractory lining or stainless steel 316 for moderate temperature exposure (downstream of partial cooling).

Baffles: Stainless steel 304/316 to direct shell-side flow.

Gaskets: Graphite or metallic type, suitable for high-temperature service.

Insulation: External mineral wool or ceramic fiber blanket.

6.2.8 Final Specification of Heat Exchanger (HX-01)

Parameter	Design Value
Type	Shell-and-tube, counter-current
Duty	1.76 MW
Heat Transfer Area	35.2 m ²
Tube Material	Inconel 625
Shell Material	Carbon steel (lined)
Tube Size	25 mm OD × 6 m length
Number of Tubes	75
Tube Passes	2
Shell Side	Hot reformer effluent
Tube Side	NG + Steam mixture
Pressure (tube side)	25–30 atm
Pressure (shell side)	Slightly > 1 atm
Velocity (tube side)	6.2 m/s
Design Temperature	Up to 900 °C (hot side)

6.3 Hydrogenator Reactor Design

The design of the Hydrogenator Reactor was based on the following:

6.3.1 Design basis & assumptions

Service: Hydrogenation to convert trace $R-S \rightarrow H_2S$ using recycled H_2 .

Operating conditions (from process): 360–385 °C; design pressure $P_d = 20 \text{ bar (g)} = 2.0 \text{ MPa} = 2.0 \text{ MN} \cdot \text{m}^{-2}$.

Basis flow/ sizing: total molar flow = $143.10 \text{ kmol} \cdot \text{h}^{-1} \rightarrow$ volumetric basis $3,207.44 \text{ Nm}^3 \cdot \text{h}^{-1}$ ($22.414 \text{ m}^3 \cdot \text{kmol}^{-1}$).

GHSV (sizing): $8,000 \text{ h}^{-1}$

Catalyst bulk density: $900 \text{ kg} \cdot \text{m}^{-3}$

Calculated catalyst volume $V_{cat} = \frac{V}{GHSV} = \frac{3,207.44}{8,000} = 0.401 \text{ m}^3$

Catalyst mass: $m_{cat} = V_{cat} \times \rho_{cat} = 0.401 \times 900 = 360.8 \text{ kgm} (= 361 \text{ kg})$

Material of construction: 316L stainless steel

Allowable stress (design standard): $f = 98.1 \text{ MNm}^{-2}$

Weld efficiency: $j = 0.85$

6.3.1.2 Kinetic Model and Justification

The Hydrogenator design is governed by the need to achieve extremely high conversion of trace organic sulfur compounds (primarily COS and CS₂). The design uses the GHSV method for physical sizing but is validated against a rigorous Chemical Reaction Engineering (CRE) model for the primary reaction:



Explicit Kinetic Parameters (Arrhenius Model)

The reaction rate constant (k_{COS}) is calculated using the Arrhenius Equation, assuming first-order dependence on the trace species (COS), which is common for highly dilute reactants over an industrial Co-Mo hydrodesulfurization catalyst:

$$k_{COS} = A \times \exp\left(\frac{-E_a}{RT}\right) \text{ where } -r'_{COS} = k_{COS} \times C_{COS} \quad \dots \text{Eqn (24)}$$

Kinetic parameter	Symbol	Value	Units	Basis
Pre-exponential factor	A	2.5×10^4	$\text{m}^3\text{gas}/(\text{kg}_{\text{cat}}\cdot\text{s})$	Literature for Co-Mo HDS
Activation Energy	Ea	65,000	J/mol	Literature for Co-Mo HDS
Gas constant	R	8.314	J/mol.K	Universal constant

($T_{\text{avg}}=372.5\text{ }^\circ\text{C}=645.65\text{ K}$):

Calculated Rate Constant (k_{COS}):

$$k_{COS} = 2.5 \times 10^4 \times \exp\left(\frac{-65000}{8.314 \times 645.65}\right) \approx 1.65 \frac{\text{m}^3\text{gas}}{\text{kg}_{\text{cat}}\cdot\text{s}}$$

6.3.2 Choice of vessel geometry (single vertical packed-bed)

Because the required catalyst volume is small (0.401 m^3), a single vertical packed-bed vessel is simpler, cheaper and consistent with the costing performed.

Selecting vessel internal diameter (D)

A compact diameter was picked to keep vessel height practical. Typical small packed-beds use D in the range 0.4–0.8 m.

We adopt: - a practical choice for this capacity and to allow multiple thermowells $D = 0.60$ m (internal) and distributor/manifold space.

Bed cross-sectional area (A)

$$A = \frac{\pi D^2}{4} = \frac{\pi (0.6)^2}{4} = 0.2827 \text{ m}^2$$

Bed height (H)

$$H = \frac{V_{cat}}{A} = \frac{0.401}{0.2827} = 1.418 \text{ m}$$

Packed bed active height: $H_{bed} = 1.5$ m.

Vessel internal length (tangent-to-tangent) includes top/bottom heads and working space; choose vessel total internal length $L_{int} = H_{bed} + 0.6$ m (cushion for distributor, gas inlet/outlet, thermowells, and freeboard).

Selected $L_{int} = 1.5 + 0.6 = 2.1$ m.

So mechanical envelope (internal):

ID = 0.60 m, TL (tangent-to-tangent) = 2.10 m.

These dimensions produce the required catalyst hold up (0.401 m^3) and leave space for internals.

6.3.3 Shell thickness calculation (cylindrical shell)

$$t_s = \frac{P \cdot D}{2 f j} \quad \dots \text{Eqn (24)}$$

Where:

P = design pressure = $2.0 \text{ MN} \cdot \text{m}^{-2}$ (20 bar)

D = internal diameter = 0.60 m

f = allowable stress = $98.1 \text{ MN} \cdot \text{m}^{-2}$

j = joint efficiency = 0.85

$$t_s = \frac{2.0 * 0.60}{2 * 98.1 * 0.85} = \frac{1.20}{166.77} = 0.00719 \text{ m}$$

Calculated shell thickness = 7.2 mm.

Outside diameter:

$$D_o = D + 2t_s = 0.60 + 2(0.01) = 0.62 \text{ m}$$

6.3.4 Head selection & thickness

A torispherical head was selected (common for pressure vessels of this size & pressure) for both top and bottom.

A conservative thickness formula for torispherical head:

$$t_h = \frac{P * D_o * C}{2 f j} \quad \dots \text{Eqn (25)}$$

where C is a stress concentration factor dependent on crown radius/knuckle geometry. For compact torispherical heads at this diameter and moderate pressure, a conservative C between 0.8–1.5 is typical - to be safe we take C = 1.0 for preliminary design.

$$t_h = \frac{2.0 * 0.62 * 1.0}{2 * 98.1 * 0.85} = \frac{1.24}{166.77} = 0.00743 \text{ m}$$

Calculated head thickness = 7.5 mm.

6.3.5 Internals - distributor, support grid, thermowells & instrumentation

Support grid / catalyst support: design to accommodate 361 kg catalyst (packed bed) and bed hydraulic loading. Use perforated support plate + hold-down appropriate for upward flow if any.

Inlet gas distributor: ensure even flow across cross-section; typical costs and internals were included in PEC.

Thermowells: at least 3 - 4 thermowells along the bed height (top, middle, bottom, and near inlet) included in instrumentation cost.

6.3.6 Final design specifications for the hydrogenator reactor

Parameter	Value / Selection
Catalyst volume	0.401 m ³
Catalyst mass	360.8 kg (= 361 kg)
Vessel internal diameter D	0.600 m
Cross-sectional area A	0.2827 m ²
Packed bed active height	1.5 m
Vessel internal length	2.10 m
Shell plate nominal thickness (selected)	10 mm
Head type	Torispherical
Head plate nominal thickness (selected)	10 mm
Design pressure	20 bar (g) = 2.0 MPa
Material	316L stainless steel
Thermowells	4 (recommended)
Orientation	Vertical fixed-bed

6.4 Pre-Reformer (PR-01) Design

The pre-reformer design was based on the following:

6.4.1 Design basis /assumptions

Basis: steady-state plant producing 10,000 m³/hr H₂.

Pre-reformer feed (after desulfurizer, and after steam addition for S/C = 3): mass flow = 2,490.51 kg/hr.

Components after pre-reformer:

CH₄ = 2,307.36 kg/hr (144.21 kmol/hr)

H₂ = 9.40 kg/hr (4.70 kmol/hr)

H₂O = 126.99 kg/hr (7.06 kmol/hr)

N₂ = 23.50 kg/hr (0.84 kmol/hr)

Inerts = 23.26 kg/hr (0.83 kmol/hr)

Total molar flow after PR: =157.64 kmol/hr (0.04379 kmol/s).

Pressure: 25 atm (design 30 atm) tube side/plant pressure.

Pre-reformer temperature (operating): 525 °C (798 K) (document reports PR exit 525 °C; use this as operating temperature).

Reactor type: Fixed-bed packed reactor (vertical tubular) with Nickel-based catalyst (pre-reforming catalyst).

Reactor operating space-velocity / sizing assumption: Volumetric GHSV (gas hourly space velocity) = $4,000 \text{ h}^{-1}$ (conservative typical industrial range for pre-reformers: $2,000\text{--}6,000 \text{ h}^{-1}$; GHSV selected to balance catalyst size and pressure drop).

Catalyst bulk density (packed): $= 800 \text{ kg/m}^3$ (typical for commercial catalyst pellets).

Catalyst particle diameter used for pressure-drop estimate: $d_p = 8\text{--}10 \text{ mm}$

Catalyst void fraction / bed porosity: $\varepsilon = 0.40$.

Gas viscosity at PR conditions: $\mu = 3 \times 10^{-5} \text{ Pa}\cdot\text{s}$

6.4.2 Volumetric flow at reactor conditions

Use ideal-gas to compute volumetric flow at PR conditions ($P = 25 \text{ atm}$, $T = 798 \text{ K}$).

$$\text{total molar flow } \dot{n} = \frac{157.64 \text{ kmol}}{\text{hr}} = 0.04379 \text{ kmol/s}$$

Ideal gas relation $\dot{V} = \frac{nRT}{P}$ with $R = 8.314 \text{ m}^3 \cdot \text{kPa}/(\text{kmol} \cdot \text{K})$

At $P = 25 \text{ atm} = 25 * 101.325 = 2,533.1 \text{ kPa}$, $T = 798 \text{ K}$

$$\dot{V} = \frac{0.1147 \text{ m}^3}{\text{s}} = 412.9 \text{ m}^3/\text{hr}$$

6.4.3 Reactor volume (GHSV method)

$\text{GHSV (h}^{-1}\text{)} = (\text{Volumetric flow at reactor conditions, m}^3/\text{hr}) \div (\text{Reactor void volume, m}^3)$

Rearranged to reactor physical volume (total bed volume):

$$V_{\text{reactor}} = \frac{\dot{V}_{\text{hr}}}{\text{GHSV}} = \frac{412.9 \text{ m}^3/\text{hr}}{4,000 \text{ hr}^{-1}} = 0.1032 \text{ m}^3$$

So required packed bed volume = 0.103 m³.

$$\text{If GHSV} = 2,000 \text{ h}^{-1}, V = 0.2065 \text{ m}^3$$

$$\text{If GHSV} = 6,000 \text{ h}^{-1}, V = 0.0688 \text{ m}^3$$

6.4.4 Catalyst weight estimate

Using typical catalyst bulk density ($\rho_{\text{bulk}} = 800 \text{ kg/m}^3$):

$$W_{\text{cat}} = V_{\text{reactor}} * \rho_{\text{bulk}} = 0.1302 * 800 = 82.6 \text{ kg}$$

So = 80–85 kg of packed catalyst is required for GHSV = 4,000 h⁻¹.

(If GHSV = 2,000 h⁻¹ → W = 165 kg. If GHSV = 6,000 h⁻¹ → W = 55 kg.)

6.4.5 Reactor geometry (practical dimensions)

For mechanical layout choose a compact vertical cylindrical packed bed with length-to-diameter ratio $L/D = 4$ (typical for small reactors).

$$V = \frac{\pi D^2}{4} * L \text{ with } L = 4D: \quad \dots \text{Eqn (26)}$$

$$V = \pi D^3, D = (V/\pi)^{\frac{1}{3}} \quad \dots \text{Eqn (27)}$$

For $V = 0.1032 \text{ m}^3$: $D = 0.32 \text{ m}$ and Length $L = 4D = 1.28 \text{ m}$

6.4.6 Pressure-drop estimate (Ergun equation, conservative)

Use Ergun equation to estimate ΔP across the packed bed:

$$\frac{\Delta P}{L} = \frac{150(1-\varepsilon)^2 \mu u_0}{\varepsilon^3 d_p^2} + \frac{1.75(1-\varepsilon) \rho u_0^2}{\varepsilon^3 d_p} \quad \dots \text{Eqn (28)}$$

Where:

$$\text{Superficial velocity } u_0 = \frac{\dot{V}}{A}$$

$$\text{For } D = 0.32 \text{ m, } A = \pi D^2/4 = 0.0805 \text{ m}^2.$$

$$\text{Volumetric flow } \dot{V} = 0.1147 \text{ m}^3/\text{s}, u_0 = 1.424 \text{ m/s}$$

$$\text{Gas density at PR conditions: mass flow } 2,490.51 \text{ kg/hr} = 0.6929 \text{ kg/s} \rightarrow \rho = 0.6929/0.1147 = 6.03 \text{ kg/m}^3.$$

$$\text{Choose particle diameter } d_p = 0.008\text{--}0.010 \text{ m (8--10mm) to reduce } \Delta P$$

$$\text{Bed porosity } \varepsilon = 0.40; \mu = 3 \times 10^{-5} \text{ Pa}\cdot\text{s}.$$

6.4.7 Materials of construction

Shell / pressure vessel: Carbon steel (SA516 Gr70) with internal stainless steel 316L cladding if corrosive; design to ASME VIII Div.1 for design pressure 35 bar (safety margin).

Internal components (distributor, support grids, thermowells): SS316/SS316L.

Nozzles / flanges: Forged steel (raised face), matching flange rating to design pressure (e.g., ANSI 300/600 as applicable).

Seals & gaskets: Graphite or metal ring gaskets (high-T).

Insulation: Mineral wool or ceramic blanket for high-temp operation.

6.4.8 Final design table of the Pre-Reformer

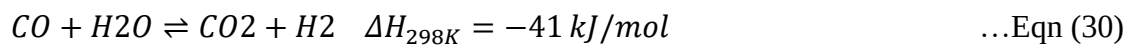
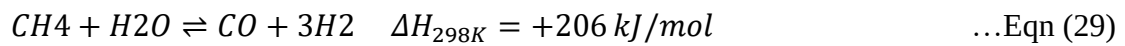
Item	Value
Unit designation	PR-01 (Pre-reformer)
Service	Pre-reforming of NG/steam ($C_2 \rightarrow CH_4$ conversion)
Operating pressure	25 bar (design 30 bar)
Operating temperature	525 °C (design 550 °C max)
Inlet volumetric flow (PR conditions)	412.9 m ³ /hr (0.1147 m ³ /s)
Packed bed volume	0.103 m ³ (GHSV = 4,000 h ⁻¹)
Reactor geometry	D = 0.32 m; L(active bed) = 1.28 m; overall vessel length = 1.6–1.8 m
Catalyst type	Ni-based pre-reforming catalyst (commercial)
Catalyst form & size	Extrudates/pellets 8–10 mm
Catalyst mass	= 83 kg
Estimated ΔP across bed	0.26–0.40 bar (depending on d_p)
Materials of construction	Shell: CS (ASME), internal SS316L for wetted internals
Internals	Gas distributor, support grid, thermowells, sample ports
Safety	Relief valves, N ₂ purge, ΔP & T alarms

6.5 Reformer Design

The steam methane reformer (SMR) was designed as follows:

6.5.1 Introduction

The steam methane reformer (SMR) is a tubular fixed-bed reactor designed to convert methane and steam into hydrogen, carbon monoxide, and carbon dioxide according to the endothermic reactions below:



The reactor operates under high temperature (850–1100°C) and moderate pressure (20–30 atm) using a nickel-based catalyst supported on alumina. The design ensures sufficient residence time, catalyst utilization, and heat transfer to sustain the endothermic reforming reaction.

6.5.2 Gas Properties

Gas	Molecular Weight (g/mol)
Hydrogen (H ₂)	2.016
Water (H ₂ O)	18.015
Methane (CH ₄)	16.043
Nitrogen (N ₂)	28.014
Argon (Ar)	39.948

6.5.3 Conversion of Mass Flowrate to Molar Flowrate

Example for hydrogen:

$$n_{H_2} = \frac{\dot{m}_{H_2}}{M_{H_2}} = \frac{9.40 \text{ kg/h}}{0.002016 \text{ kg/mol}} = 4663.49 \text{ mol/h}$$

Total molar flowrate:

$$\dot{n}_{total} = 5.814 \times 10^5 \text{ mol/h} = 161.50 \text{ mol/s}$$

6.5.4 Reactor Operating Conditions

Parameter	Symbol	Value	Unit
Temperature	T	850°C (1123.15 K)	K
Pressure	P	27 atm (2.736×10^6)	Pa

6.5.5 Volumetric Flowrate (Ideal Gas Law)

$$PV = nRT \quad \dots \text{Eqn (31)}$$

$$V = \frac{nRT}{P} = \frac{(5.814 \times 10^5)(8.314)(1123.15)}{2.736 \times 10^6} = 0.55 \text{ m}^3/\text{s} = 1984.59 \text{ m}^3/\text{h}$$

6.5.6 Gas Hourly Space Velocity (GHSV)

$$GHSV = \frac{Q}{V_{cat}} \quad \dots \text{Eqn (32)}$$

Assuming GHSV = 5000 h⁻¹ (typical for Ni-based SMR catalysts at 800–900°C):

$$V_{cat} = \frac{1984.54}{5000} = 0.397 \text{ m}^3$$

6.5.7 Catalyst Tube Design Parameters

Parameter	Symbol	Range / Value	Unit
Tube internal diameter D		0.10 – 0.115	m
Tube length	L	6 – 12	m
Tube material	—	253 MA Stainless Steel	—
Cross-sectional area	$A = \pi D^2/4$	0.00785 – 0.01039	m ²
Tube volume	$V = A \times L$	0.0943 – 0.1247	m ³

6.5.8 Number of Tubes

(a) Volume Constraint

$$N_{vol} = \frac{V_{cat}}{A \times L} \quad \dots \text{Eqn (33)}$$

Diameter (m)	Area (m ²)	Volume (m ³)	Tubes Required
0.100	0.00785	0.09425	4
0.115	0.01039	0.12466	3

(b) Velocity Constraint ($u \leq 2.0 \text{ m/s}$)

$$N_{vel} = \frac{Q}{\mu_{max} A} \quad \dots \text{Eqn (34)}$$

Diameter (m)	Area (m ²)	Tubes Required
0.100	0.00785	36
0.115	0.01039	27

Hence, 27–36 tubes are required depending on the selected diameter.

6.5.9 Void Fraction Calculations

Parameter	Symbol	Typical Value
Inter-particle void fraction	$\varepsilon_{\text{inter}}$	0.40
Intra-particle porosity	$\varepsilon_{\text{intra}}$	0.25
Total void fraction	$\varepsilon_{\text{total}} = \varepsilon_{\text{inter}} + \varepsilon_{\text{intra}}(1 - \varepsilon_{\text{inter}}) = 0.55$	—
$V_{\text{void total}} = \varepsilon_{\text{total}} V_{\text{bed}}$		
Tube Diameter (m)	Vbed (m ³)	Vvoid (m ³)
0.100	0.3969	0.2183
0.115	0.3969	0.2183

6.5.10 Packed Bed Characteristics

Parameter	Formula	0.100 m Tube	0.115 m Tube
Number of tubes	μ	36	27
Tube area (m ²)	$\pi D^2/4$	0.00785	0.01039
Packed length (m)	$V_{\text{bed}} / (\mu A)$	1.404	1.415
Bed void volume (m ³)	$\varepsilon_{\text{inter}} \times V_{\text{bed}}$	0.1588	0.1588
Intra-particle void (m ³)	$\varepsilon_{\text{intra}}(1 - \varepsilon_{\text{inter}}) V_{\text{bed}}$	0.0595	0.0595
Total void volume (m ³)	$\varepsilon_{\text{total}} \times V_{\text{bed}}$	0.2183	0.2183

6.5.11 Reactor Flow Characteristics

Parameter	Symbol	0.100 m Tube	0.115 m Tube	Unit
Total bed volume	V _{bed}	3.3929	3.3654	m ³
Total void volume	V _{void}	1.866	1.851	m ³
Reactor total volume	V _R	0.3969	0.3969	m ³
Volume of solids (packing)	V _{solid}	0.1786	0.1786	m ³
Space velocity	GHSV	584.9	589.7	h ⁻¹
Cross-sectional area of voids —		13.866	13.851	m ²
Flow velocity through voids —		0.03976	0.03980	m/s
Superficial velocity	u _{max}	2.0	2.0	m/s

6.5.12 Final Design specifications for the Reformer

Parameter	Symbol / Range	Value / Range	Unit
Reactor Type	—	Tubular (Fixed Bed)	—
Operating Temperature	T	850–1100	°C
Operating Pressure	P	27 (2.736×10^6)	atm (Pa)
Flowrate	Q	0.5513 (1984.59)	m ³ /s (m ³ /h)
Tube Length	L	6–12	m
Tube Diameter (ID)	D	0.100–0.115	m
Tube Material	—	253 MA Stainless Steel	—
Number of Tubes	N	27–36	—
Catalyst Volume	V _{cat}	0.397	m ³
Reactor Volume	V _R	0.3969	m ³
Space Velocity	GHSV	5000	h ⁻¹
Packed Length	L _{packed}	1.404–1.415	m
Total Void Fraction	$\varepsilon_{\text{total}}$	0.55	—
Total Void Volume	V _{void}	0.2183	m ³
Speed Through Voids	—	0.0398	m/s

6.5.13 Kinetic Parameters of Steam Methane Reforming

The kinetic rate of methane reforming over a Ni-based catalyst can be expressed by a Langmuir–Hinshelwood–Hougen–Watson (LHHW) type equation (Xu and Froment, 1989):

$$r_{CH_4} = \frac{k_1 P_{CH_4} P_{H_2O} - \frac{p_{H_2}^3 p_{CO}}{K_{eq1}}}{(1 + K_{CH_4} P_{CH_4} + K_{H_2O} P_{H_2O} + K_{CO} P_{CO} + K_{H_2} P_{H_2})^2} \quad \dots \text{Eqn (35)}$$

Where:

r_{CH_4} = rate of methane reforming ($\text{mol} \cdot \text{kg}^{-1} \cdot \text{s}^{-1}$)

k_1 = rate constant ($\text{mol} \cdot \text{kg}^{-1} \cdot \text{s}^{-1} \cdot \text{Pa}^{-2}$)

K_{eq1} = equilibrium constant of the reforming reaction

K_i = adsorption equilibrium constants for component i

Kinetic Constants (Xu & Froment, 1989)

Parameter	Symbol	Expression	Value Range
Rate constant	k_1	$k_1 = 1.07 \times 10^{15} \exp(-240100/RT)$	$\text{mol} \cdot \text{kg}^{-1} \cdot \text{s}^{-1} \cdot \text{Pa}^{-2}$
Adsorption constant for CH_4	K_{CH_4}	$= 6.65 \times 10^{-4} \exp\left[\frac{f_0}{RT}\right](38280/RT)$	Pa^{-1}
Adsorption constant for H_2O	K_{H_2O}	$= 1.77 \times 10^5 \exp\left[\frac{f_0}{RT}\right](88680/RT)$	Pa^{-1}
Adsorption constant for CO	K_{CO}	$= 8.23 \times 10^{-5} \exp\left[\frac{f_0}{RT}\right](70650/RT)$	Pa^{-1}
Adsorption constant for H_2	K_{H_2}	$= 6.12 \times 10^{-9} \exp\left[\frac{f_0}{RT}\right](82900/RT)$	Pa^{-1}
Activation Energy	E_a	240.1×10^3	J/mol
Temperature Range	—	823–1123	K

Source:

Xu, J. and Froment, G.F. (1989). *Methane steam reforming, methanation and water-gas shift: I. Intrinsic kinetics*. *AIChE Journal*, 35(1), 88–96.

6.5.14 Engineering Remarks

The design meets industrial SMR standards, operating effectively between 850–1100°C.

Selected GHSV = 5000 h^{-1} ensures optimum catalyst performance and conversion.

Tube material (253 MA stainless steel) provides excellent resistance to carburization and creep at high temperature.

The kinetic parameters provide the basis for reaction rate modeling and reactor performance simulation.

6.6 High Temperature Shift Converter Design

The design for the High temperature shift converter is as follows:

6.6.1 Reaction kinetics & thermodynamics

Water-Gas Shift Reaction:



$$\Delta H = -41.1 \text{ kJ/mol (exothermic)}$$

Equilibrium Constant at 400°C

Using thermodynamic data: $K_{eq} = 9.4$ (at 400°C)

Reaction Kinetics (Fe-Cr Catalyst)

Power-law rate expression:

$$-R_{CO} = k P_{CO}^{0.9} P_{H_2O}^{0.25} P_{H_2}^{-0.6} [\text{mol}/(\text{g-cat} \cdot \text{h})] \quad \dots \text{Eqn (37)}$$

Where P is the partial pressures in atm.

with $k = 0.23 \text{ mol}/(\text{g-cat} \cdot \text{h} \cdot \text{atm}^{1.55})$ at 400°C

6.6.2 Reactor sizing & performance

Design Basis

Feed: 115.18 kmol/h CO, 258.74 kmol/h H₂O (S/C = 2.25)

Target Conversion: 95% CO conversion

Conditions: 400°C, 27 atm

Catalyst Mass Calculation

Using numerical integration of the power-law rate equation:

$$W = F_{CO} \int_0^X \frac{dX}{-r_{CO}} \quad \dots \text{Eqn (38)}$$

from 0 to 0.95 conversion, catalyst weight (W) = 359 kg

6.6.3 Reactor Geometry

Optimized dimensions for good flow distribution:

Diameter = 0.5 m

Bed height = 2.62 m

L/D ratio = 5.24 (optimal for plug flow)

6.6.4 Hydraulic design

Superficial Velocity: $u_s = 1.082$ m/s (at reactor conditions)

Pressure Drop: Using Ergun equation:

$\Delta P = 1.28$ bar (acceptable for industrial operation)

6.6.5 Mechanical design

Vessel Specifications

Material: SA-516 Grade 70 Carbon Steel

Design Pressure: 450 psig (10% over operating)

Design Temperature: 750°F (400°C)

Shell Thickness: 0.5 inches

Heads: Hemispherical, 0.75 inches

6.6.5.1 Weight Analysis

Empty vessel weight: 194 kg

Catalyst weight: 359 kg

Total operating weight: 553 kg

6.6.6 High Temperature Shift Converter Final Design Specifications

Parameter	Value	Unit	Reference
Catalyst Type	Fe-Cr (promoted)	-	Twigg, 1996
Catalyst Mass	359	Kg	Calculated
Reactor Diameter	0.5	M	Calculated
Bed Height	2.62	M	Calculated
Operating Temperature	400	°C	Hla et al., 2009
Operating Pressure	27	Atm	This work
CO Conversion	95	%	This work
Pressure Drop	1.28	Bar	Calculated
L/D Ratio	5.24	-	Towler & Sinnott, 2013

6.7 PSA Design

The design for the pressure swing adsorber design is as follows:

6.7.1 Basis and Capacity

The hydrogen plant is designed to produce 10,000 m³/hr of high-purity hydrogen (99.9 %) from steam methane reforming. The off-gas from the condenser (Unit 8) is used as the feed to the PSA system. The PSA operates at steady state and recovers 85 % of hydrogen in the feed.

6.7.2 Materials of Construction for PSA Unit

In industrial PSA hydrogen purification systems, the vessels and associated equipment are required to operate under cyclic high-pressure hydrogen service. For this reason, the adsorption columns are typically fabricated from low-alloy steels such as 2.25Cr-1Mo or SA-372, which provide sufficient mechanical strength and resistance to hydrogen embrittlement. In order to improve corrosion resistance and minimize the risk of hydrogen attack, the inner surface of the vessel is normally clad or lined with austenitic stainless steel (e.g. SS304 or SS316). The same stainless-steel alloys are also used for the switching valves, piping manifolds and support plates, as these components are directly exposed to the corrosive feed stream and undergo frequent pressure cycling. The combination of low-alloy structural steel and stainless-steel lining represents the most cost-effective and reliable material configuration for PSA hydrogen purification units.

All calculations below assume STP = 1 atm, 25 °C (22.414 m³ per kmol of gas).

Basis and Feed Composition.

Hydrogen product requirement = 10,000 m³/hr

At STP:

$$H_2 \text{ molar flow} = \frac{10,000 \text{ m}^3/\text{hr}}{22.414 \text{ m}^3/\text{kmol}} = 446.13 \text{ kmol/hr}$$

Since **85 % of H₂ in the feed is recovered**, the feed must contain:

$$446.13/0.85 = 524.86 \text{ kmol/hr}$$

6.7.3 PSA Operating conditions and assumptions

Parameter	Value (assumed)
Adsorption pressure	15 bar
Desorption pressure	1 bar
Temperature	30 °C (isothermal)
Adsorbent	Zeolite 13X
Working capacity for H ₂	1.3 mol H ₂ /kg adsorbent
Void fraction	0.35
Particle density (ρ_p)	1,100 kg/m ³
Target H ₂ purity	99.9 %
Number of beds	4 beds

Working Capacity of Adsorbent

Working capacity (Δq) = 1.3 mol H₂ / kg adsorbent

$$\Delta q = 1.3/1000 = 0.0013 \text{ kmol } H_2 \text{ per kg adsorbent}$$

Mass of Adsorbent Required

H₂ adsorbed per cycle per bed must match the amount of H₂ processed per bed.

H₂ to be purified = 524.86 kmol/hr

With a 4-bed system, assume 25 % load per bed:

$$H_2 \text{ per bed} = \frac{524.86}{4} = 131.215 \text{ kmol/hr}$$

Let τ = adsorption time per bed (assume 15 min = 0.25 hr)

$$H_2 \text{ processed per bed per cycle} = 131.215 \text{ kmol/hr} \times 0.25 \text{ hr} = 32.80 \text{ kmol}$$

Required adsorbent mass per bed:

$$Mass = \frac{32.80}{\Delta q} = \frac{32.80}{0.0013} = 25,230.77 \text{ kg}$$

Size of Each Bed

Adsorbent bulk density (ρ_{bulk}) = (1 – void fraction) × particle density

$$\rho_{\text{bulk}} = (1 - 0.35) \times 1100 = 715 \text{ kg/m}^3$$

Required bed volume:

$$V_{\text{bed}} = \frac{25,230.77}{715} = 35.28 \text{ m}^3$$

Assume bed diameter = 2 m → cross-sectional area:

$$A = \frac{\pi D^2}{4} = \frac{\pi (2)^2}{4} = 3.142 \text{ m}^2 \quad \dots \text{Eqn (39)}$$

Bed length:

$$L = \frac{V}{A} = \frac{35.28}{3.142} = 11.23 \text{ m} \quad \dots \text{Eqn (40)}$$

Bed size: 2 m diameter × 11.2 m length

Bed Productivity

$$\text{Productivity} = \frac{H^2 \text{ processed per bed}}{V_{bed}} = \frac{32.80}{35.28} 0.93 \text{ kmol } H^2 \text{ per m}^3 \text{ adsorbent per cycle}$$

Total Adsorbent in PSA

$$\text{Total mass of adsorbent} = 4 \times 25,230.77 = 100,923.1 \text{ kg}$$

PSA Cycle Description (4-Bed System)**Pressurization (=10 % of cycle)**

Bed is brought from 1 bar to 15 bar using product or feed gas.

Purpose: prepare bed for adsorption.

Adsorption (=40 %)

Feed gas enters at the top at 15 bar; H_2 exits at the bottom as high-purity product.

Impurities (CO_2 , CO, etc.) remain adsorbed.

Co-Current Depressurization (=15 %)

Pressure is reduced in the *same* direction as feed flow and the gas released is used to pressurize another bed.

Purpose: recover residual H₂.

Counter-Current Blowdown (=15 %)

Pressure is further reduced *opposite* to feed direction, driving impurities toward the feed end.

Purpose: remove adsorbed impurities.

Purge (=20 %)

Low pressure H₂ is used to purge the bed from the feed end.

All remaining impurities are removed and the bed is ready for the next cycle at any given time.

Bed Step

- 1 Pressurization → Adsorption
- 2 Adsorption → Co-Current Depressurization
- 3 Blowdown
- 4 Purge

Biochar CO₂ Adsorption Unit (Secondary Stage)

Tail gas to biochar contains:

- $CO_2 = 126.26 \text{ kmol/hr} \times (1 - 0.85 \text{ } H_2 \text{ recovery}) = 18.94 \text{ kmol/hr } CO_2$

90 % capture with biochar: $CO_2 \text{ adsorbed} = 0.90 \times 18.94 = 17.05 \text{ kmol/hr}$

Assume a biochar working capacity of $1.0 \text{ mol CO}_2/\text{kg} = 0.001 \text{ kmol/kg}$

Required adsorbent mass:

$$(17.05)/(0.001) = 17,050 \text{ kg biochar}$$

Assume bulk density of biochar = 450 kg/m^3

$$V = \frac{17,050}{450} = 37.89 \text{ m}^3$$

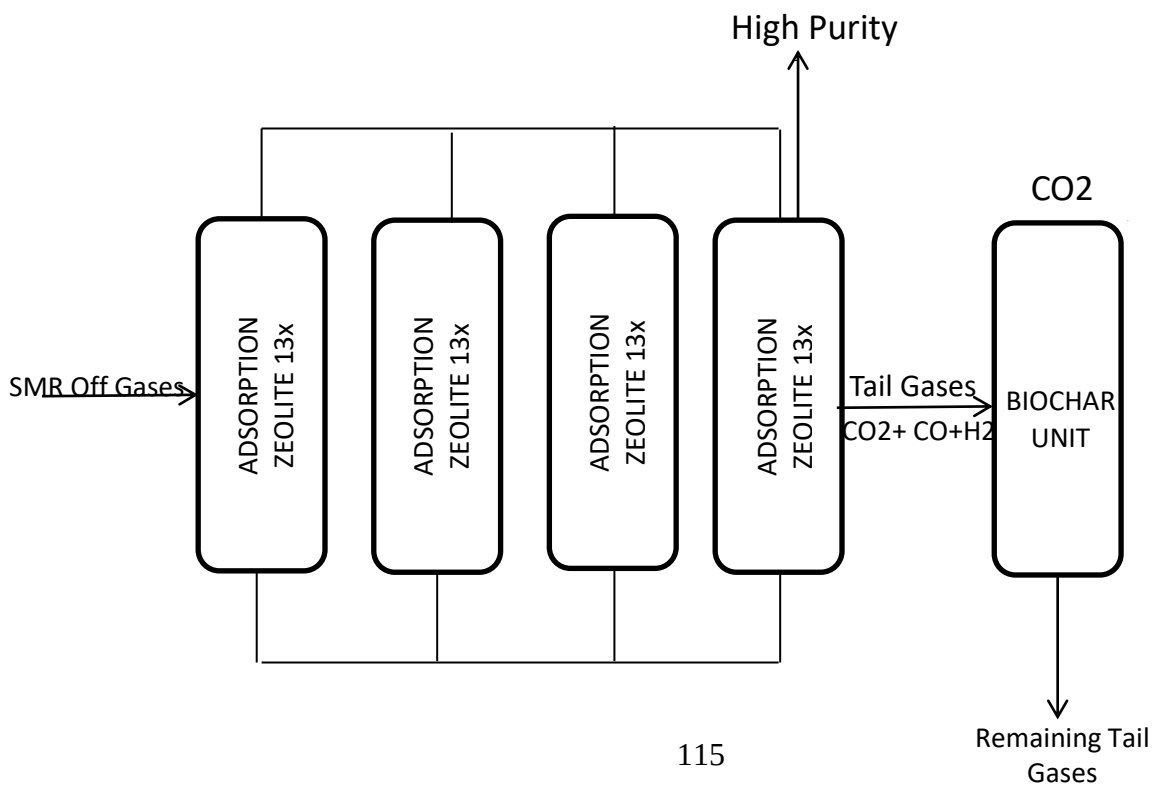
For a 2 m diameter vessel:

$$L = \frac{37.89}{3.142} = 12.06 \text{ m}$$

Biochar bed: 2 m Ø × 12.1 m length

6.7.5 Final design specifications for the Pressure Swing Adsorber

Item	Result
Number of PSA beds	4
Adsorption pressure	15 bar
Desorption pressure	1 bar
Adsorbent (PSA)	Zeolite 13X
Adsorbent mass per bed	25,230 kg
Total adsorbent (PSA)	100,923 kg
Bed size (PSA)	2 m Ø × 11.2 m length
Adsorption time per bed	15 min
Biochar for CO ₂ capture	17,050 kg
Biochar bed size	2 m Ø × 12.1 m length



6.8 Design Specification of Pump (P-01)

Parameter	Value
Function	Circulates water and fluids
Type	Centrifugal pump
Flow rate	8 m ³ /h
Head	30 m
Material	Stainless steel
Power	5.5 Kw
Operating temperature	100°C

6.9 Design Specification of Water Tank (TK-01)

Parameter	Value
Function	Storage of process and feed water
Volume	5 m ³
Material	Mild steel with epoxy lining
Height	2.5 m
Diameter	1.6 m
Orientation	Vertical, cylindrical
Design pressure	Atmospheric

CHAPTER SEVEN

7.0 SITE LOCATION AND PLANT LAYOUT

7.1 Introduction

The successful implementation of any chemical process design does not end with the selection of process routes, material balances, and equipment design alone. The physical location of the plant and the manner in which the individual units are arranged within the site strongly determine the safety, operability, environmental compliance, and overall economic viability of the project. This chapter discusses the proposed site for the hydrogen production plant integrated with biochar-based carbon capture, the criteria for site selection, the estimated land area required, and the conceptual layout of the plant.

7.2 Site Selection Criteria

The choice of an appropriate site is critical because it directly influences plant construction cost, operational reliability, and the level of environmental and social acceptance. The following factors guided the selection of the site:

1. Proximity to Feedstock Supply:

The primary feedstock for the plant is natural gas, which is sourced from the Nigerian Liquefied Natural Gas (NLNG) grid. Therefore, the site must be close to an existing natural gas pipeline to reduce the cost of feedstock transport and minimize line losses.

2. Availability of Utilities:

The plant requires a reliable supply of electricity, process water, and steam. Proximity to a substation and the possibility of drilling boreholes or connecting to industrial water networks are important considerations.

3. Transportation and Accessibility:

Accessibility by road for raw material delivery, equipment transport, and product dispatch is essential. A site with proximity to a major highway or an industrial corridor is most suitable. If possible, access to a rail network or port facility is an added advantage.

4. Topography and Soil Conditions:

The site must be relatively flat with stable soil conditions to reduce foundation and earthwork costs. It must not be located in a flood-prone area.

5. Environmental and Social Considerations:

The location must comply with environmental regulations and avoid ecologically sensitive areas. Adequate buffer zones should exist between the plant and nearby residential areas to minimize the impact of emissions, noise, and vibrations.

6. Safety and Emergency Access:

The site must allow for adequate setbacks from the property boundary and neighboring facilities in accordance with NFPA and API standards. Provision for firewater supply, emergency access roads, and muster points must also be ensured.

7. Potential for Expansion:

The chosen site must be spacious enough to accommodate future expansion of plant capacity or the addition of auxiliary processing units.

Based on the above considerations, the most suitable site is an industrial zone located in proximity to the NLNG gas pipeline corridor in southern Nigeria, which satisfies feedstock availability, utility access, and transport logistics.

7.3 Site Size Estimation

The proposed hydrogen production capacity of 10,000 m³/h requires significant space for process units, storage, utilities, and safety buffers. By considering comparable hydrogen reforming plants and standard spacing requirements, the following estimate is proposed:

Process Units (Reformer, Pre-reformer, HT Shift, PSA, Boilers): **12,000 m²**

Feedstock Handling and Storage: **5,000 m²**

Biochar CO₂ Capture Unit and Storage: **6,000 m²**

Utilities (Cooling Tower, Substation, Boiler House, Water Treatment): **~8,000 m²**

Administration, Laboratory, and Control Room: **~3,000 m²**

Internal Roads and Emergency Access: **7,000 m²**

Future Expansion and Green Buffer Zone: **10,000 m²**

Total Estimated Land Area: 51,000 m² (≈5 hectares)

This ensures adequate provision for process units, utilities, and safety requirements.

7.4 Conceptual Plant Layout

The conceptual layout of the hydrogen production plant is designed to achieve a logical sequence of operations, safe working conditions, and efficient use of space. The arrangement of equipment ensures smooth material flow, minimizes piping runs, and enhances operability and maintainability, while meeting safety, environmental, and regulatory standards.

The major functional blocks of the plant are described as follows:

7.4.1 Feed and receiving section (North-West Corner)

The plant receives natural gas from the NLNG grid at the northwest side of the site. This area houses the gas metering, pressure-reducing, and conditioning facilities, along with the hydrogenator and desulfurization reactors. Positioning this section at the entry point minimizes piping length before the gas enters the pre-reformer. Adequate safety spacing and isolation valves are included to handle potential leaks or contaminants in the incoming feed.

7.4.2 Pre-Reformer and reformer section (Central Core)

The pre-reformer and fired tubular reformer are located at the central core of the facility since they are the heart of the hydrogen generation process. Their placement at the center allows for optimal heat integration and short piping runs to downstream equipment. The reformer stack is positioned with sufficient clearance for safe flue gas discharge. Adjacent to the reformer, the process gas boiler is installed to capture and recover heat from flue gases, improving the overall plant energy efficiency.

7.4.3 High-temperature shift converter (downstream of reformer)

The effluent gas from the reformer passes directly to the high-temperature shift converter, positioned immediately downstream. This arrangement ensures minimal heat losses and effective utilization of the high-temperature gas stream. The HT shift converter is followed by waste heat exchangers, which condition the syngas by recovering thermal energy for use in other utility systems.

7.4.4 Pressure swing adsorption (psa) and compression section (east end)

The PSA unit, together with the hydrogen compression and storage facilities, is located on the eastern boundary of the plot. This location near the product dispatch point minimizes hydrogen transport distances. Furthermore, the placement ensures that potential impurities or leaks from the PSA do not backflow into the process area, thereby maintaining product purity and safety.

7.4.5 Biochar CO₂ capture area (Southern Zone)

The biochar adsorption beds and CO₂ handling facilities are located at the southern end of the site. This section is spaced adequately from the hydrogen purification and compression areas to prevent cross-contamination. Storage silos and biochar handling equipment are also installed here, with appropriate containment and dust suppression systems.

7.4.6 Utilities and support systems (Southern Periphery):

Key utility systems such as cooling towers, water treatment plants, boiler feedwater units, firewater tanks, air compressors, and electrical substations are located along the southern periphery. This arrangement prevents interference with critical process units while allowing easy access for operation, monitoring, and maintenance.

7.4.7 Administration and control buildings (Upwind Side)

To safeguard personnel, the control room, central administration block, maintenance workshops, and laboratory are situated upwind of the process units at a safe setback distance. This ensures protection from potential accidental releases, fire hazards, or toxic gas exposure. Emergency access roads connect these buildings directly to external exit routes.

7.4.8 Emergency and safety facilities

The plant is equipped with a flare stack for controlled venting of process gases, strategically positioned downwind and at a safe distance from occupied areas. A firewater ring-main system, gas detection sensors, emergency shutdown systems, and muster points are integrated into the layout. Multiple ingress and egress routes ensure safe evacuation and quick emergency response.

7.5 Layout Considerations

The layout has been developed with the following design principles:

1. Linear flow arrangement from feedstock reception to hydrogen purification and storage.
2. Adequate spacing (10–25 m) between major equipment blocks to ensure maintenance access and fire safety.
3. Two main internal roads (8–10 m wide) running east–west and north–south, with secondary roads (4–6 m wide) for maintenance access.

4. Separation of high-risk process units (reformer, furnace, compressors) from administrative and control buildings by at least 100–200 m.
5. Allocation of sufficient space for stormwater drainage, flare stack placement, and firewater facilities.

7.6 Safety and Environmental Measures

1. A dedicated flare system is provided for safe disposal of waste gases.
2. Gas detectors are distributed across the reformer, PSA, and compressor areas.
3. All stormwater run-off is directed to an oil–water separator before discharge.
4. Biochar handling areas are covered and dust-controlled.
5. Fire hydrants are installed at 50 m intervals along the firewater ring-main.

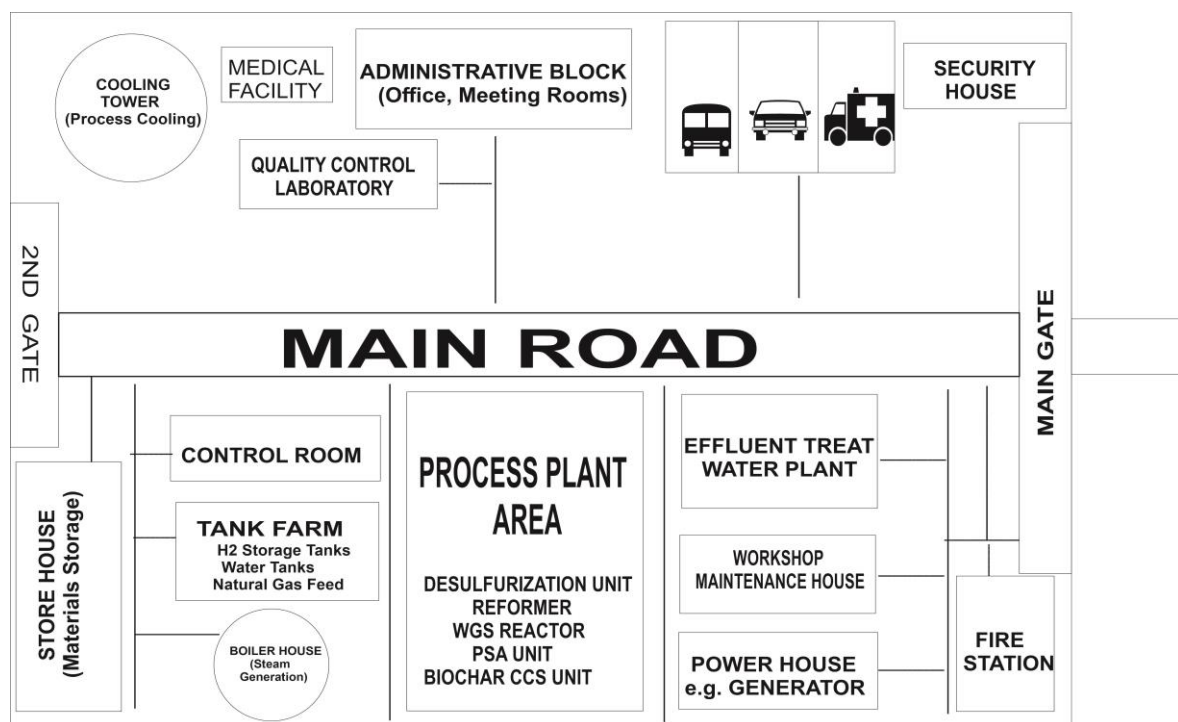


Fig. 7.1: Plant Layout of Hydrogen Plant

CHAPTER EIGHT

8.0 PROCESS CONTROL AND INSTRUMENTATION

8.1 Introduction

Process control and instrumentation are vital components in the design and operation of chemical process plants. In a hydrogen production plant operating via Steam Methane Reforming (SMR) integrated with a biochar-based carbon capture system, maintaining accurate control of process parameters is crucial for achieving optimal hydrogen yield, product purity, safety, and environmental compliance. The process involves multiple high-temperature and high-pressure operations, such as reforming, shift conversion, and adsorption, all of which are sensitive to fluctuations in temperature, pressure, and flow rate. Effective control systems ensure that these variables remain within safe and optimal operating limits, thereby stabilizing the process and preventing hazardous conditions (Seborg et al., 2020).

Instrumentation serves as the sensory component of the control system, providing real-time measurements of critical process variables such as temperature, pressure, flow, level, and gas composition. These signals are transmitted to controllers—typically Proportional-Integral-Derivative (PID) controllers or advanced Distributed Control Systems (DCS)—which analyze deviations from set points and send corrective actions to final control elements like valves, pumps, and heaters (Marlin, 2021).

This chapter discusses the control philosophy adopted for the hydrogen production plant, detailing the control loops for major units, the instrumentation strategy, automation architecture, and safety interlocks. The design approach ensures process efficiency, product consistency, and plant safety.

8.1 Objectives of Process Control

The objectives of process control in the hydrogen production plant are centered on ensuring the safe, stable, and economically efficient operation of all process units. Given the complexity and sensitivity of hydrogen production via steam methane reforming (SMR), coupled with integrated carbon capture using biochar, precise control of process variables is essential for optimal performance. The control system must respond rapidly to disturbances, maintain set operating conditions, and guarantee product consistency while preventing hazardous situations. The key objectives are discussed in detail below.

8.1.1 Process stability and continuity of operation

One of the fundamental objectives of process control is to maintain steady-state conditions throughout the plant despite inevitable disturbances, such as fluctuations in feed composition, pressure drops, or temperature variations. In the hydrogen production plant, stable operation ensures continuous flow of natural gas through the preheater, reformer, and PSA units without oscillations that could cause inefficiencies or equipment stress. Properly tuned PID controllers stabilize variables like pressure, temperature, and flow rate, ensuring smooth operation and avoiding plant shutdowns. A stable process also minimizes catalyst degradation and mechanical wear of compressors and pumps, thereby extending equipment life.

8.1.2 Product quality and consistency

Product quality control is critical, particularly for achieving the desired hydrogen purity of 99.9% or higher at the PSA outlet. Variations in feed rate, reaction temperature, or adsorption cycle timing can directly affect the hydrogen composition and carbon monoxide slip. The control system ensures that temperature and pressure conditions in each reactor

and PSA column are maintained within narrow tolerances. Gas analyzers continuously monitor product composition and send feedback to composition controllers, which make necessary adjustments to maintain purity standards. Maintaining consistent product quality not only meets process design specifications but also satisfies industrial and energy sector requirements for fuel-cell-grade hydrogen (Atilhan *et al.*, 2021).

8.1.3 Process safety and equipment protection

Given that hydrogen is a highly flammable and explosive gas, safety remains a core objective of the control system. Process control minimizes risks by preventing hazardous deviations such as overpressure, overtemperature, or flame instability in the reformer furnace. Safety Instrumented Systems (SIS), including pressure relief valves, emergency shutdown systems, and fire and gas detectors, are integrated with the main control system to automatically isolate or shut down affected units in case of abnormal conditions. For example, a rise in reformer tube temperature beyond the safe limit triggers an automatic fuel cutoff and initiates emergency cooling. The control system thereby acts as both a preventive and protective barrier against accidents (Seborg *et al.*, 2020).

8.1.4 Energy efficiency and resource optimization

Energy management is a key objective in hydrogen production plants because the reforming and steam generation steps are energy-intensive. Efficient control of steam-to-carbon ratios, fuel flow, and heat exchanger duties can significantly reduce fuel consumption and carbon emissions. Temperature control in heat exchangers, reformers, and boilers ensures optimal heat recovery, while flow control loops regulate the exact amount of steam required for reforming without wastage. Advanced control strategies such as model predictive control (MPC) can be used in industrial applications to further

optimize energy balance and minimize operational costs (Marlin, 2021). Proper control also allows efficient integration of recovered heat for use in other parts of the plant, promoting sustainability.

8.1.5 Environmental protection and emission control

Modern process plants are required to operate within strict environmental regulations. In this design, the integration of a biochar-based carbon capture system adds a new dimension to process control, where maintaining CO₂ capture efficiency and managing regeneration cycles become key objectives. The control system regulates feed gas flow to the adsorption unit, monitors outlet CO₂ concentration using gas analyzers, and adjusts regeneration conditions to maintain adsorbent performance. Additionally, flare systems are controlled to ensure safe combustion of vent gases, thereby minimizing greenhouse gas emissions. Through these mechanisms, the control system contributes to environmental sustainability and compliance with emission standards (Zhou *et al.*, 2023).

8.1.6 Reliability, availability, and maintainability

Another important objective of process control is to improve plant reliability and uptime. Automation systems are designed with redundancy (dual transmitters, backup power supply, and redundant DCS servers) to minimize downtime during maintenance or equipment failure. Continuous monitoring and alarm systems allow early detection of faults such as sensor drift or valve malfunction, enabling preventive maintenance actions. The inclusion of diagnostic and self-checking capabilities in modern transmitters further enhances maintainability and operational reliability (Sinnott & Towler, 2020).

Reliable process control ensures that hydrogen production meets economic targets with minimal interruptions, contributing to long-term plant profitability.

8.2 Key Process Variables and Measurement Instruments

Process control systems rely on the continuous measurement and regulation of four key variables: flow, temperature, pressure, and composition. Each variable requires a specific type of instrument for accurate detection and feedback.

Variable	Measured By	Controlled By	Function/Location
Temperature (T)	Thermocouples, RTDs	Temperature (TIC) → Control Valve	Reformer, Preheater, WGS, Boilers
Pressure (P)	Pressure Transmitter (PT)	Pressure Controller (PIC) → Valve/Compressor	PSA unit, Reactors, Steam lines
Flow (F)	Orifice or Flow Meter	Mass Flow Controller (FIC) → Control Valve	Natural gas, steam, and water feeds
Level (L)	Level Transmitter (LT)	Level Controller (LIC) → Control Valve	Boilers, Condensers, Steam drum
Composition (C)	Gas Analyzer (H ₂ , CO ₂ , CO)	Composition (AIC)	PSA outlet, Biochar unit exhaust

Each transmitter sends a 4–20 mA analog or digital signal to the controller in the DCS system, which compares the measured value to the set point and adjusts the manipulated variable accordingly.

8.3 Control Philosophy and System Architecture

The hydrogen production plant employs a Distributed Control System (DCS) integrated with Programmable Logic Controllers (PLCs) to manage both process and safety operations. The DCS handles continuous process variables (temperature, flow, pressure, level), while the PLC system manages discrete control tasks such as valve sequencing in the PSA unit and the adsorption/desorption cycles in the biochar column.

The control system architecture includes:

Field Level: Sensors (transmitters) and actuators (valves, drives).

Control Level: PID controllers within DCS/PLC for automatic regulation.

Supervisory Level: Human–Machine Interface (HMI) for operator monitoring and control.

Safety Level: Emergency Shutdown (ESD) system and Fire & Gas (F&G) system for safety protection.

Signal transmission is carried out through shielded industrial cables using standard communication protocols (e.g., HART, Foundation Fieldbus, or Modbus TCP/IP). All signals converge in a Central Control Room (CCR) equipped with display panels, alarm systems, and manual override functions.

8.4 Control Systems in Major Process Units

8.4.1 Feed preheater (HX-01)

Controlled Variable: Outlet temperature of natural gas.

Manipulated Variable: Flow rate of hot flue gas or heating medium.

Control System: A temperature control loop (TIC) regulates the outlet gas temperature. A thermocouple measures the gas temperature and transmits the signal to the temperature controller, which adjusts the control valve on the hot flue gas line.

Purpose: To ensure that the natural gas is preheated to approximately 360°C before entering the hydrogenator reactor, optimizing reforming efficiency.

8.4.2 Hydrogenation reactor (HT-01)

Controlled Variables: Reactor outlet temperature and hydrogen flow.

Manipulated Variables: Heating medium temperature and H₂ feed rate.

Control Scheme: Cascade temperature flow control ensures stable operation by using a master temperature controller to adjust hydrogen feed flow.

Purpose: To achieve complete conversion of organic sulfur compounds to hydrogen sulfide (H₂S) while preventing overheating and catalyst damage.

8.4.3 Desulfurization reactor (DS-01)

Controlled Variable: Outlet sulfur concentration.

Manipulated Variable: Inlet gas temperature or bypass flow rate.

Instrumentation: Gas analyzers (for H₂S content) and temperature transmitters.

Control Strategy: Feedback control loop with a sulfur analyzer linked to a temperature controller ensures sulfur removal to below 0.1 ppm.

Purpose: To protect nickel catalysts in the reformer from sulfur poisoning.

8.4.4 Pre-reformer (PR-01)

Controlled Variable: Reactor temperature.

Manipulated Variable: Steam-to-carbon ratio.

Control System: **Temperature control loop** with a secondary flow control for steam.

Purpose: To convert higher hydrocarbons into methane and prevent coke formation on the reformer catalyst.

8.4.5 Primary steam methane reformer (R-01)

Controlled Variables: Tube outlet temperature and flue gas oxygen content.

Manipulated Variables: Fuel gas flow rate and air-to-fuel ratio.

Control System: Multi-point temperature control using thermocouples along reformer tubes, coupled with oxygen trim control to optimize combustion.

Purpose: To maintain reformer temperature between 850–900°C and ensure efficient methane conversion.

Safety Interlocks: Automatic shutdown on high tube metal temperature or low furnace draft pressure.

8.4.6 High Temperature Shift Reactor (HTCO)

Controlled Variables: Reactor outlet temperature and CO concentration.

Manipulated Variables: Steam flow rate and cooling water rate.

Control System: **Cascade control** with master temperature controller and slave steam-flow controller.

Purpose: To maintain optimal shift reaction conditions, ensuring maximum CO conversion to hydrogen.

8.4.7 Process gas boiler (B-01) and steam drum (ST-01)

Controlled Variables: Steam pressure and drum water level.

Manipulated Variables: Fuel gas flow and feedwater flow.

Control Strategy: Three-element drum level control (measuring steam flow, feedwater flow, and drum level) to stabilize boiler operation under varying loads.

Purpose: To ensure a constant supply of high-pressure steam for reforming and power generation.

8.4.8 Pressure swing adsorption (PSA-01)

Controlled Variables: Product hydrogen purity and bed pressure.

Manipulated Variables: Valve sequencing and purge flow rate.

Control System: Sequential control executed by PLC based on time, pressure, and composition inputs.

Purpose: To ensure continuous hydrogen purification and achieve 99.9% purity through automated cyclic adsorption and regeneration stages.

Instrumentation: Pressure transmitters, hydrogen analyzers, solenoid valves.

8.4.9 Biochar adsorption column (BC-01)

Controlled Variables: CO₂ concentration at the outlet and adsorption bed temperature.

Manipulated Variables: Gas flow rate and regeneration heating rate.

Control System: **Feedback control loop** using CO₂ gas analyzers and thermocouples.

Purpose: To maintain efficient CO₂ capture and ensure biochar regeneration cycles are triggered automatically.

Safety Interlocks: Pressure relief valves and temperature cutoffs to prevent thermal degradation of biochar.

8.4.10 Compressors and pumps

Controlled Variables: Discharge pressure and flow rate.

Manipulated Variables: Motor speed or valve position.

Control Strategy: Pressure–flow cascade control with safety interlock for overpressure protection.

Purpose: To maintain stable gas and liquid flows throughout the process.

8.5 Automation and Control Strategy

The plant employs a hierarchical automation system consisting of three layers:

1. **Regulatory Control Layer:** Handles all PID-based control loops (TIC, FIC, PIC, LIC).
2. **Supervisory Control Layer:** Oversees coordination between units, energy optimization, and performance monitoring.
3. **Safety and Shutdown Layer:** Dedicated to preventing hazardous incidents through automatic isolation, venting, and emergency trip systems.

The control system is configured for automatic mode during steady-state operation, with manual override capability during startup, shutdown, or maintenance. Alarms are categorized into three levels:

- **High Priority:** Triggers automatic shutdown (e.g., hydrogen leak).
- **Medium Priority:** Alerts operators to deviations (e.g., pressure drift).
- **Low Priority:** Advises minor corrections or monitoring.

The control room interface displays real-time trends, process diagrams, and alarms, providing operators with an integrated overview of the entire plant's status.

8.6 Safety Instrumented Systems (SIS)

Safety is ensured through the integration of Safety Instrumented Systems (SIS) which operate independently from the DCS. These include:

- **Emergency Shutdown (ESD):** Automatically shuts down the process on detection of high pressure, temperature, or hydrogen leakage.
- **Fire and Gas Detection System (F&G):** Equipped with infrared flame detectors, hydrogen gas detectors, and heat sensors.

- **Pressure Relief Valves (PRVs):** Installed on all pressure vessels, heat exchangers, and reformer tubes.
- **Flare System (FL-01):** Burns off excess or relief gases under safe conditions.
- **Inert Gas Purging:** Nitrogen purging during startup and shutdown to prevent explosive gas mixtures.

8.7 Instrumentation Diagram (P&ID Overview)

A Process and Instrumentation Diagram (P&ID) provides a graphical representation of the control philosophy and instrumentation used throughout the hydrogen production plant. It serves as a critical design document that integrates the process flow diagram (PFD) with detailed control and safety features, showing how instruments, controllers, and actuators interact to achieve automatic process control.

In this design, the P&ID outlines how temperature, pressure, flow, level, and composition control loops are configured across major process units, as well as how interlocks and safety systems are connected. Each control loop is represented by standard instrument identification codes such as TIC (Temperature Indicating Controller), PIC (Pressure Indicating Controller), FIC (Flow Indicating Controller), and LIC (Level Indicating Controller).

8.7.1 Description of the P&ID

The P&ID for the hydrogen production plant is divided into three primary control zones:

Zone 1: Reforming and Feed Preparation Section

This zone includes the feed preheater (HX-01), hydrogenation reactor (HT-01), desulfurization unit (DS-01), and pre-reformer (PR-01).

- Each reactor has temperature transmitters (TT) connected to temperature controllers (TIC) that manipulate control valves regulating heating or cooling media.
- Flow controllers (FIC) regulate natural gas and steam feed rates to maintain the steam-to-carbon ratio.
- Pressure transmitters (PT) monitor inlet and outlet pressures, linked to safety interlocks that trip compressors in the event of overpressure.

Zone 2: Reforming and Shift Conversion Section

This includes the primary reformer (R-01), high-temperature shift (WGS-01) reactor.

- The reformer uses multiple TICs to monitor tube outlet temperatures and a fuel gas control valve (FCV) connected to a combustion air control valve (ACV) through an oxygen analyzer (OAI) for combustion optimization.
- The shift converters are equipped with cascade temperature–flow loops; the primary controller adjusts cooling water or steam flow to maintain outlet temperature.
- Gas analyzers (AI) downstream of each reactor measure CO concentration and send feedback to maintain maximum CO conversion efficiency.

Zone 3: Separation and Carbon Capture Section

This zone covers the process gas boiler (B-01), steam drum (ST-01), pressure swing adsorption (PSA-01), **and** biochar adsorption column (BC-01).

- The boiler uses a three-element level control system (LIC) with signals from drum level, feedwater flow, and steam flow transmitters.

- In the PSA unit, PLC-based sequential control handles valve timing for adsorption, equalization, and regeneration. Gas analyzers (AI-H₂) monitor hydrogen purity.
- The biochar adsorption column includes temperature (TIC) and composition control (AIC-CO₂) loops to maintain high CO₂ capture efficiency.
- A pressure control loop (PIC) ensures the bed operates within the desired adsorption pressure range, while the regeneration system is governed by a temperature control loop.

8.7.2 Control interconnections and signal flow

All instrument signals whether analog or digital are transmitted to the Distributed Control System (DCS) located in the Central Control Room (CCR). The DCS interfaces with field instruments via smart transmitters using HART or Foundation Fieldbus protocols. The DCS is also integrated with the Safety Instrumented System (SIS), which operates independently for emergency shutdown functions.

- **Analog signals (4–20 mA):** Used for transmitters (temperature, pressure, flow, level).
- **Digital signals (ON/OFF):** Used for solenoid valves, limit switches, and alarms.
- **Communication links:** Modbus TCP/IP for PLC–DCS integration, ensuring seamless real-time data exchange between process and safety systems.

The signal flow for each control loop follows this sequence: Sensor (measurement) → Transmitter → Controller (DCS/PLC) → Final Control Element (valve, pump, heater) → Process Variable adjustment.

8.7.3 Safety and alarm integration

All control loops are supported by alarm set points and interlocks.

- **High–High (HH)** or **Low–Low (LL)** signals trigger automatic plant trips or isolation valves.
- Hydrogen detectors in the reformer area send immediate shutdown commands to the ESD system.
- Pressure Safety Valves (PSVs) and Pressure Relief Valves (PRVs) discharge gases to the flare stack (FL-01) to prevent equipment rupture.
- Each zone has emergency push buttons for manual shutdown in addition to automatic control functions.

8.7.4 Standard instrumentation symbols used

Symbol	Meaning
TIC	Temperature Indicating Controller
FIC	Flow Indicating Controller
PIC	Pressure Indicating Controller
LIC	Level Indicating Controller
AIC	Analyzer Indicating Controller
TT	Temperature Transmitter
FT	Flow Transmitter
PT	Pressure Transmitter
LT	Level Transmitter
CV	Control Valve
PSV/PRV	Pressure Safety/Relief Valve
AI	Analyzer Indicator

These standard ISA (Instrumentation Society of Automation) symbols are used to maintain consistency and international standardization across the instrumentation drawings (ISA S5.1 Standard, 2020).

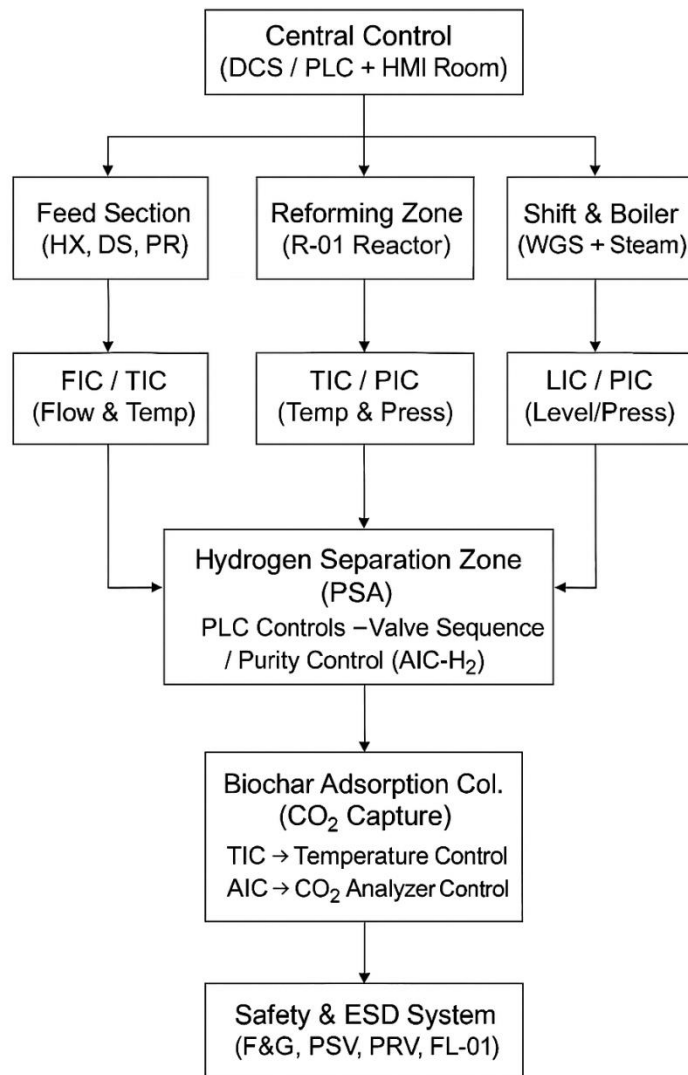


Figure 8.1: Simplified Process Control Block Diagram for Hydrogen Production Plant Integrated with Biochar-Based Carbon Capture System

CHAPTER NINE

9.0 ENVIRONMENTAL MANAGEMENT AND SAFETY

9.1 Introduction

The hydrogen production plant with biochar-based carbon capture involves several processes that operate at high temperatures, elevated pressures, and handle flammable and toxic substances. These conditions pose risks both to the environment and human health if not properly managed. Consequently, the plant integrates environmental management and safety strategies that conform to international best practices such as ISO 14001:2015 for environmental management, ISO 45001:2018 for occupational health and safety, and Occupational Safety and Health Administration (OSHA) standards.

The philosophy adopted is preventive rather than corrective, emphasizing risk identification, minimization, mitigation, and continuous monitoring. This chapter highlights the environmental and safety measures integrated into the design, construction, and operation of the plant.

9.2 Environmental Impact Assessment (EIA)

9.2.1 Air emissions

The primary sources of air emissions are the fired reformer furnace, process vent gases, and the PSA unit purge streams. These emissions contain carbon dioxide (CO₂), carbon monoxide (CO), oxides of nitrogen (NO_x), and traces of methane or hydrogen. Without control, these emissions contribute to air pollution and climate change. The use of low-NO_x burners, optimized combustion systems, and biochar-based CO₂ capture reduces these emissions significantly, ensuring compliance with regulatory limits.

9.2.2 Water usage and discharge

Large volumes of water are required for steam generation, cooling, and cleaning operations. Wastewater arises from boiler blowdown, cooling tower blowdown, and water treatment rejects. These effluents may contain high dissolved solids, chemicals, and thermal load, which could pollute water bodies if discharged untreated. To mitigate this, the plant incorporates a water treatment system where effluents undergo neutralization, clarification, and recycling, thereby reducing freshwater demand.

9.2.3 Solid waste generation

The plant produces solid wastes such as spent ZnO catalysts from desulfurization, exhausted biochar from CO₂ adsorption, filter residues, and packaging materials. Some of these wastes are hazardous and require special handling. Recycling, safe disposal, and regeneration methods are implemented, in line with the Basel Convention on hazardous waste management.

9.2.4 Noise pollution

Compressors, pumps, blowers, and other rotating equipment generate noise that may affect workers and nearby communities. Noise suppression measures, such as acoustic enclosures, silencers, and green buffer zones, are incorporated to keep noise levels within regulatory limits.

9.2.5 Land and ecosystem impact

The plant occupies land space that may disrupt local ecosystems. Careful site selection, landscaping, and environmental offsets (such as tree planting) are included to restore ecological balance and minimize biodiversity loss.

9.3 Emission and Waste Management

9.3.1 Air pollution control

The integration of a biochar-based CO₂ capture system ensures that carbon dioxide from the reformer and PSA purge streams is effectively adsorbed, thus contributing to greenhouse gas reduction. Flare stacks are used to safely combust vented gases during process upsets, while continuous monitoring of stack emissions ensures compliance with limits set by NESREA and IFC guidelines.

9.3.2 Wastewater treatment

Effluent treatment plants (ETPs) are designed to handle process water, boiler blowdown, and cooling water discharges. Treated water is either recycled for reuse in the plant or discharged in compliance with environmental standards. This minimizes water wastage and reduces environmental impact.

9.3.3 Solid waste handling

Spent catalysts are returned to suppliers for regeneration and metal recovery. Exhausted biochar is either regenerated on-site or disposed of as non-hazardous waste. Other general wastes are segregated at the source and recycled where possible.

9.4 Occupational Health and Safety (OHS)

9.4.1 Hazard identification and risk assessment (HIRA)

Comprehensive HIRA is carried out to assess risks associated with hydrogen leakage, catalyst handling, high-pressure steam, and rotating machinery. Identified hazards are

categorized, and preventive as well as protective measures are integrated into plant operations.

9.4.2 Personal protective equipment (PPE)

All staff are required to wear PPE appropriate to their roles. These include flame-retardant clothing, helmets, ear protection, gloves, goggles, safety boots, and portable gas detectors for hydrogen and H₂S.

9.4.3 Permit-to-work (PTW) system

A strict PTW system governs all high-risk activities such as hot work, confined space entry, electrical work, and maintenance. Each permit includes hazard assessment, safety measures, and authorization before work commences.

9.4.4 Training and awareness

Regular training programs are conducted for staff, including induction training, refresher safety courses, and emergency response drills. This ensures that all personnel are familiar with plant safety protocols.

9.5 Process Safety Management (PSM)

9.5.1 Inherently safer design (ISD)

The plant is designed with corrosion-resistant materials, optimized layout for minimum leak paths, and redundancy in critical systems to reduce inherent risks.

9.5.2 Safety instrumented systems (SIS)

Safety interlocks, shutdown valves, pressure relief valves, and redundant control systems are installed. These systems ensure automatic plant shutdown during unsafe conditions, minimizing the chance of escalation.

9.5.3 Hazard and operability studies (HAZOP)

Regular HAZOP studies are carried out to identify deviations from normal operating parameters. Recommendations from these studies are implemented to continuously improve plant safety.

9.5.4 Layer of protection analysis (LOPA)

LOPA methodology is applied to ensure that each critical hazard scenario has multiple layers of protection, such that failure of one barrier does not result in catastrophic incidents.

9.6 Fire and Explosion Safety

9.6.1 Hydrogen fire detection

Since hydrogen flames are invisible, infrared and ultraviolet flame detectors are used for quick identification of leaks and fires.

9.6.2 Explosion-proof equipment

All electrical installations within classified hazardous areas are explosion-proof and compliant with IECEx/ATEX standards.

9.6.3 Ventilation systems

Adequate ventilation in enclosed spaces prevents accumulation of flammable gases.

Positive-pressure ventilation is used in control rooms to prevent gas ingress.

9.6.4 Fire protection systems

The plant is equipped with fire hydrants, sprinklers, foam systems, and portable extinguishers. A dedicated firewater tank and ring-main system ensure reliable supply during emergencies.

9.7 Environmental and Safety Monitoring

9.7.1 Emission monitoring

Continuous Emission Monitoring Systems (CEMS) track CO₂, CO, NO_x, and O₂ levels from stack gases.

9.7.2 Effluent monitoring

Effluent discharge quality is continuously monitored for parameters such as pH, COD, BOD, and TSS.

9.7.3 Workplace monitoring

Workplace hydrogen, H₂S, and noise levels are monitored using fixed and portable devices to ensure compliance with occupational exposure limits.

9.7.4 Safety audits

Annual audits and third-party inspections are conducted to identify gaps and recommend corrective actions.

9.8 Regulatory Compliance

9.8.1 National regulations

The plant complies with Nigerian Environmental Standards and Regulations Enforcement Agency (NESREA) rules and Department of Petroleum Resources (DPR) safety guidelines.

9.8.2 International standards

Operations are aligned with OSHA (2018) standards, IFC (2017) EHS Guidelines, and CCPS (2007) risk-based safety principles.

9.8.3 ISO standards

ISO 14001:2015 – Environmental management systems.

ISO 45001:2018 – Occupational health and safety.

CHAPTER TEN

10.0 COST ESTIMATION AND ECONOMIC ANALYSIS

10.1 Introduction

This chapter presents the capital cost estimation, operating cost evaluation, and economic analysis of a hydrogen production plant designed to produce 10,000 m³/h of hydrogen ($\approx$ 899 kg/h) via Steam Methane Reforming (SMR) integrated with a biochar-based carbon capture system (CCS).

The estimation follows the factorial method as outlined in Coulson and Richardson's Chemical Engineering Design, 6th Edition, Chapter 6, which provides a reliable basis for preliminary cost evaluation of chemical process plants. Calculations were adjusted to October 2025, using an exchange rate of ₦1,480 per USD, and all costs are expressed in 2025 U.S. dollars and Nigerian Naira.

The objective of this analysis is to determine the total capital investment (TCI), annual operating cost, annual revenue, and economic indicators such as payback period, which collectively define the project's economic feasibility.

10.2 Basis and Assumptions

Parameter	Value	Remark
Plant Capacity	10,000 m ³ /h H ₂	Equivalent to 899 kg/h
Operating Availability	85%	7,875 tonnes H ₂ /year
H ₂ Lower Heating Value (LHV)	120 MJ/kg	Power equivalent $\approx$ 30 MW
Exchange Rate	¥1,480 per USD	October 2025
Estimation Method	Coulson & Richardson (Factorial Method)	Chapter 6
Cost Year	2025 (adjusted with CEPCI)	CEPCI 2025 $\approx$ 815
Working Capital	15% of Fixed Capital	Standard assumption
OPEX	40% of Hydrogen Revenue	Benchmark for SMR plants
Hydrogen Selling Price	USD 3.00/kg	Industrial hydrogen price
Carbon Credit	USD 100/tonne CO ₂ captured	Biochar CCS benefit
CO ₂ Captured	70,875 tonnes/year	9 kg CO ₂ per kg H ₂
Accuracy Level	$\pm$ 30% (AACE Class 4)	Conceptual design level

10.3 Methodology

The factorial method (Coulson & Richardson, 6th Ed.) estimates total plant investment using a base purchased equipment cost (PCE) and multiplying factors for direct and indirect costs.

The relationships are as follows:

$$PPC = PCE \times (1 + f_1 + f_2 + \dots + f_9) \quad \dots \text{Eqn (40)}$$

$$CF = PPC \times (1 + f_{10} + f_{11} + f_{12}) \quad \dots \text{Eqn (41)}$$

$$TCI = CF + (0.15 \times CF) \quad \dots \text{Eqn (42)}$$

Where:

PPC = Total physical plant cost

CF = Fixed capital

TCI = Total capital investment

$f_1 - f_9$ represent direct cost factors (erection, piping, buildings, etc.)

$f_{10} - f_{12}$ represent indirect cost factors (design, contractor's fee, contingency)

For fluid-process plants like SMR, the following standard values are used:

$$(f_1 + f_2 + \dots + f_9) = 2.40, (f_{10} + f_{11} + f_{12}) = 0.45$$

Thus:

$$PPC = 3.40 \times PCE, CF = 1.45 \times PPC, TCI = 1.15 \times CF \quad \dots \text{Eqn (43)}$$

10.4 Purchased Equipment Cost (PCE)

The purchased equipment cost covers all major process units necessary for hydrogen production via SMR with integrated carbon capture.

S/N	Equipment Unit	Function	Installed Cost (USD Million)
1	Feed Desulfurization Unit	Removes sulfur compounds from natural gas	0.30
2	SMR Reformer Furnace	Main methane reforming reactor	5.00
3	Water–Gas Shift (WGS) Reactor	Converts $\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + \text{H}_2$	0.45
4	PSA Unit	Purifies hydrogen to $\geq 99.9\%$	1.00
5	Compressors and Pumps	Compresses gases and circulates fluids	1.70
6	Heat Exchangers and Boilers	Steam generation and cooling	0.60
7	Biochar Carbon Capture Unit	CO_2 adsorption using biochar	4.80
8	Cooling Tower	Removes excess process heat	0.20
9	Storage Tanks	Stores hydrogen and feedstocks	0.15
Total Purchased Equipment Cost (PCE)			14.20

Therefore,

$$PCE = 14.20 \text{ million USD}$$

10.5 Capital Cost Estimation

Using the factorial relationships:

1. **Total Physical Plant Cost (PPC):**

$$PPC = 14.20 \times 3.40 = 48.28 \text{ million USD}$$

2. **Fixed Capital (CF):**

$$CF = 48.28 \times 1.45 = 70.01 \text{ million USD}$$

3. **Working Capital (15% of CF):**

$$CW = 0.15 \times 70.01 = 10.50 \text{ million USD}$$

4. **Total Capital Investment (TCI):**

$$TCI = CF + CW = 70.01 + 10.50 = 80.51 \text{ million USD}$$

5. **Conversion to Nigerian Naira:**

$$80.51 \times 1,480 = \text{₦}119.16 \text{ billion}$$

Total Capital Investment (TCI): USD 80.51 million (₦119.16 billion)

10.6 Operating Cost and Revenue Analysis

Item	Calculation Basis	Value (USD Million/Year)
Hydrogen Revenue	7,875,000 kg × \$3.00/kg	23.63
Carbon Credit Revenue	70,875 tCO ₂ × \$100/t	7.09
Total Annual Revenue	—	30.72
Operating Cost (OPEX)	40% of H ₂ revenue	9.45
Net Annual Income (Before Tax)	Revenue – OPEX	21.27

10.7 Economic Evaluation

Payback Period (PBP):

$$PBP = \frac{TCI}{Net\ Annual\ Income} = \frac{80.51}{21.27} = 3.78\ years$$

10.8 Summary of Cost and Economic Indicators

Parameter	Symbol	Value (USD Million)	Equivalent (₦ Billion)
Purchased Equipment Cost	PCE	14.20	21.02
Total Physical Plant Cost	PPC	48.28	71.47
Fixed Capital	CF	70.01	103.61
Working Capital	CW	10.50	15.55
Total Capital Investment (TCI)	—	80.51	119.16
Annual Hydrogen Revenue	—	23.63	34.97
Carbon Credit Revenue	—	7.09	10.49
Total Annual Revenue	—	30.72	45.46
Operating Cost (OPEX)	—	9.45	13.98
Net Annual Income	—	21.27	31.48
Payback Period (PBP)	—	3.78 years	—

CHAPTER ELEVEN

11.0 SUMMARY, CONCLUSION AND RECOMMENDATIONS

11.1 Summary

This project presented the design of a hydrogen production plant integrated with a biochar-based carbon capture system, targeting a production capacity of 10,000 m³/h of hydrogen through the Steam Methane Reforming (SMR) process. The design integrates a biochar adsorption unit for carbon dioxide capture, providing a sustainable and low-cost alternative to conventional amine scrubbing methods for emission reduction.

The process begins with natural gas feed from the Nigerian Liquefied Natural Gas (NLNG) facility, which is preheated in a heat exchanger and passed through a hydrogenation reactor to convert organic sulfur compounds to hydrogen sulfide. The gas then enters the desulfurization reactor where zinc oxide adsorbent removes the H₂S. The desulfurized feed is mixed with steam and sent to a pre-reformer, where higher hydrocarbons are converted into methane to prevent coking in the main reformer.

In the steam methane reformer (R-01), methane reacts with steam over a nickel catalyst at high temperature (850–900°C) and moderate pressure (28 atm) to form a synthesis gas (syngas) rich in hydrogen and carbon monoxide. The syngas then flows to the High-Temperature Shift Converter (HTCO), where carbon monoxide reacts with steam to form additional hydrogen and carbon dioxide. The shift reaction enhances the overall hydrogen yield while reducing CO content.

The hydrogen-rich gas is then cooled and sent to a Pressure Swing Adsorption (PSA) unit, where high-purity hydrogen ($\geq 99.9\%$) is obtained by removing CO₂, CH₄, and other trace

gases through cyclic adsorption and desorption. The separated CO₂ stream is directed to a biochar adsorption column for capture and temporary storage. The biochar serves as a renewable and regenerative adsorbent with high affinity for CO₂, thereby reducing the plant's overall carbon footprint.

Material and energy balance calculations were performed for all units, and stainless steel (AISI 316) was selected for major equipment due to its excellent resistance to high temperature and corrosion. The plant's control philosophy employs a Distributed Control System (DCS) integrated with Programmable Logic Controllers (PLCs) for real-time monitoring and automation. PID control loops maintain stability of temperature, pressure, flow, and composition, while safety systems such as emergency shutdown (ESD), relief valves, and hydrogen detectors protect both personnel and equipment.

The overall plant design demonstrates the technical and environmental viability of integrating hydrogen production with a biochar-based carbon capture system. The process achieves high hydrogen yield, reduced CO₂ emissions, and improved energy efficiency through effective heat integration between process streams.

11.2 Conclusion

1. Objective 1: Process Design Development

A comprehensive process design for the production of 10,000 m³/hr of hydrogen was successfully developed. The design integrates major process units, including feed preheating, hydrogenation, desulfurization, pre-reforming, steam methane reforming, high-temperature shift conversion, hydrogen purification through Pressure Swing Adsorption (PSA), and biochar-based carbon capture. Each unit operation was designed to operate under optimal conditions of temperature, pressure, and flow to ensure efficient hydrogen

generation and minimal emissions. The overall process flow diagram (PFD) and material balance confirm the technical feasibility of the plant configuration.

2. Objective 2: Material and Energy Balance Analysis

Detailed material and energy balances were performed across each process unit to determine feedstock requirements, energy consumption, and hydrogen yield. The calculations established the appropriate steam-to-carbon ratio for reforming and quantified the thermal energy recovery through heat integration between reformer flue gases and process streams. Results indicate that the plant can achieve high conversion efficiency and energy utilization while maintaining steady-state operation under design conditions.

3. Objective 3: Integration of Biochar into Carbon Capture System

The integration of biochar as a CO₂ adsorbent within the carbon capture section was successfully achieved. Biochar demonstrated high CO₂ adsorption capacity and operational feasibility under process gas conditions. The potential for producing biochar locally from Nigerian agricultural residues such as corn cobs and rice husks offers a sustainable and low-cost solution to emission control. This innovation enhances the environmental performance of the plant while promoting resource circularity and local value addition.

4. Objective 4: Material Selection and Equipment Design

Appropriate materials of construction were selected based on operating temperature, pressure, and chemical environment. Stainless steel AISI 316 was chosen for high-temperature and corrosive sections such as the reformer, heat exchangers, and piping systems, while carbon steel and alloy steel were selected for auxiliary components where

suitable. The equipment sizing and material specifications ensure mechanical integrity, safety, and longevity under continuous industrial operation.

5. Objective 5: Environmental Benefits of Biochar-Assisted Carbon Capture

The inclusion of the biochar-based CO₂ capture unit significantly improves the environmental profile of the hydrogen production process. Compared to conventional SMR plants without capture systems, the integrated design reduces CO₂ emissions substantially while utilizing a renewable adsorbent. The captured CO₂ can be stored, reused, or applied in enhanced agricultural applications, further supporting a circular economy. This integration aligns with Nigeria's commitment to global carbon reduction goals and the United Nations Sustainable Development Goals (SDGs).

6. Objective 6: Economic Feasibility Evaluation

Preliminary economic analysis indicates that the designed hydrogen plant is economically viable when operated at optimal conditions. The use of biochar as a low-cost, locally available carbon capture medium lowers operational expenses compared to imported amine systems. Heat recovery and process optimization further reduce fuel consumption and energy costs. When benchmarked against global hydrogen production systems, the design demonstrates competitive production costs and favorable long-term sustainability, especially when scaled up or integrated with renewable energy sources.

11.3 Recommendations

1. The designed hydrogen production process should be implemented at a pilot scale to validate the design data and assess real operating performance.

2. Further optimization of the process should be carried out using simulation software such as Aspen HYSYS or CHEMCAD to improve efficiency and energy integration.
3. Locally produced biochar from Nigerian biomass residues should be developed and characterized to determine its CO₂ adsorption capacity and regeneration potential.
4. The plant should be integrated with renewable or waste heat recovery systems to reduce fossil fuel consumption and improve overall thermal efficiency.
5. A comprehensive techno-economic analysis and life cycle assessment should be conducted to evaluate the cost-effectiveness and environmental impact of the process.
6. Advanced control systems such as model predictive control (MPC) should be introduced to enhance process stability, safety, and performance.
7. Government and industry should support hydrogen development through policies that encourage clean technologies and biochar-based carbon capture systems.
8. Continuous improvement of plant safety systems, including leak detection, automatic shutdowns, and HAZOP studies, should be ensured.
9. Research should be extended to explore potential uses of captured CO₂ for value-added applications such as enhanced oil recovery and synthetic fuels.
10. Nigeria should establish a national hydrogen strategy to promote sustainable hydrogen production and integrate biochar carbon capture into future energy projects.

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